Boundary-response closure and controlled intermittency tests for Casimir-pressure fluctuations

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(Dated: September 16, 2026)

The mean Casimir force does not determine time-averaged vacuum-pressure statistics. We ask which nonlinear boundary data suffice to predict these statistics and whether their scale dependence is intermittent. For a quantized distributed transmission line with internal boundary resonators, we retain local-pressure operator ordering and ground-state dynamics. Two microscopic boundaries can share, at one reference geometry, the harmonic model, the first-order two-point response, and the leading zero-frequency quartic coefficient, yet produce opposite first-order pressure-variance corrections. An analytic continuum mode-tail bound preserves this separation: these static data do not close the pressure observable. We calculate ordered cumulants through eighth order at first order, the complete next cosine coefficient in an occupation-refined three-mode regulator, and unexpanded-cosine statistics in finite projected Hamiltonians. These resolve nonlinear pressure-shape changes relative to the harmonic device. A fourteen-resonance scale test, independent estimator controls, and finite-strength depth comparisons show changing crossover slopes rather than a persistent anomalous exponent. At fixed finite coordinate count, an all-excitation result shows that ultimate long-window moments are controlled by discrete energy-gap paths, exponentially dependent on squared Gaussian averaging width. Rare-event stability can require stronger controls than moment agreement: under one occupation-and-selection refinement, the eighth increment moment changes by 0.0052% while a fixed hard-threshold probability changes by 23.6%. The results establish incomplete static boundary closure and finite-regulator nonlinear pressure-shape changes; they do not establish persistent Casimir-pressure intermittency or an undisturbed record of vacuum bursts.

Keywords: Casimir effect; Fluctuation phenomena; Correlation functions; Mesoscopic systems

I. INTRODUCTION

The Casimir effect is usually characterized by its mean interaction force, but the pressure in a bounded quantum field also fluctuates in its ground state. Our central aim is to determine whether this vacuum stress can have intermittent structure, which boundary mechanisms could produce it, and what a microscopic model must retain to predict it. We examine three distinct possibilities: rare large pressure outcomes, persistent anomalous dependence of pressure moments on the averaging time, and temporally clustered outcomes under a specified measurement. They require different tests; evidence for one is not sufficient to establish the others. Vacuum force correlations and smeared stress distributions establish that fluctuations beyond the mean are meaningful observables [1–4]. We retain the pressure laws and their scale dependence while separately testing the changes produced by boundary nonlinearity. A prediction of registered bursts additionally requires the measurement dynamics and its disturbance of the field.

We begin with a static nonlinear boundary in its ground state, using a distributed transmission line terminated by internal resonators as a controlled Casimir geometry. Superconducting circuits provide a physical setting in which a nonlinear element interacts with many quantum modes [5, 6]. Quantizing the line and boundary together allows us to test how the resonance hierarchy, nonlinear strength, and averaging time affect the pressure statistics. The observable calculated here is the time-averaged pressure just inside the boundary. Subtracting its harmonic vacuum mean fixes an additive reference. For ordinary central moments at finite coupling, we additionally subtract the interacting ground-state mean; higher cumulants and width increments are unaffected by this scalar shift. Neither subtraction constructs the net force on a plate or subtracts two pressure probability distributions. Our closure question is which boundary-response functions, together with the correct pressure observable, suffice to reproduce the microscopic model's ordered cumulants and selected relations between averaging scales. Agreement is tested for the same geometry, state, and averaging windows. Finite temperature and a declared detector are additional physical tests, while driven boundaries constitute a separate extension of the static vacuum question.

Higher cumulants make the statistical question more precise. The third cumulant measures asymmetry about the mean; the fourth measures a departure from Gaussian shape after contributions fixed by the variance are removed. Sixth and eighth cumulants test further information that these lower orders do not determine. A crucial baseline is that a Gaussian field need not have Gaussian pressure. Pressure contains products of field operators, and its higher cumulants can already be nonzero for a completely harmonic boundary. Exact stress distributions, high-moment calculations, and spectral diagonalization illustrate this distinction [3, 4, 7–10]. We there-

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fore use two explicit comparisons: the pressure statistics of the original harmonic device, and a Gaussian field reproducing the retained interacting two-point response. The second comparison needs its own missing-transition controls. Non-Gaussian pressure by itself is evidence for neither boundary nonlinearity nor intermittency.

The closure question arises when the internal boundary coordinates are eliminated. Their influence on the line can then be represented by frequency-dependent boundary coefficients. The quadratic coefficient Γ_2 specifies the harmonic response; Γ_4 couples four boundary-field arguments, and Γ_6 couples six. These coefficients are often called boundary vertices. A useful reduced description need not retain every microscopic coordinate, but it must retain enough of this information to reproduce the pressure statistics under consideration. We test a closure by comparing its predictions with the microscopic model for the same local pressure, averaging width, geometry, and nonlinear order. The reduction must also preserve the pressure observable and its quantum ordering. Generalized dynamical boundary conditions and nonlinear fluctuation-response relations provide the broader setting for this comparison [11, 12].

The distinction between static and frequency-dependent boundary data gives a direct test. We construct boundaries that have the same harmonic model and the same leading four-field coefficient at zero frequency, but distribute their quartic coupling differently among internal resonators. A quadratic stiffness adjustment fixes the two-point response through first order at one reference geometry. Thus the comparison isolates information in the nonlinear frequency dependence while the calibrated linear response remains common. The resulting first-order pressure-variance corrections can have opposite signs, and an analytic bound on omitted modes preserves their separation in the continuum coefficient. For larger branch families, we quantify the remaining range of pressure predictions when progressively more frequency-dependent boundary data are supplied. Nonlinear corrections to Casimir energies and to fluctuational electrodynamics have established precedents [13–16]. Those works concern nonlinear boundary or medium effects on mean energies and fields in models distinct from the present circuit. The question here is which specified boundary data determine an ordered pressure statistic.

Increasing the cumulant order and increasing the nonlinear order answer different questions. A single quartic field interaction can change pressure cumulants well beyond fourth order, because each pressure factor is itself quadratic in the field. We derive its first-order contributions through the eighth pressure cumulant, including both the change of ground state and the change of time evolution. We also examine the next power of a dimensionless coupling multiplier β for a stable cosine boundary potential. Its second-order pressure coefficient contains both two quartic insertions and the next term in the microscopic potential. Expressing that sixth-degree term in normal-ordered form produces quadratic and quartic contributions that must accompany the sextic contribution. These complete second-order coefficients are checked at a three-mode regulator. The first-order two-point calibration is held fixed; second-order changes of that response remain part of the prediction.

To study scale dependence, we compare pressure averages over two widths. If O_τ denotes the pressure averaged over width τ, their difference is the Hermitian operator $X_\tau = O_\tau - O_{b\tau}$, with $b > 1$. Its even moments are nonnegative. Dividing the moment of order p by the variance to power $p/2$ removes the overall fluctuation amplitude; dividing again by the same quantity in the harmonic model isolates the nonlinear change in distribution shape. The harmonic pressure's own non-Gaussian statistics and scale dependence remain part of the physical question and are examined separately. Because pressures at different times need not commute, all moments are computed as powers of this one averaged operator, with that ordering retained throughout.

A change of shape at one averaging width does not establish intermittency. Resonances can also produce changes over a finite range of widths. For a scaling anomaly induced by the nonlinear boundary relative to its harmonic reference, the test used here requires nonzero shape exponents for several moment orders over a common resolved interval, with that interval persisting or widening as the physical resonance range is extended. We therefore distinguish extending a frequency hierarchy from adding resonances within a fixed band, and both from increasing the number of retained line modes. We also vary continuously how the nonlinear coupling is allocated across the resonances. The completed first-order calculations reach fourteen branches spanning a frequency factor of 8192. They resolve allocation-dependent shape corrections whose slopes change with averaging width, rather than establishing a persistent anomalous exponent. Mode refinement, an additional increment ratio, and known-law controls test this interpretation; separate arithmetic checks exclude unresolved ratios of very small moments. These conclusions concern the first-order hierarchy coefficient. We then retain the full cosine at finite coupling, so that the next test does not assume the perturbative remainder is small. Independent occupation, local-orbital, bath, and line-mode refinements check the numerical model. A second family holds the reference phase variance fixed on every branch as the resonance hierarchy grows. This separates a weak-coupling infrared limit imposed by the original parameter choice from the question of finite-strength scale dependence. Increasing depth defines a family of different circuits: existing branch capacitances and stiffnesses change with the number of resonances R. Equal reference phase variance does not fix the complete linear response or the counterterm values across that family. This change of device must be distinguished from numerical refinement at fixed calibration, and from the first-order length comparison that keeps the microscopic boundary fixed within each spatial regulator.

A fixed device and an indefinitely extending resonance hierarchy need not share the same long-time law. Gaussian averaging suppresses transitions whose energy differences remain nonzero as the window widens. We prove that, for the confining full-cosine Hamiltonian at any fixed finite coordinate representation, every fixed even pressure moment is ultimately controlled by discrete energy-gap paths, even after all oscillator excitations are restored. Its leading term is an exponential in τ^2. This describes dependence on averaging width; the pressure distribution need not become Gaussian. Adding progressively slower resonances changes the relevant gaps and crossover widths, so the fixed-model theorem does not establish a uniform result along that sequence. It also does not show that the numerical windows sampled here have reached the ultimate regime.

The term *spectral* fixes the meaning of this intermittency test. At each width, we study the probability law associated with one Hermitian pressure operator. These laws do not by themselves define a joint classical process over all widths or a sequence of time-resolved bursts. A detector counting experiment can also impose its own operator ordering and backaction [17–20]. We first study the specified local-pressure spectral laws and ordered cross-scale products. We then introduce a declared Gaussian measurement instrument to calculate joint event probabilities and the energy deposited by the measurement. Its clustering is an operational result for that instrument; it is not identified with an undisturbed classical record of vacuum pressure.

A rare-event test must also be checked directly rather than inferred from moment convergence. In the finite-strength comparison below, increasing the excitation allowance and changing the retained spectral selection alters the eighth increment moment by only 0.0052%, but changes a specified large-outcome probability by 23.6%. Independently checked spectral sums resolve both quantities. The example shows why a stable high moment cannot certify a tail probability, even before asking whether the event law scales across a widening resonance hierarchy. Finite readout resolution is treated as part of the observable, and its effect is tested rather than used to assume convergence.

The present calculation stops short of a simultaneous limit in resonance depth, retained line modes, local occupation, pressure-reachable rank, and averaging width. That common-interior finite-strength hierarchy requires a separate numerical construction. We state the missing controls explicitly rather than treating the observed finite-window crossovers as evidence for or against an asymptotic exponent.

Sections II and III define the model, pressure, and calibration. Section IV develops the higher-cumulant calculation, followed by the variance closure criterion in Sec. V, the next nonlinear order in Sec. VI, and the scale test in Sec. VII. Sections VIII and IX present the central closure, geometry, finite-strength, scale, rare-tail, and measurement results. The appendices give the detailed finite-coupling construction and controls, distinguish fixed-window from moving-window limits, prove the fixed-coordinate all-excitation statement, and retain the extended convergence inventories. Appendix F gives the continuum variance bound.

II. DISTRIBUTED MODEL AND REDUCED BOUNDARY DATA

Let $\Phi(x,t)$ be the transmission-line flux on $0 < x < a$, with $\Phi(0,t) = 0$ and endpoint flux $\phi(t) = \Phi(a,t)$. The endpoint couples to internal branch fluxes $q_r(t)$, $r = 1,\dots,R$. The harmonic Lagrangian is

$$\mathcal{L}_0 = \frac{1}{2}\int_0^a \mathrm{d}x \left[C_0\dot{\Phi}^2 - \frac{(\partial_x\Phi)^2}{L_0}\right] + \frac{C_b\dot{\phi}^2 - K_b\phi^2}{2} + \sum_r \left[\frac{C_r\dot{q}_r^2}{2} - \frac{K_{cr}(\phi - q_r)^2}{2} - \frac{(K_{gr}+K_{Jr})q_r^2}{2}\right]. \tag{1}$$

Here C_0, L_0 are capacitance and inductance per unit length, C_b, C_r are positive boundary capacitances, and the K coefficients are inverse-inductance stiffnesses. The line and all boundary coordinates are quantized together. The wave speed is $v = (C_0L_0)^{-1/2}$. Josephson boundary circuits and weakly anharmonic circuit quantization motivate this type of model [5, 21]; no device bandwidth or detector model is assumed.

At imaginary frequency $i\xi$, harmonic branch elimination gives

$$\begin{aligned} D_r(i\xi) &= C_r\xi^2 + K_{cr} + K_{gr} + K_{Jr}, \\ h_r(i\xi) &= \frac{K_{cr}}{D_r(i\xi)}, \\ \Gamma_2(i\xi) &= C_b\xi^2 + K_b + \sum_r \left(K_{cr} - \frac{K_{cr}^2}{D_r(i\xi)}\right). \end{aligned} \tag{2}$$

Including the line then gives the endpoint Green function

$$G_{\phi\phi}(i\xi) = \left[\Gamma_2(i\xi) + \frac{\xi}{vL_0}\coth\frac{\xi a}{v}\right]^{-1}. \tag{3}$$

The line term tends to $1/(L_0a)$ at zero frequency. The frequency-domain covariance is $\hbar G_{\phi\phi}$; the equal-time endpoint variance is $\langle\phi^2\rangle_0 = (\hbar/\pi)\int_0^\infty G_{\phi\phi}(i\xi)\,\mathrm{d}\xi$.

The transfer function h_r maps endpoint flux to branch flux in the harmonic model. For the normal-ordered quartic branch perturbation defined below, the coefficient of the leading four-field term is

$$\Gamma_4(\xi_1,\xi_2,\xi_3,\xi_4) = \sum_r u_r \prod_{j=1}^4 h_r(i\xi_j), \qquad \sum_j \xi_j = 0. \tag{4}$$

Its effective-action prefactor is $\beta/4!$; the external harmonic propagators have not been attached. Thus Γ_4 is

neither a pressure cumulant nor, by itself, a measured nonlinear susceptibility. At subsequent powers of β, corrections to Γ_2 and Γ_4 arise together with higher-field vertices.

The mean interaction has a separate scattering definition. With a Dirichlet left end, define $k = \xi/v$ and $r_R = (k - L_0\Gamma_2)/(k + L_0\Gamma_2)$. The harmonic Casimir interaction energy is

$$E_C(a) = \frac{\hbar}{2\pi}\int_0^\infty \mathrm{d}\xi\, \log\left[1 + r_R(i\xi)e^{-2ka}\right], \qquad F_C = -\partial_a E_C. \tag{5}$$

The derivative holds the physical line and boundary coefficients fixed. These are harmonic interaction quantities, defined by subtracting the separated-boundary vacuum reference; F_C is not the mean of the harmonically centered local pressure P defined below. For the two-branch parameters of Sec. VIII A, the reproduced harmonic benchmark gives $E_C = -0.0796338853$ and $F_C = -0.0528854008$ at $a = 1$. The higher local-pressure cumulants below answer a different, complementary question: which boundary information determines the fluctuations of the specified inside stress in this same geometry? Their definition does not apply the energy subtraction to a probability distribution.

A closure specifies which of these field vertices and which pressure-source terms are retained. Its adequacy is tested by comparing the resulting $\kappa_p(\tau)$ with the microscopic boundary model, at the same perturbative order, geometry, averaging width τ, and operator ordering. At a complete finite spatial regulator, eliminating only the branches while retaining the line coordinates, their conjugate momenta, and pressure insertions preserves this comparison. For example, Gaussian conditioning gives $q_r = h_r * \phi + \eta_r$, where ϕ is the endpoint flux and η_r is the residual branch field. If $v_r = \langle q_r^2\rangle_0 = w_r + d_r$, with w_r and d_r the transmitted and residual variances, then

$$\left\langle :q_r^4: _{v_r} \right\rangle_{\eta_r} =: (h_r * \phi)^4 :_{w_r} . \tag{6}$$

Equation (6) is the Gaussian conditioning identity, written here in Euclidean notation. To apply branch elimination to the pressure moments, the corresponding contour kernels must retain the real-time order of each pressure product and the imaginary-time preparation of the ground state. If line modes are also eliminated, their contributions to the pressure source must be retained to preserve the same equality of pressure moments. Simply projecting the pressure onto a subset of modes defines a regulated approximation whose convergence is a separate question.

III. OBSERVABLE AND CALIBRATED NONLINEAR EXPANSION

We use the local pressure on the inside of the line endpoint at $x = a$,

$$P_{\mathrm{in}} = \frac{C_0}{2}\dot{\Phi}(a)^2 + \frac{1}{2L_0}\left[\partial_x\Phi(a)\right]^2, \qquad P = P_{\mathrm{in}} - \langle P_{\mathrm{in}}\rangle_0, \tag{7}$$

where C_0, L_0 are capacitance and inductance per unit length and the subscript 0 denotes the harmonic vacuum at the same geometry and regulator. The subtraction fixes the additive reference for the mean; all cumulants of order $p \geq 2$ remain those of the total local inside pressure. They are not an interaction-only probability law obtained by subtracting fluctuations at another separation. Throughout the nonlinear expansion, the canonical operator P is held fixed. The ground state and its Heisenberg evolution change with the Hamiltonian. For centered ordinary raw moments at finite coupling, we additionally use $O_\tau - \langle O_\tau\rangle_\beta I$ in the finite-model ground state, where O_τ is the Gaussian pressure average defined below. This scalar shift leaves increments and cumulants of order two and higher unchanged. The endpoint slope is evaluated directly from the line profile; replacing it by an eliminated nonlinear boundary equation would require the corresponding pressure-source insertions. At a finite spatial regulator this local pressure is also distinct from a generalized length force $-\partial H/\partial a$ defined at fixed normalized mesh coordinates. A geometry comparison must state which operator and which fixed calibration it uses.

For M real harmonic modes with positive frequencies ω_m, write

$$H_0 = \hbar\sum_{m=1}^{M}\omega_m a_m^\dagger a_m, \qquad q_r = \sum_m c_{rm}(a_m + a_m^\dagger), \qquad c_{rm} = T_{rm}\sqrt{\frac{\hbar}{2\omega_m}}. \tag{8}$$

The zero of energy is the harmonic vacuum. The line expansion is $\Phi(x) = \sum_m T_m(x)\sqrt{\hbar/(2\omega_m)}(a_m + a_m^\dagger)$. The mode profiles $T_m(x)$ and branch amplitudes T_{rm} are normalized using the full capacitance form,

$$C_0\int_0^a T_m(x)T_n(x)\,\mathrm{d}x + C_b T_m(a)T_n(a) + \sum_r C_r T_{rm}T_{rn} = \delta_{mn}. \tag{9}$$

With $d_m = T_m'(a)\sqrt{\hbar/(2\omega_m)}$ and $z_m = T_m(a)\sqrt{\hbar\omega_m/2}$,

the centered pressure is

$$P = \boldsymbol{a}^\dagger A_0 \boldsymbol{a} + \frac{1}{2}\left(\boldsymbol{a}^\dagger B_0 (\boldsymbol{a}^\dagger)^T + \boldsymbol{a}^T B_0 \boldsymbol{a}\right),$$
$$A_0 = \frac{dd^T}{L_0} + C_0 z z^T, \tag{10}$$
$$B_0 = \frac{dd^T}{L_0} - C_0 z z^T.$$

Here $\boldsymbol{a}$ is the column of annihilation operators.

The perturbative Hamiltonian is

$$H(\beta) = H_0 + \beta V + \beta^2 W + O(\beta^3), \qquad V = \sum_r \frac{u_r}{24} : q_r^4 :_0, \tag{11}$$

where $v_r = \sum_m c_{rm}^2$ and $: q_r^4 :_0 = q_r^4 - 6 v_r q_r^2 + 3 v_r^2$. Normal ordering implements the first-order stiffness calibration $\delta K_r = -u_r v_r/2$: its quadratic potential is $\beta \delta K_r q_r^2/2$. It removes the first-order tadpole correction to the harmonic two-point function at this reference geometry and regulator. No second-order response matching is imposed. Equivalently, a first-order insertion of the normal-ordered quartic into a linear-field two-point function vanishes: two external fields cannot saturate its four legs without a removed self-contraction. A change of length must keep this chosen physical calibration fixed; it must not silently replace v_r by a newly calibrated value.

For a stable microscopic completion with $u_r \le 0$, use

$$U_r(q;\beta) = \frac{K_{Jr}}{\vartheta_r(\beta)^2}[1 - \cos(\vartheta_r(\beta) q)], \qquad \vartheta_r(\beta)^2 = -\frac{\beta u_r}{K_{Jr}}, \tag{12}$$

including the same first-order stiffness calibration. The harmonic term $K_{Jr} q^2/2$ is already part of H_0. Expanding the residual potential through degree six and normal ordering gives, up to a scalar,

$$W = \sum_r \frac{u_r^2}{K_{Jr}} \left[\frac{v_r^2}{16} : q_r^2 :_0 + \frac{v_r}{48} : q_r^4 :_0 + \frac{1}{720} : q_r^6 :_0\right]. \tag{13}$$

All three terms belong to the same second-order cosine coefficient. Keeping only the normal-ordered sextic term would change the Hamiltonian. For negative u_r, the quartic Hamiltonian in Eq. (11) is used as a perturbative expansion of this stable model, not as a globally stable negative-quartic potential.

IV. ORDERED CUMULANTS THROUGH EIGHTH ORDER

Let $f_\tau(t) = e^{-t^2/(2\tau^2)}/(\sqrt{2\pi}\tau)$ and $g_\tau(\nu) = e^{-\tau^2\nu^2/2}$. We study the spectral statistics of the Hermitian time-smeared operator

$$O_\tau(\beta) = \int_{\mathbb{R}} f_\tau(t) e^{iH(\beta)t/\hbar} P e^{-iH(\beta)t/\hbar}\, \mathrm{d}t. \tag{14}$$

Its spectral cumulants are defined by the formal source series

$$K(s,\beta) = \log\langle\Omega_\beta| e^{s O_\tau(\beta)} |\Omega_\beta\rangle, \tag{15}$$
$$\kappa_p(\tau,\beta) = p!\,[s^p] K(s,\beta)$$
$$= \kappa_p^{(0)} + \beta c_{p,1} + \beta^2 c_{p,2} + O(\beta^3), \tag{16}$$

Here $|\Omega_\beta\rangle$ is the ground state or its perturbative expansion, and $[s^p]$ selects an ordinary power-series coefficient. We also write $\delta\kappa_p = c_{p,1}$ for the first-order coefficient, particularly in the variance formulas and benchmark tables below. This use of a formal series does not assume a finite exponential moment for positive s after the continuum limit. The products are powers of one smeared operator, rather than time-ordered products of instantaneous pressures. A detector counting protocol introduces additional coupling and ordering information [19, 20].

A. Harmonic Gaussian series

Write $O_\tau = S + \beta D + O(\beta^2)$. The quadratic harmonic operator S has the form of Eq. (10), with

$$A_{mn} = (A_0)_{mn} g_\tau(\omega_m - \omega_n),$$
$$B_{mn} = (B_0)_{mn} g_\tau(\omega_m + \omega_n). \tag{17}$$

For real symmetric A, B, its action on the vacuum has the Gaussian form

$$e^{sS}|0\rangle = c(s) \exp\left[\frac{1}{2}\boldsymbol{a}^\dagger Z(s) (\boldsymbol{a}^\dagger)^T\right]|0\rangle,$$
$$K_0(s) = \log c(s). \tag{18}$$

The squeezing matrix and vacuum amplitude satisfy

$$Z' = B + AZ + ZA^T + ZBZ,$$
$$K_0' = \tfrac{1}{2}\,\mathrm{Tr}(BZ), \tag{19}$$
$$Z(0) = 0, \qquad K_0(0) = 0.$$

Writing $Z(s) = \sum_{j\ge1} s^j Z_j$ yields the recurrence

$$Z_1 = B, \qquad (n+1) Z_{n+1} = A Z_n + Z_n A^T + \sum_{j=1}^{n-1} Z_j B Z_{n-j}. \tag{20}$$

Thus $Z_1, \ldots, Z_7$ determine the harmonic cumulants through κ_8. The recurrence retains the operator commutators; a determinant for commuting Gaussian variables generally gives different cumulants.

B. Ground-state preparation and real-time evolution

At first order, the source correction separates as

$$K(s,\beta) = K_0(s) + \beta\big[K_{1,\mathrm{state}}(s) + K_{1,\mathrm{dyn}}(s)\big] + O(\beta^2). \tag{21}$$

Writing $|\Omega_\beta\rangle = |0\rangle + \beta|\psi_1\rangle + O(\beta^2)$, the corresponding first-order raw moment has the explicitly ordered form

$$\delta\mu_n = 2\,\mathrm{Re}\langle\psi_1|S^n|0\rangle + \sum_{j=0}^{n-1}\langle 0|S^j D S^{n-1-j}|0\rangle. \tag{22}$$

Its two terms cannot in general be combined by replacing the field operators with commuting random variables.

Because V is normal ordered, its action on the vacuum creates four quanta. The first ground-state correction is $|\psi_1\rangle = -\hbar^{-1}\int_0^\infty e^{-tH_0/\hbar}V|0\rangle\,\mathrm{d}t$. Defining $c_r(t)_m = c_{rm}e^{-\omega_m t}$ gives

$$K_{1,\mathrm{state}}(s) = -\sum_r \frac{u_r}{4\hbar}\int_0^\infty \left[c_r(t)^T Z(s) c_r(t)\right]^2 \mathrm{d}t. \tag{23}$$

The two ground-state insertions and the three pairings of four creation operators give the factor 1/4. Each coefficient reduces to finite sums with denominators $\omega_m + \omega_n + \omega_k + \omega_l > 0$.

The evolution correction is

$$D = \frac{i}{\hbar}\int_{\mathbb{R}} \mathrm{d}t\, f_\tau(t)\int_0^t \mathrm{d}v\,[V(v), P(t)], \tag{24}$$

where both operators inside the commutator evolve with H_0. For component frequencies α, γ, its time kernel is

$$\begin{aligned} &i\int \mathrm{d}t\, f_\tau(t)\int_0^t \mathrm{d}v\, e^{i\alpha v + i\gamma t} = g_\tau^{[1]}(\alpha+\gamma, \gamma),\\ &g^{[1]}(x,y) = \frac{g(x)-g(y)}{x-y}, \end{aligned} \tag{25}$$

with $g^{[1]}(x,x) = g'(x)$. This continuous divided difference includes degenerate transition frequencies without an arbitrary choice of eigenvectors within their subspaces.

For completeness, the polynomial coefficients of D can be written directly in mode amplitudes. Let $R = (a_1, \ldots, a_M, a_1^\dagger, \ldots, a_M^\dagger)^T$, with signed frequencies $\nu = (-\omega, +\omega)$ and signs $\sigma = (-1, +1)$. Duplicate c_r, d, z on the two sets of signed indices, and define

$$Q_{r\ell a} = -c_{ra}\left[\frac{\sigma_a d_\ell d_a}{L_0} + \sigma_\ell C_0 z_\ell z_a\right]. \tag{26}$$

If $\mathcal{W}$ denotes complete Weyl symmetrization, then

$$D = \sum_r u_r \mathcal{D}_r, \tag{27}$$

$$\mathcal{D}_r = \sum_{ijk\ell} T^{(4)}_{r,ijk\ell}\mathcal{W}(R_i R_j R_k R_\ell) + \sum_{i\ell} T^{(2)}_{r,i\ell}\mathcal{W}(R_i R_\ell), \tag{28}$$

$$T^{(4)}_{r,ijk\ell} = \frac{c_{ri}c_{rj}c_{rk}}{6\hbar}\sum_a Q_{r\ell a} \times g_\tau^{[1]}(\nu_i+\nu_j+\nu_k+\nu_\ell, \nu_\ell - \nu_a), \tag{29}$$

$$T^{(2)}_{r,i\ell} = -\frac{v_r c_{ri}}{2\hbar}\sum_a Q_{r\ell a} \times g_\tau^{[1]}(\nu_i+\nu_\ell, \nu_\ell - \nu_a). \tag{30}$$

The quadratic term follows from the normal-ordering subtraction in V and must accompany the quartic term. These tensors use the bare pressure amplitudes d, z; only A, B in the Gaussian source are already smeared.

Differentiating the source exponential gives

$$K_{1,\mathrm{dyn}}(s) = \int_0^s \mathrm{d}\eta\,\frac{\langle 0|e^{(s-\eta)S}De^{\eta S}|0\rangle}{c(s)}. \tag{31}$$

Define the commutator generator and normalized Weyl covariance

$$\begin{aligned} &= \mathsf{K}R,\\ \mathsf{K} = \begin{pmatrix} -A & -B \\ B & A^T \end{pmatrix}, &\qquad C_L(s) = \begin{pmatrix} 0 & I/2 \\ I/2 & Z(s) \end{pmatrix}. \end{aligned} \tag{32}$$

The covariance at the insertion is

$$C(s,\eta) = e^{-\eta\mathsf{K}}C_L(s)e^{-\eta\mathsf{K}^T}. \tag{33}$$

Wick contraction then yields

$$\frac{\langle 0|e^{(s-\eta)S}\mathcal{D}_r e^{\eta S}|0\rangle}{c(s)} = 3\sum_{ijk\ell} T^{(4)}_{r,ijk\ell}C_{ij}C_{k\ell} + \sum_{i\ell} T^{(2)}_{r,i\ell}C_{i\ell}. \tag{34}$$

The three pairings are equal because $T^{(4)}$ is symmetric in its first three indices. This covariance is an algebraic Gaussian contraction and need not be a positive classical covariance.

There is a finite exact quadrature for each truncated source series. With $\eta = ts$, the coefficient of s^m in $C(s, ts)$ is a polynomial in t of degree at most m. The same bound holds for the contraction in Eq. (34). Since $\mathrm{d}\eta = s\,\mathrm{d}t$, the coefficient of s^p in $K_{1,\mathrm{dyn}}$ integrates a polynomial of degree at most $p-1$ on $[0,1]$. Four Gauss–Legendre nodes therefore integrate every coefficient through $p = 8$ exactly. This is exactness of the formal coefficients, not a four-node approximation to the finite-s source function. Together with Eq. (23), it gives all first-order cumulants through eighth order at fixed M, without an occupation or Fock-energy cutoff. The remaining mode limit must be examined for each requested cumulant.

V. VARIANCE FUNCTIONAL AND STATIC CLOSURE

A. Ordered mode-pair functional

The order-two specialization makes the loss of boundary information explicit and admits a separate continuum estimate. For the real mode amplitudes of this circuit and a real even differentiable filter, let $p = (m, n)$ and $q = (k, l)$ be ordered mode pairs; use the shorthand

$$\Omega_p = \omega_m + \omega_n, \qquad W_{rp} = c_{rm} c_{rn} (B_0)_{mn}. \tag{35}$$

All ordered pairs are included; no extra restriction $m \le n$ is imposed. The harmonic variance is

$$\kappa_2^{(0)}(\tau) = \frac{1}{2} \sum_{mn} (B_0)_{mn}^2 g_\tau(\omega_m + \omega_n)^2. \tag{36}$$

We denote by $\delta\kappa_2$ the derivative of κ_2 with respect to β at $\beta = 0$, holding the geometry, averaging window and coefficients u_r fixed. This first-order coefficient is

$$\boxed{\begin{aligned} \delta\kappa_2(\tau) &= \sum_r u_r \mathcal{C}_r(\tau), \\ \mathcal{C}_r(\tau) &= \frac{1}{2\hbar} \sum_{p,q} W_{rp} W_{rq} K_g(\Omega_p, \Omega_q). \end{aligned}} \tag{37}$$

Here

$$K_g(x, y) = \frac{y g(x)^2 - x g(y)^2}{x^2 - y^2}, \quad x \neq y, \tag{38}$$

$$K_g(x, x) = g(x) g'(x) - \frac{g(x)^2}{2x}. \tag{39}$$

The state term is obtained by replacing $K_g(x, y)$ with $-g(x)g(y)/(x + y)$; the dynamics term is the difference. For the ordinary Gaussian window at zero averaging time, $g = 1$ and the dynamics term vanishes. The remaining kernel is $-1/(x + y)$, as ordinary ground-state perturbation theory requires. For the increment filter, both widths tend to zero together and $g_\tau - g_{2\tau}$ instead vanishes at fixed mode count. The pair-matrix derivation is detailed in Appendix E. Although this equal-time control is well defined at finite mode count, the continuum harmonic variance need not remain finite as $\tau \to 0$.

For a direct derivation, the quartic vertex contains four branch fields, which contract with the four fields supplied by the two quadratic pressure insertions. Applying Eq. (22) at $n = 2$, the ground-state correction yields the kernel $-g(x)g(y)/(x+y)$. The real-time commutator terms use the divided differences in Eq. (E2); combining the two pressure positions and symmetrizing p, q gives Eq. (39). Only the pressure pair coefficient B_0 remains. Number-preserving pressure terms cannot saturate the four quartic legs in this two-insertion vacuum contraction. Equation (37) is independently checked against the unreduced ordered-monomial algebra, which includes every allowed contraction.

Equation (37) contains no occupation or Fock-energy cutoff. For N harmonic modes it involves N^2 pair weights and an $\mathcal{O}(N^4)$ arithmetic sum, evaluated in blocks rather than as a stored N^4 tensor. The kernel has dimensions of time, so $u_r W_{rp} W_{rq} K_g/\hbar$ has dimensions of pressure squared. The removable diagonal singularity is evaluated analytically. For the Gaussian window and its scale difference used here, the first-order variance correction and its state and dynamics terms converge absolutely under the stated positive circuit parameters. A general differentiable filter requires a separate convergence check. Appendix F derives an explicit $\mathcal{O}(N^{-3})$ omitted-mode bound and explains why Gaussian averaging does not exponentially suppress every virtual high-frequency contribution.

B. Static-closure criterion

Fix the complete harmonic model, mode regulator or convergent limit, and sampling window. Put $a_r = h_r(0)^4$, so the static quartic coefficient is $G = \sum_r a_r u_r$. Here static refers specifically to the zero-frequency value of the vertex in Eq. (4). Relating it to a measured nonlinear susceptibility also requires the external propagators and a response convention.

Proposition 1 (Static closure within a branch family)**.** *On a family of allowed perturbative coupling variations $\mathcal{U}$, knowledge of G determines $\delta\kappa_2$ if and only if*

$$\boldsymbol{a} \cdot \boldsymbol{v} = 0 \quad \Longrightarrow \quad \boldsymbol{\mathcal{C}}(\tau) \cdot \boldsymbol{v} = 0 \tag{40}$$

for all differences $\boldsymbol{v}$ of allowed couplings. If $\mathcal{U}$ contains an open subset of the full coupling space, this is equivalent to $\boldsymbol{\mathcal{C}}(\tau)$ being proportional to $\boldsymbol{a}$.

The proof is immediate from Eq. (37): two coupling vectors with equal G differ by an element of the nullspace of $\boldsymbol{a}$. The criterion says precisely when that nullspace is also invisible to the specified pressure variance. Its content for the present system lies in the ordered functional $\mathcal{C}_r(\tau)$, not in the elementary linear algebra alone. For a constrained family, only its allowed directions are tested.

For two branches, construct a family labeled by η while keeping the harmonic model and a nonzero scalar G fixed. Choose

$$u_1(\eta) = \frac{\eta G}{a_1}, \qquad u_2(\eta) = \frac{(1 - \eta) G}{a_2}, \qquad 0 \le \eta \le 1. \tag{41}$$

Every member then satisfies $\Gamma_{4,\eta}(0, 0, 0, 0) = G$. The quartic vertices agree with one another at zero frequency; their values at other frequency combinations generally differ. The predicted curve is affine:

$$\delta\kappa_2(\eta, \tau) = G \left[\eta \frac{\mathcal{C}_1(\tau)}{a_1} + (1 - \eta) \frac{\mathcal{C}_2(\tau)}{a_2} \right]. \tag{42}$$

The separating coefficient is $\mathcal{S}(\tau) = \mathcal{C}_1/a_1 - \mathcal{C}_2/a_2$. A nonzero $\mathcal{S}$ falsifies static closure at that window within

this family. A zero at an isolated window would not establish closure at other windows.

If the harmonic branch parameters and averaging time are known, G and one ideal value of $\delta\kappa_2$ give two linear equations for u_1, u_2. Their solution is unique whenever $\Delta = a_1\mathcal{C}_2 - a_2\mathcal{C}_1 \neq 0$:

$$u_1 = \frac{G\mathcal{C}_2 - a_2\delta\kappa_2}{\Delta}, \qquad u_2 = \frac{a_1\delta\kappa_2 - G\mathcal{C}_1}{\Delta}. \tag{43}$$

This inversion establishes uniqueness for exact input data within the specified two-branch model. Experimental reconstruction would additionally require an uncertainty analysis, particularly because the nonlinear correction can be small compared with the harmonic variance.

VI. COMPLETE SECOND-ORDER COSINE COEFFICIENTS

Let $F_p[H]$ denote the spectral cumulant obtained from the ground state of H and Eq. (14), keeping P and τ fixed. In a finite matrix representation with a nondegenerate ground state,

$$\begin{aligned} c_{p,2} &= \frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}\epsilon^2}F_p[H_0 + \epsilon V]\Big|_{\epsilon=0} \\ &\quad + \frac{\mathrm{d}}{\mathrm{d}\epsilon}F_p[H_0 + \epsilon W]\Big|_{\epsilon=0} \\ &\equiv c_{p,2}^{VV} + c_{p,1}[W]. \end{aligned} \tag{44}$$

The first term includes two quartic insertions. The second includes all contractions of the cosine sextic potential in Eq. (13). Both derivatives include ground-state preparation and Hamiltonian-dependent smearing. In particular, $c_{p,2}$ is the coefficient of β^2, not the unhalved second derivative.

For eigenstates of a finite Hamiltonian, smearing is evaluated as $(O_\tau)_{ab} = P_{ab}g_\tau((E_a - E_b)/\hbar)$, followed by ordered matrix powers in the ground state. We obtain $c_{p,1}[W]$ from the first matrix derivative, using the continuous divided differences of the smearing function. An independent symmetric calculation gives

$$\begin{aligned} D_p(h) &= \frac{1}{2h^2}\big[F_p[H_0 + hV] \\ &\qquad + F_p[H_0 - hV] - 2F_p[H_0]\big], \\ \frac{4D_p(h/2) - D_p(h)}{3} &= c_{p,2}^{VV} + O(h^4). \end{aligned} \tag{45}$$

Successive step refinements resolve truncation against cancellation error. The full cosine at positive β supplies a distinct comparison with the retained first- and second-order series. Its finite-basis matrix is the projection of the cosine operator, $\Pi\cos(\vartheta_r q_r)\Pi$; it is not the cosine of the projected coordinate. The projection can be evaluated from

$$e^{i\vartheta_r q_r} = e^{-\vartheta_r^2 v_r/2}e^{i\vartheta_r\sum_m c_{rm}a_m^\dagger}e^{i\vartheta_r\sum_m c_{rm}a_m}. \tag{46}$$

This comparison retains the first-order calibration specified above, so changes of the physical two-point response at order β^2 are part of the microscopic prediction. Mode convergence and occupation convergence remain separate from convergence in h and β.

A. The sixth-order boundary vertex and lower corrections

The second nonlinear order also refines the boundary closure itself. In this subsection the action and time variables are Euclidean and the pressure source is zero. At fixed retained line field, let $X_r = h_r * \phi$ and write $q_r = X_r + \eta_r$ with $\langle\eta_r(t)\eta_s(t')\rangle_\eta = \hbar\delta_{rs}\chi_r(t-t')$, where $\chi_r(i\xi) = 1/D_r(i\xi)$. Here the star denotes time convolution. If $\mathcal{V}$ and $\mathcal{W}$ are the time integrals of V and W, conditional Gaussian elimination gives

$$\begin{aligned} S_{\mathrm{eff}} &= S_{0,\mathrm{eff}} + \beta\langle\mathcal{V}\rangle_\eta \\ &\quad + \beta^2\left[\langle\mathcal{W}\rangle_\eta - \frac{\langle\mathcal{V}\mathcal{V}\rangle_{\eta,c}}{2\hbar}\right] + O(\beta^3). \end{aligned} \tag{47}$$

Define $w_r = \langle X_r^2\rangle_0 = v_r - \hbar\chi_r(t=0)$. The connected conditional product of two normal-ordered quartics is exactly

$$\begin{aligned} &\langle :q_r(t)^4: _{v_r} : q_r(t')^4 : _{v_r}\rangle_{\eta,c} \\ &\quad = \sum_{j=1}^{4}\binom{4}{j}^2 j!\,[\hbar\chi_r(t-t')]^j \\ &\qquad \times : X_r(t)^{4-j}: _{w_r} : X_r(t')^{4-j}: _{w_r}. \end{aligned} \tag{48}$$

The coefficients are $16, 72, 96, 24$. Each colon subtracts its own equal-time variance; retained-line contractions between the two times are not removed. Together with $\langle\mathcal{W}\rangle_\eta$, this identity retains all lower-degree terms as well as the six-field term.

Let $\mathcal{P}_{3|3}$ denote the ten unordered partitions of six external frequencies into two triples, and let A_P be either triple in partition P. The coefficient of the six-field term, with action prefactor $\beta^2/6!$, is

$$\begin{aligned} \Gamma_6^{(2)}(\xi_1,\ldots,\xi_6) &= \sum_r u_r^2\prod_{j=1}^{6}h_r(i\xi_j) \\ &\quad \times\left[\frac{1}{K_{J_r}} - \sum_{P\in\mathcal{P}_{3|3}}\chi_r\left(i\sum_{j\in A_P}\xi_j\right)\right], \\ \sum_j \xi_j &= 0. \end{aligned} \tag{49}$$

The first term is the direct cosine contribution. The second comes from one residual-branch contraction between two quartics. The two choices of triple are equivalent by frequency conservation and evenness of χ_r. An indepen-

dent static minimization gives

$$\Gamma_6^{(2)}(0,\dots,0) = \sum_r h_r(0)^6 u_r^2 \left(\frac{1}{K_{Jr}} - \frac{10}{S_r}\right), \qquad S_r = K_{cr} + K_{gr} + K_{Jr}. \tag{50}$$

The full coupled branch propagator must not replace χ_r here: its line-mediated part is already generated by exchange between the retained quartic vertices. Such a replacement would count that propagation twice. Equations (47)–(50) make explicit why the next closure contains more than a static sextic number: it also includes frequency-dependent lower-vertex corrections and the pressure-source terms required by the chosen observable. The microscopic second-order calculation includes these effects without asserting that $\Gamma_6^{(2)}$ alone determines a pressure cumulant. An independent reduced real-time pressure calculation would additionally need the ordered source and ground-state preparation-contour insertions; the zero-source action identities displayed here do not constitute that calculation.

VII. POSITIVE INCREMENT MOMENTS AND THE SPECTRAL-INTERMITTENCY TEST

Moment-order spectra characterize scale-dependent probability laws, as illustrated by multifractal random walks [22]. Magnitude-cumulant analysis instead studies cumulants of logarithmic magnitudes [23]; these differ from the pressure cumulants computed here. To test how the shape of pressure fluctuations depends on averaging scale, define one Hermitian increment operator at each width,

$$X_\tau = O_\tau - O_{b\tau}, \qquad M_p(\tau,\beta) = \langle \Omega_\beta | X_\tau^p | \Omega_\beta \rangle, \qquad b > 1, \qquad p = 2,4,6,8. \tag{51}$$

The reference ratio is $b = 2$. Both operators use the same Hamiltonian and state, and all preceding formulas apply with $g_\tau \mapsto g_\tau - g_{b\tau}$, including the divided difference in D. Stationarity and normalization of the two windows imply $\langle X_\tau \rangle = 0$. The even moments are nonnegative spectral moments of X_τ; a signed cumulant correction, or its absolute value, cannot replace them.

Let a superscript (0) denote the matched harmonic model with the same geometry, branch parameters, pressure, window, and regulator. Where the moments are positive, define

$$R_p = \frac{M_p}{M_2^{p/2}}, \qquad Q_p = \frac{R_p}{R_p^{(0)}}, \qquad A_p = \log Q_p. \tag{52}$$

R_p removes the fluctuation amplitude, while Q_p compares the remaining shape with the already non-Gaussian harmonic pressure. $Q_2 = 1$ identically. If changing the number of branches changes the harmonic dynamics, each such model requires its own harmonic reference; agreement of a static loading parameter does not establish agreement of the complete harmonic response across different branch counts.

The raw moments follow from the same source series, $\langle e^{sX_\tau} \rangle = \exp K_X(s,\beta)$, retaining every term at the claimed perturbative order. Write $M_p = M_p^{(0)} + \beta m_{p,1} + \beta^2 m_{p,2} + O(\beta^3)$, $r_p = m_{p,1}/M_p^{(0)}$, and $s_p = m_{p,2}/M_p^{(0)}$. Then

$$A_p = \beta\left(r_p - \frac{p}{2} r_2\right) + \beta^2 \left[s_p - \frac{r_p^2}{2} - \frac{p}{2}\left(s_2 - \frac{r_2^2}{2}\right)\right] + O(\beta^3). \tag{53}$$

We denote the first-order shape coefficient by $A_{p,1} = \partial_\beta A_p|_0 = r_p - p r_2/2$, so $A_p = \beta A_{p,1} + O(\beta^2)$. Taking an unexpanded logarithm of first-order moments would instead include an incomplete set of higher powers of β.

To distinguish a sustained scaling regime from a change of shape over a finite resonance crossover, we examine the logarithmic shape slope

$$\Delta_p = \frac{\mathrm{d}A_p}{\mathrm{d}\log\tau} = \left(\zeta_p - \frac{p}{2}\zeta_2\right) - \left(\zeta_p^{(0)} - \frac{p}{2}\zeta_2^{(0)}\right), \qquad \zeta_p = \frac{\mathrm{d}\log M_p}{\mathrm{d}\log\tau}. \tag{54}$$

The corresponding first-order slope is $\Delta_{p,1} = \mathrm{d}A_{p,1}/\mathrm{d}\log\tau$; $\Delta_p = \beta\Delta_{p,1} + O(\beta^2)$ at fixed regulator and width. A nonzero A_p at one width establishes a shape change at that width. A nonzero local Δ_p establishes scale-dependent shape, but does not by itself distinguish an extended scaling law from a resonance crossover. A common amplitude exponent shift $\zeta_p - \zeta_p^{(0)} = p\delta h$ gives $\Delta_p = 0$ for every p. For a scaling anomaly attributable to the nonlinear boundary relative to its harmonic reference, we require approximately constant, nonzero Δ_p for several moment orders on a common resolved interval. The interval must persist or grow as the physical resonance hierarchy is extended, and the slopes must survive the stated numerical and nonlinear-order checks. This relative criterion does not exhaust the physical intermittency question: Δ_p can vanish if the nonlinear and harmonic models share the same curved raw moment-order spectrum. Their separate ζ_p spectra and event laws must therefore also be retained. Increasing the number of retained line modes refines a numerical approximation; widening the boundary resonance hierarchy changes the physical model. Increasing the density of resonances in a fixed band is a third, distinct comparison.

Accordingly, local slopes are compared with fits over common shifted windows and with the corresponding harmonic curves. The relevant numerical uncertainty and omitted nonlinear terms must be smaller than the proposed effect in A_p and its slope, which can involve strong

cancellation. Known-law controls distinguish three possibilities: a non-Gaussian distribution with one amplitude exponent and constant R_p; a lognormal scale family with $\Delta_p = -\sigma^2 p(p-2)/2$; and a finite-resonance mixture whose changing local slopes mimic scaling on a short interval. These controls test the estimator, not the quantum mechanism.

Finite regulators also have identifiable endpoint behavior. At a fixed finite Hilbert cutoff,

$$X_\tau = \frac{b^2-1}{2\hbar^2}\tau^2[H,[H,P]] + O(\tau^4). \tag{55}$$

If its leading even moment is nonzero, $M_p \propto \tau^{2p}$ as $\tau \to 0$ and R_p tends to a constant. At large widths, the finite set of nonzero transition gaps gives exponential suppression. These limits explain how finite spectral crossovers can occur without an extended scaling regime. They do not determine the continuum short-width limit, since removing the cutoff and taking $\tau \to 0$ need not commute. Finally, the test concerns the spectral distribution of each X_τ. A common classical process, time-resolved bursts, a cross-scale joint law, or a detector counting distribution would require additional constructions beyond these one-observable moments. The finite-coupling tensor construction, excited-pressure diagnostics, and operational measurement definitions are collected in the appendices. This keeps the central closure and scale tests visible while retaining the full technical specification in the same article.

VIII. NUMERICAL COMPARISONS

A. Two-branch reference and numerical cutoffs

We use dimensionless units $\hbar = C_0 = L_0 = a = 1$, with $C_b = K_b = 0.5$ and

$$(C_r, K_{cr}, K_{gr}, K_{Jr}) = \begin{cases} (1, 0.5, 0.25, 0.25), & r = 1, \\ (0.25, 0.8, 1.2, 2), & r = 2. \end{cases} \tag{56}$$

The uncoupled branch resonance frequencies are 1 and 4, with $h_1(0) = 0.5$ and $h_2(0) = 0.2$. We set $G = -10^{-4}$ in Eq. (41); $\eta = 0$ gives $(u_1, u_2) = (0, -0.0625)$, whereas $\eta = 1$ gives $(-0.0016, 0)$. All family members use the same calibration geometry and harmonic modes.

Independent continuum modes are obtained from sine profiles and the monotone secular equation between its impedance poles,

$$\frac{k\cot(ka)}{L_0} = C_b\omega^2 - K_b - \sum_r K_{cr} + \sum_r \frac{K_{cr}^2}{K_{cr} + K_{gr} + K_{Jr} - C_r\omega^2}, \tag{57}$$

where $k = \omega/v$. The positive capacitance norm fixes the amplitudes and the derivative is evaluated directly from the sine profile. The chosen parameters have no coincident poles requiring dark-mode nullspace treatment.

A consistent-mass finite-element discretization independently checks these profiles and slopes. Mode-count convergence, spatial convergence, total occupation, Fock energy, and weak nonlinearity are distinct tests. In particular, increasing the mode count above a fixed Fock energy ceiling is not sufficient convergence evidence. The pair sum avoids this ceiling entirely for the variance; our fourth-cumulant matrix calculations retain and report it.

B. Finite-coupling comparison

The stable cosine in Eq. (12) supplies a finite-coupling comparison for the same allocation $u_r(\eta)$. At each β, all allocations share the leading static coefficient βG, while varying β tests the perturbative expansion. Define the residuals after retaining one or two nonlinear orders as

$$\mathcal{E}_p^{(1)}(\beta) = \kappa_p^{\text{cosine}}(\beta) - \kappa_p^{(0)} - \beta c_{p,1}, \tag{58}$$

$$\mathcal{E}_p^{(2)}(\beta) = \mathcal{E}_p^{(1)}(\beta) - \beta^2 c_{p,2}. \tag{59}$$

They are $O(\beta^2)$ and $O(\beta^3)$, respectively, at a fixed regulator where the expansion applies. These residuals are distinct from the standardized increment moments R_p in Eq. (52).

The endpoint response comparison uses the maximum of $|G_{\phi\phi}^{\text{cosine}}(i\xi;\beta)/G_{\phi\phi}^{(0)}(i\xi) - 1|$ over $\xi = 0, 0.5, 1, 2, 4, 8$. The baseline cosine controls use three retained continuum modes and independently vary the occupation cutoff. They test the stable weak-coupling expansion at that regulator.

C. Matched-static separation with a bounded mode tail

Figure 1 compares members of the family with the same zero-frequency quartic coefficient. The closed variance sum was evaluated at 24, 48, 72, and 96 continuum modes, without an occupation or energy ceiling, at ten averaging times from 0.1 to 1.6. Both O_τ and the increment $O_\tau - O_{2\tau}$ were evaluated. The corrections at the allocation endpoints $\eta = 0$ and $\eta = 1$ have opposite signs at several resolved windows. The five allocation values follow the predicted first-order affine dependence.

At $\tau = 0.4$, the difference between the allocations $\eta = 1$ and $\eta = 0$ is 3.75606×10^{-7}, whereas its omitted-mode bound is 3.86629×10^{-9}. The separate bounds for $\eta = 0$ and $\eta = 1$ are 3.86610×10^{-9} and 1.84217×10^{-13}, each smaller than its corresponding correction magnitude. Thus both opposite signs and the nonoverlap of the continuum coefficient intervals survive the omitted-mode bound. Relative to the common harmonic variance, the separation is 6.2402 parts per million for $G = -10^{-4}$. This establishes separation of the defined perturbative

TABLE I. Continuum-mode pressure variance at 96 modes. The last column bounds the omitted-mode error in the difference between allocations $\eta = 1$ and $\eta = 0$ by the sum of the two analytic bounds in Eq. (F6). It is not a statistical standard error or a floating-point certificate.

τ	$\kappa_2^{(0)}$	$\delta\kappa_2(\eta = 0)$	$\delta\kappa_2(\eta = 1)$	separation tail bound
0.2	8.17555×10^{-1}	4.11670×10^{-7}	1.76378×10^{-8}	4.74157×10^{-9}
0.4	6.01911×10^{-2}	-3.54746×10^{-7}	2.08607×10^{-8}	3.86629×10^{-9}
0.8	4.73491×10^{-4}	7.04413×10^{-9}	-1.57710×10^{-9}	1.74282×10^{-9}

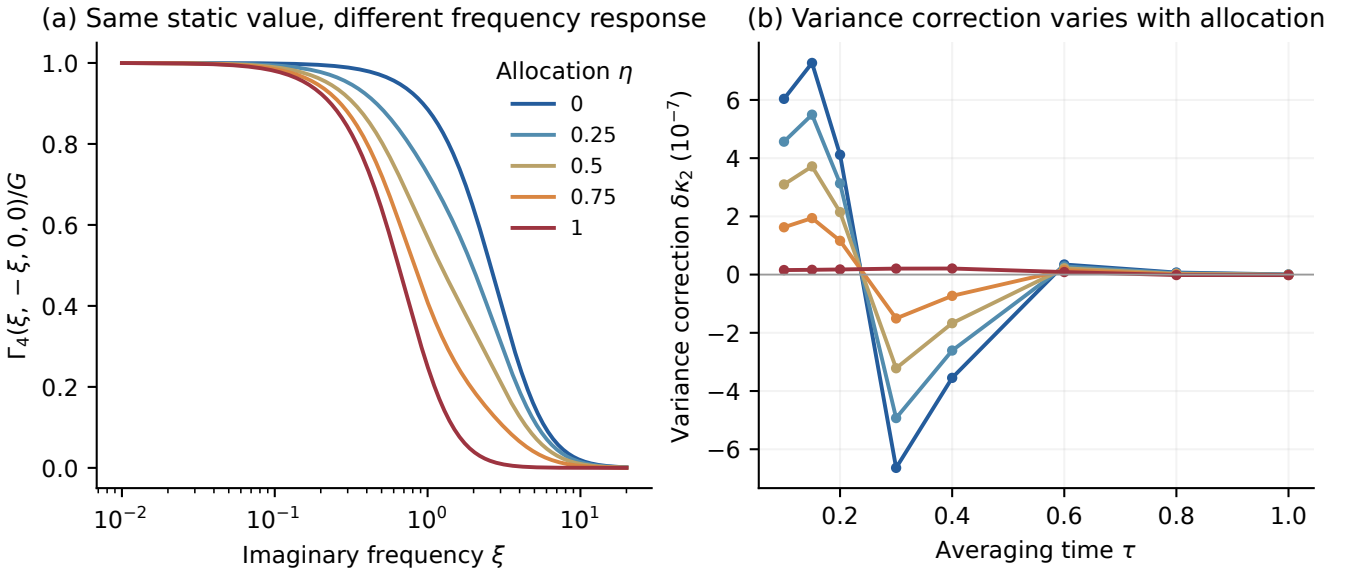


FIG. 1. Boundary models with a common two-point response through first order and the same leading static quartic coefficient. (a) The normalized vertex slice $\Gamma_4(\xi, -\xi, 0, 0)/G$ tends to one at zero frequency for every allocation; the common unnormalized value is G. (b) The first-order spectral pressure-variance correction $\delta\kappa_2$ changes with allocation and averaging time. Lines connect evaluated windows; they do not assert an interpolation error bound.

observable; it does not by itself establish practical detector sensitivity.

D. Higher-cumulant and microscopic validation

The fourth-cumulant calculation retains basis states with at most six harmonic excitations in total and excitation energy no greater than 40 or 48, in the stated dimensionless units. At 20 modes the results at ceiling 48 are shown in Table II. They are complementary finite-regulator evidence; the variance convergence proof is not asserted for these fourth cumulants.

Increasing the energy ceiling from 40 to 48 changes the fourth-cumulant corrections at the two allocation endpoints by 0.8745% and 0.0139% at $\tau = 0.2$; 0.0217% and 0.0146% at $\tau = 0.4$; 0.0250% and 0.0112% at $\tau = 0.8$. The percentages are $100|\delta\kappa_4(48) - \delta\kappa_4(40)|/|\delta\kappa_4(48)|$, in allocation order $\eta = 0, 1$. They are empirical energy-cutoff differences, not proven fourth-cumulant error bounds.

The three-mode stable-cosine controls at $\tau = 0.4$ use $\eta = 0, 1/2, 1$ and $\beta = 1, 1/4, 1/16$. Reducing β by four reduces the residual fourth-cumulant error by factors from 15.9963 to 16.0000; the variance residual decreases by factors from 15.9912 to 16.0062. Both are consistent with the expected factor 16 for an $\mathcal{O}(\beta^2)$ remainder. Raising the total-occupation ceiling from 10 to 12 changes the checked fourth cumulants by at most 7.93809×10^{-15}. The independently evaluated relative two-point response mismatch decreases approximately as β^2 (Fig. 3). These are fixed-three-mode controls, separate from the continuum variance proof.

TABLE II. First-order correction $\delta\kappa_4$ to the fourth pressure cumulant at 20 modes and excitation-energy cutoff 48.

τ	$\eta = 0$	$\eta = 1$
0.2	4.23030×10^{-5}	4.08826×10^{-6}
0.4	-4.93110×10^{-6}	2.01919×10^{-8}
0.8	6.31771×10^{-9}	3.17180×10^{-10}

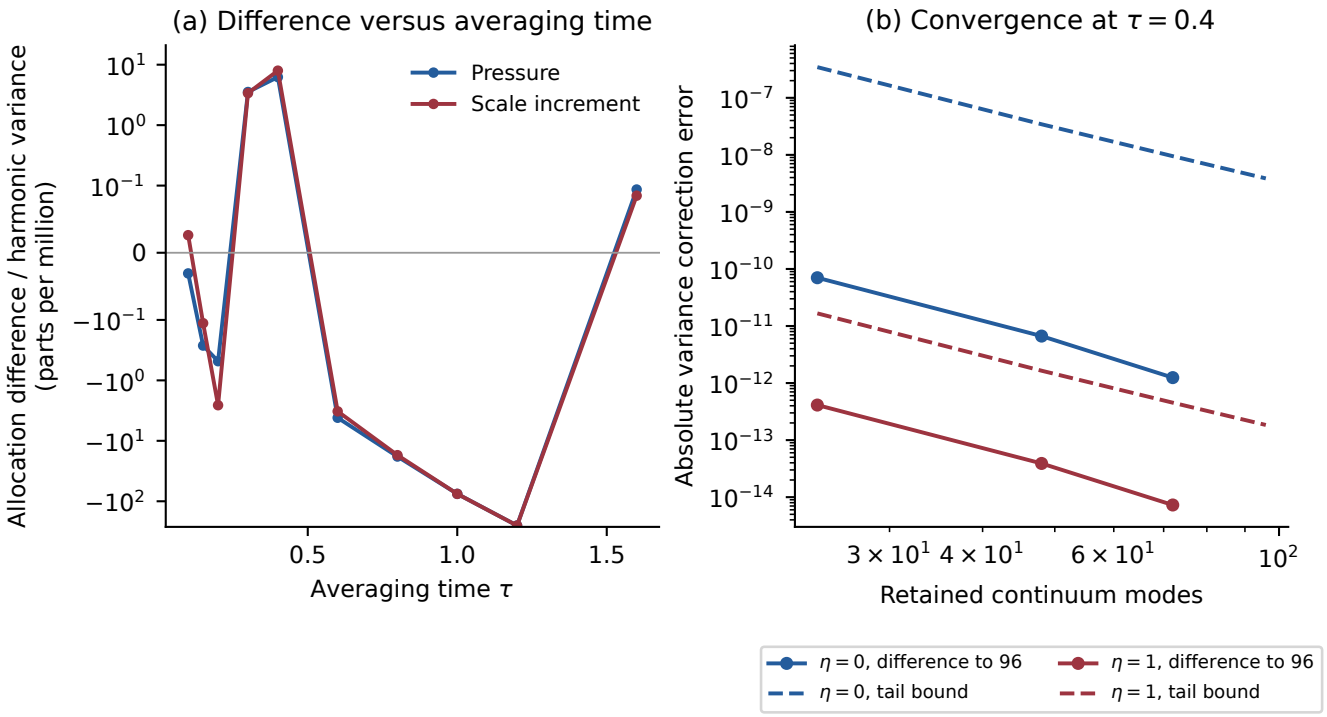


FIG. 2. (a) Signed difference between allocations $\eta = 1$ and $\eta = 0$, normalized by the corresponding harmonic pressure or increment variance. The symmetric-logarithmic vertical scale preserves signs near zero. (b) Mode refinement at $\tau = 0.4$: absolute differences from the 96-mode value and conservative analytic omitted-mode bounds are shown separately. The reference value is not labeled as an exact continuum answer.

E. Higher cumulants and the next cosine order

The Gaussian-insertion calculation was compared with independent one-, two-, and three-mode Fock calculations through κ_8. Twelve cases tested the preparation and dynamics contributions separately, the variance pair functional, four versus five insertion nodes, and oscillator controls with nonunit $\hbar$. These comparisons test the ordered coefficients at the specified regulators. The Gaussian calculation itself has no occupation or excitation-energy cutoff; increasing the number of line modes remains necessary for a continuum statement about higher cumulants.

For the second-order oscillator control, take $\hbar = \omega = K_J = 1$, $q = (a + a^\dagger)/\sqrt{2}$, $P =: q^2 :$, and the zero-width limit. A repeated-quartic calculation with $V =: q^4 : /24$ gives $c_{2,2}^{VV} = 47/3072$. This coefficient is unchanged by reversing the quartic sign. For the stable cosine with $u = -1$, the next potential in Eq. (13) gives $c_{2,1}[W] = -3/128$, so the complete coefficient is $c_{2,2} = -25/3072$. The independent numerical evaluation differs from these rational values by less than 3.8×10^{-11}. Retaining only the sextic normal-ordered monomial would fail this comparison.

The completed cosine refinement retains three continuum modes and compares total-occupation ceilings $Q = 14$ and $Q = 16$. The circuit is that of Sec. VIII A, with static quartic coefficient $G = -10^{-2}$. This nonlinear budget differs from the $G = -10^{-4}$ used for the bounded continuum variance example. The physical two-point response is calibrated only through first order; its second-order change remains part of the microscopic prediction.

Table III selects the ordinary pressure observable at $\eta = 0$ and $\tau = 0.2$. It separates the repeated-quartic contribution $c_{p,2}^{VV}$ from $c_{p,1}[W]$, which includes the quadratic, quartic, and sextic terms in Eq. (13). Their sum is the coefficient of β^2 in Eq. (16).

Figure 4 compares the residuals after the first- and complete second-order expansions over $\beta = 1/64, 1/32, 1/16, 1/8, 1/4$. The displayed powers refer to refinement of the coupling multiplier at fixed averaging time, not to intermittency or a law in averaging scale. The $Q = 14$ and $Q = 16$ curves are shown separately even where they nearly coincide. The cubic guide is anchored at the largest-β second-order residual for $Q = 16$; it is not a fitted exponent. The selected comparison does not establish uniform cubic convergence over other allocations, averaging widths, cumulant orders, or mode limits.

The finite-coupling spectral shape can also be compared directly. At $\beta = 1/4$, define the skewness $S_3 = \kappa_3/\kappa_2^{3/2}$ and standardized fourth cumulant $S_4^{(c)} = \kappa_4/\kappa_2^2$. The latter equals the excess kurtosis and is distinct from

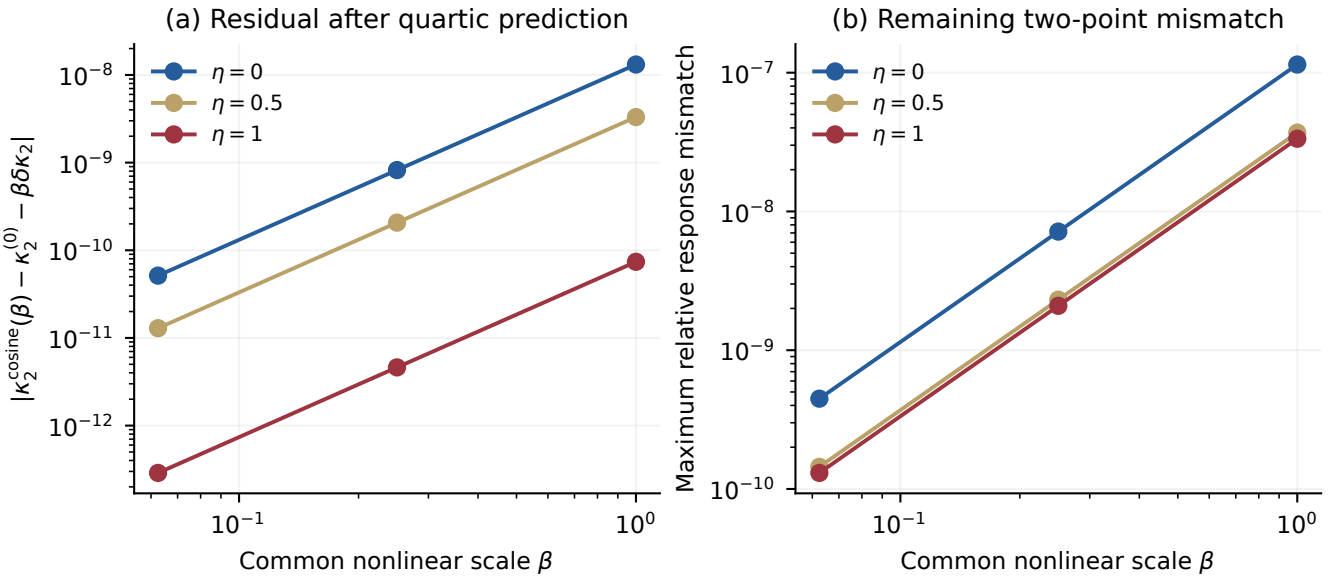


FIG. 3. Stable microscopic control at three retained continuum modes and $\tau = 0.4$. (a) Variance residual $|\mathcal{E}_2^{(1)}(\beta)|$ after subtracting the harmonic value and $\beta\delta\kappa_2$. (b) Maximum relative endpoint two-point mismatch over imaginary frequencies $0, 0.5, 1, 2, 4, 8$. Both decrease at approximately second order in the common quartic scale β.

TABLE III. Finite-three-mode pressure coefficients at $\eta = 0$, $\tau = 0.2$, $G = -10^{-2}$, and occupation ceiling $Q = 16$. The coefficient $c_{p,2} = c_{p,2}^{VV} + c_{p,1}[W]$ multiplies β^2; it is not the unhalved second derivative. The absolute differences Δ_h and Δ_Q describe successive Richardson refinements and the change from $Q = 14$ to $Q = 16$, respectively. They are empirical diagnostics, not statistical error bars or continuum bounds.

p	$c_{p,1}$	$c_{p,2}^{VV}$	$c_{p,1}[W]$	$c_{p,2}$	Δ_h	Δ_Q
2	-2.60848×10^{-4}	-2.17326×10^{-5}	-1.46894×10^{-4}	-1.68626×10^{-4}	2.57202×10^{-12}	1.96880×10^{-12}
3	-1.74151×10^{-3}	7.83947×10^{-5}	-4.30939×10^{-4}	-3.52544×10^{-4}	1.47734×10^{-11}	5.15143×10^{-12}
4	3.18727×10^{-3}	9.87046×10^{-4}	-4.58730×10^{-3}	-3.60025×10^{-3}	8.02025×10^{-11}	1.81188×10^{-11}
6	1.27688×10^{0}	9.77168×10^{-2}	-3.62209×10^{-1}	-2.64492×10^{-1}	2.87722×10^{-9}	3.37271×10^{-9}
8	1.47514×10^{2}	1.48999×10^{1}	-3.53192×10^{1}	-2.04193×10^{1}	2.92403×10^{-7}	1.59829×10^{-7}

the fourth standardized raw moment. Table IV compares the harmonic pressure with the full projected cosine at the same η, τ, and mode set. These are non-Gaussian-shape quantities at one width, not intermittency diagnostics; the harmonic pressure already has nonzero higher cumulants.

These results extend the comparison in nonlinear order at a fixed finite mode set. Agreement between occupation cutoffs, agreement of finite-difference steps, and convergence as line modes are added are separate requirements; the continuum variance bound does not supply a bound for the higher-order coefficients in this table.

At a fixed harmonic regulator, $u_r(\eta)$ is affine in the allocation parameter. Both the repeated-quartic term and W are quadratic in these couplings, so every exact coefficient $c_{p,2}(\eta)$ is a quadratic polynomial. This determines the full allocation curve from $\eta = 0, 1/2, 1$; it is not a polynomial fit in cumulant order. As an independent check, we evaluated $\eta = 1/4$ at $Q = 14$, $\tau = 0.2, 0.4, 0.8$, for ordinary averages and increments, through $p = 8$. The predicted value is

$$c_{p,2}(1/4) = \frac{3}{8}c_{p,2}(0) + \frac{3}{4}c_{p,2}(1/2) - \frac{1}{8}c_{p,2}(1). \qquad (60)$$

The largest relative discrepancy for $p = 2, \ldots, 8$ is 8.96×10^{-8}; all absolute discrepancies and independent step changes are retained with the numerical record. The polynomial identity concerns the second-order coefficient, not the unexpanded finite-coupling cosine result.

F. Boundary resonance hierarchies

To keep the total static harmonic loading fixed as the resonance hierarchy is extended, we use $S_r = 1/R$ and

$$C_r = \frac{S_r}{\Omega_r^2}, \quad K_{cr} = \frac{S_r}{4}, \quad K_{gr} = \frac{S_r}{4}, \quad K_{Jr} = \frac{S_r}{2}. \qquad (61)$$

Here Ω_r is the uncoupled resonance of branch r. Every branch has $h_r(0) = 1/4$, and its contribution to the

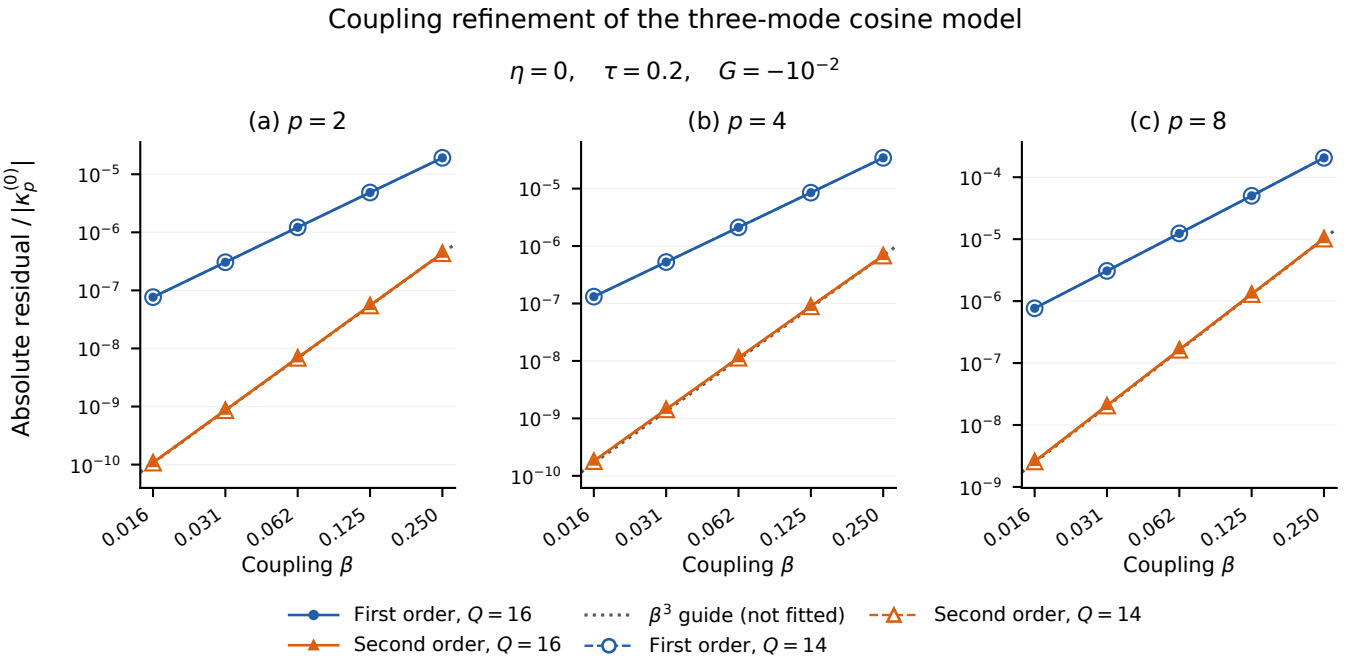


FIG. 4. Coupling refinement of the ordinary smeared pressure at three retained modes, $\eta = 0$, $\tau = 0.2$, and $G = -10^{-2}$. Panels show $p = 2, 4, 8$. Blue curves display $|\mathcal{E}_p^{(1)}|/|\kappa_p^{(0)}|$ and orange curves display $|\mathcal{E}_p^{(2)}|/|\kappa_p^{(0)}|$, with the residuals defined in Eq. (59). Hollow markers and dashed lines use $Q = 14$; smaller filled markers and solid lines use $Q = 16$. Dotted curves are anchored β^3 guides. Lines join calculated coupling values. No statistical error bars or universal residual exponent are inferred.

TABLE IV. Standardized pressure cumulants at $\eta = 0$, $\tau = 0.2$, three retained modes, and full-cosine coupling $\beta = 1/4$. The signed change is the $Q = 16$ cosine value minus the harmonic value. The last column is the absolute difference between cosine values at $Q = 16$ and $Q = 14$, not an uncertainty bound.

Quantity	harmonic	cosine, $Q = 16$	signed change	occupation change
S_3	4.506702819	4.506490091	-2.12728×10^{-4}	1.95399×10^{-14}
$S_4^{(c)}$	23.172724820	23.181205164	8.48034×10^{-3}	1.52767×10^{-13}

static harmonic kernel is $3S_r/16$. Consequently the total branch loading is $3/16$ for every R. We retain the same line and endpoint parameters as above, but these branch parameters define a new family rather than the original two-branch reference circuit.

The initial comparison uses $\Omega_r = 1.1\,\rho^{r-1}$ and $R = 2, 4, 6$. The choice $\rho = 2$ extends the upper frequency as branches are added; $\rho = 8^{1/(R-1)}$ instead fills the fixed interval $[1.1, 8.8]$ more densely. To allocate the quartic budget $G = -10^{-7}$, define

$$\begin{aligned} \ell_r &= \frac{\Omega_r^{-1}}{\sum_s \Omega_s^{-1}}, \\ h_r^{\mathrm{alloc}} &= \frac{\Omega_r}{\sum_s \Omega_s}, \\ u_r(\eta) &= \frac{G}{(1/4)^4}\left[(1-\eta)\ell_r + \eta h_r^{\mathrm{alloc}}\right]. \end{aligned} \tag{62}$$

The allocation weight h_r^{alloc} is distinct from the transfer function $h_r(i\xi)$. We evaluate $\eta = 0, 1/2, 1$ and the additional uniform allocation $u_r = G/[R(1/4)^4]$. These four allocations share the full harmonic model and static G at each R. Across different R, only the stated static loading is common; the entire harmonic frequency response changes.

The scale estimator was checked on twelve deterministic known-law controls using the same positive-moment analysis. For a non-Gaussian single-scaling family $Y_\tau = (\tau/T)^h Z$ with fixed Z, all standardized moments are constant. Here T is a reference width, and the control uses $10^{-3} \le \tau/T \le 1$. The lognormal control is defined by $M_p = (p-1)!!(\tau/T)^{ph-\sigma^2 p(p-2)/2}$ for even p, with the Gaussian single-scaling reference $M_p^{(0)} = (p-1)!!(\tau/T)^{ph}$. For $h = 1/2$ and $\sigma^2 = 0.02$, the analytic exponents at $p = 2, 4, 6, 8$ are $1, 1.92, 2.76, 3.52$, and the reference-normalized shape slopes at $p = 4, 6, 8$ are $-0.08, -0.24, -0.48$. A discrete two-resonance mixture supplies a control with changing local slopes but

TABLE V. Exploratory slopes of $\log A_{p,1}$ against $\log\tau$ at $R = 14$, twenty-four modes, and $12.8 \le \tau \le 204.8$. These describe the positive nonlinear correction itself. They are not the moment exponents ζ_p or the intermittency diagnostic Δ_p. No statistical confidence interval is assigned to the deterministic fit.

s	$p = 4$	$p = 6$	$p = 8$
-1	0.0741	0.0646	0.0625
-0.5	-0.4437	-0.4541	-0.4558
0	-0.9602	-0.9698	-0.9707
0.5	-1.4747	-1.4820	-1.4818
1	-1.9865	-1.9905	-1.9890

no hierarchy limit. These tests check that the analysis distinguishes moment order, non-Gaussian shape, and a scale law; they do not supply a physical mechanism for the boundary system.

We extended the physical hierarchy toward lower frequencies at fixed upper pole 8.8, using $\Omega_r = 8.8\,2^{r-R}$ and $R = 2, 6, 10, 14$. The deepest boundary spans a factor 8192 in bare resonance frequency. The averaging grid runs from 0.4 to 1638.4 in factors of two. The saved numerical records retain unresolved tail values; the discussion below uses the finite intermediate window where independent mode comparisons are available. This extension increases the total branch capacitance and changes the harmonic response; each boundary is compared with its own harmonic pressure law.

The saved branch insertion coefficients also permit a continuous allocation family without repeating the field contractions:

$$w_r(s) = \frac{\Omega_r^s}{\sum_j \Omega_j^s}, \qquad u_r(s) = \frac{G\, w_r(s)}{(1/4)^4}, \qquad -1 \le s \le 1. \tag{63}$$

All these boundaries have the same harmonic reference and the same static quartic coefficient G at fixed R. The choices $s = -1, 0, 1$ reproduce the low-frequency, uniform, and high-frequency allocations. Evaluating $s = -1/2$ and $s = 1/2$ samples the interior of this continuous allocation family.

Figure 5 shows actual first-order shape changes through the eighth moment. For uniform allocation at $R = 14$, $A_{8,1}$ decreases from 1.67622×10^{-5} to 1.14479×10^{-6} as τ increases from 12.8 to 204.8. Across this window, changing twenty retained modes to twenty-four changes $A_{p,1}$ by at most 0.0002004% for $p = 4, 6, 8$ in the uniform family. This is an empirical mode-refinement result, not an analytic error bound. Smaller widths have larger relative changes, especially near cancellations, and are not assigned the same numerical accuracy.

A nearly constant shape offset for the low-frequency allocation and a decaying shape correction for the other allocations do not supply a persistent nonzero $\Delta_{p,1}$. The dyadic grid was supplemented by six widths displaced by half a step in $\log_2\tau$. For $12.8 < \tau < 204.8$, the compensated quantity $\tau A_{8,1}$ differs by up to 5.21% from linear interpolation in $\log\tau$ between the neighboring dyadic values. These deviations resolve structure that the five-point decay fit cannot describe; discrete resonance spacing and finite endpoints both remain relevant to its interpretation. The large-width ratios of exponentially small moments require their own arithmetic and nonlinear-remainder checks. The evidence here therefore describes the first-order scale dependence of the specified family; it neither establishes vacuum intermittency nor excludes a different fully nonlinear regime or boundary family.

TABLE VI. Finite-window pressure-increment moment slopes, $R = 14$, $M = 24$, $b = 2$, uniform allocation, and $G = -10^{-7}$. All columns are dimensionless; the last two multiply the nonlinear parameter β. A nonzero window-average $\widehat{\Delta}_{p,1}$ does not imply that its local slope is constant.

p	$\widehat{\zeta}_p^{(0)}$	$\widehat{\zeta}_{p,1}$	$\widehat{\Delta}_{p,1}$
2	-2.09102	-2.6916×10^{-7}	0
4	-4.15343	-1.3170×10^{-6}	-7.7866×10^{-7}
6	-6.22436	-3.3598×10^{-6}	-2.5523×10^{-6}
8	-8.29846	-6.4915×10^{-6}	-5.4148×10^{-6}

a. The resolved moment-order spectrum. To report the moment exponents themselves separately from the correction-envelope fits, let $\mathcal{S}[y]$ denote the least-squares slope of y against $\log\tau$ at the five widths $12.8, 25.6, 51.2, 102.4, 204.8$. Define

$$\begin{aligned} \widehat{\zeta}_p^{(0)} &= \mathcal{S}[\log M_p^{(0)}], \\ \widehat{\zeta}_{p,1} &= \mathcal{S}[m_{p,1}/M_p^{(0)}], \\ \widehat{\Delta}_{p,1} &= \widehat{\zeta}_{p,1} - \tfrac{p}{2}\widehat{\zeta}_{2,1}. \end{aligned} \tag{64}$$

The finite-window slope of $\log M_p$ is therefore $\widehat{\zeta}_p^{(0)} + \beta\widehat{\zeta}_{p,1} + O(\beta^2)$. Table VI gives these quantities at the same uniform fourteen-branch, twenty-four-mode regulator. The near-linear harmonic spectrum and the small nonlinear corrections are both retained. The hats mark finite-window summaries, not proven asymptotic exponents: the adjacent slopes vary with width, as already shown for $\Delta_{p,1}$ in Fig. 5(c). No continuous multifractal spectrum is reconstructed from these four orders.

The collapsed-resonance endpoint supplies a physical closure control. Set every pole equal to $\Omega = 8.8$ while retaining $S_r = 1/R$ and the same static budgets. The bright coordinate $Q = R^{-1}\sum_r q_r$ is equivalent to one branch with $(C, K_c, K_g, K_J) = (\Omega^{-2}, 1/4, 1/4, 1/2)$; the other $R - 1$ coordinates are dark harmonic oscillators. Their contribution to each branch variance is $v_{\mathrm{dark}} = (R-1)\hbar\Omega/2$. Conditional averaging gives

$$\langle :q_r^4: _{v_Q + v_{\mathrm{dark}}} \rangle_{\mathrm{dark}} =: Q^4 :_{v_Q}, \qquad \sum_r u_r = G/(1/4)^4. \tag{65}$$

Consequently every first-order line-pressure functional is independent of R and of the allocation at fixed total u.

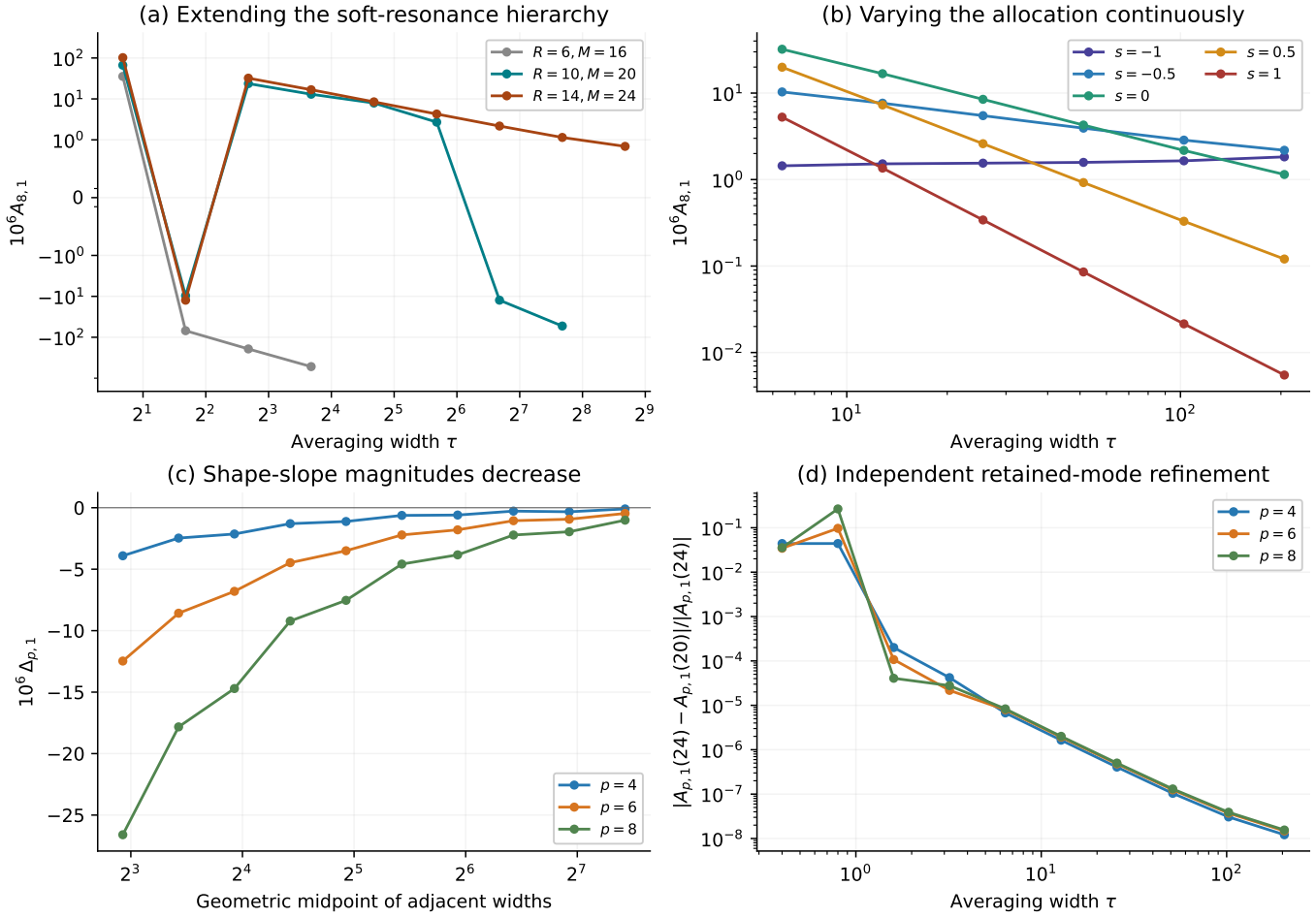


FIG. 5. First-order nonlinear pressure-increment shape diagnostics for the soft-resonance hierarchy. (a) Extending the hierarchy moves its slowest resonance crossover to longer averaging widths. (b) At fixed $R = 14$, changing s in Eq. (63) changes the scale dependence of the shape correction. (c) Finite differences $\Delta_{p,1} = \Delta A_{p,1}/\Delta \log \tau$ for the uniform family at twenty modes include the independently added between-grid widths and vary across the interval. (d) Retained-mode changes are shown separately; they are not statistical error bars. Lines join evaluated widths. The plotted quantities are derivatives at zero nonlinear coupling.

Complete four-element finite-element line regulators with $R = 2, 6, 10, 14$ reproduce this control for ordinary averages and increments at $\tau = 0.2, 0.4, 0.8$ through κ_8. The largest absolute difference in the nonlinear cumulant coefficients is 2.406×10^{-14}. As a positive sensitivity check, omitting v_{dark} from the microscopic calibration generates the predicted nonzero bright quadratic perturbation $\gamma_{\mathrm{wrong}} : Q^2 :$, where $\gamma_{\mathrm{wrong}} = (\sum_r u_r)(R-1)\hbar\Omega/8$. The same calculations recover that added response. The first-order invariance does not imply invariance of the full cosine under arbitrary branch replication. We also tested numerical cancellation directly. Four and five Gaussian insertion nodes both integrate the order-eight formal polynomial exactly. For the uniform $R = 14$, twenty-mode hierarchy at $\tau = 12.8, 51.2, 204.8$, the resulting $A_{p,1}$ values for $p = 4, 6, 8$ differ relatively by at most 2.5×10^{-14}. The variance also agrees with the independent pair sum. These comparisons measure numerical sensitivity rather than a rigorous rounding bound. In contrast, two-branch values at $\tau = 3.2$ and 6.4 change by order one under the same node change: the enormous ratios obtained there from tiny moments are numerically unresolved and are excluded from the physical scale interpretation. Thus neither a finite floating-point ratio nor a large raw correction is accepted as evidence for intermittency.

b. A second increment ratio. The choice of two averaging widths is part of the observable, so a change in b need not preserve numerical amplitudes. We independently formed $X_\tau = O_\tau - O_{3\tau}$, including its first-order dynamical correction, and checked one-, two-, and three-mode calculations through order eight against a Fock implementation. All six controls passed; the $b = 2$ specialization reproduced the original calculation exactly. For the uniform fourteen-branch model with twenty retained

modes, the new ratio gives

τ	$A_{8,1}$ $(b=3)$
12.8	1.43367×10^{-5}
25.6	7.22049×10^{-6}
51.2	3.63593×10^{-6}
102.4	1.85022×10^{-6}

(66)

Its adjacent derivative with respect to $\log\tau$ changes from -1.02665×10^{-5} to -2.57624×10^{-6} across these intervals. Thus the changed increment ratio preserves a decreasing correction and a slope whose magnitude also decreases. It does not turn the finite correction into a constant anomalous exponent. This window comparison uses twenty modes; the twenty-four-mode refinement reported above concerns $b=2$.

G. A conditional soft-branch scaling law

For the power-weight allocations in Eq. (63), write $Z_s=\sum_r\Omega_r^s$. The coefficient $A_{p,1}=\partial_\beta A_p|_0$ is defined by $A_p=\beta A_{p,1}+O(\beta^2)$. Each pole family uses its own harmonic reference.

A quasistatic model gives a conditional explanation of the possible scale dependences. To display units, introduce a stiffness scale S_*, set to one in the numerical family, and take $S_r=S_*/R$, $K_{cr}=K_{gr}=S_*/(4R)$, $K_{Jr}=S_*/(2R)$, and $C_r=S_*/(R\Omega_r^2)$. Neglecting the line and endpoint inertia defines

$$B=K_b+\frac{1}{L_0a}+\frac{S_*}{4},\qquad \phi=\frac{S_*}{4RB}\sum_r q_r. \tag{67}$$

The resulting harmonic Hamiltonian and leading gradient pressure are

$$H_{\mathrm{qs},0}=\sum_r\left[\frac{R\Omega_r^2p_r^2}{2S_*}+\frac{S_*q_r^2}{2R}\right]-\frac{S_*^2}{32R^2B}\left(\sum_r q_r\right)^2, \tag{68}$$

$$P_{\mathrm{qs}}=\frac{S_*^2}{32R^2B^2L_0a^2}\left(\sum_r q_r\right)^2. \tag{69}$$

This model retains the collective branch coupling. Its harmonic potential is positive when $B>S_*/16$.

Set $t=\tau z$, $x_r=\tau\Omega_r$, $q_r=\sqrt{\hbar R/(S_*\tau)}\,y_r$, and $p_r=\sqrt{\hbar S_*\tau/R}\,\pi_r$. The harmonic Hamiltonian is then $\hbar/\tau$ times a dimensionless Hamiltonian depending on $\mathbf{x}$, R, and S_*/B; the pressure has prefactor $\hbar S_*/(32R\tau B^2L_0a^2)$ multiplying $(\sum_r y_r)^2$. With the harmonic normal-ordering covariance transformed by the same map, the quartic becomes

$$V=\frac{\hbar}{\tau}\frac{32}{3}\epsilon_s\sum_r x_r^s:y_r^4:,\qquad \epsilon_s=\frac{G\hbar R^2}{S_*^2Z_s\tau^{s+1}}. \tag{70}$$

The window becomes a fixed dimensionless window. The canonical change transforms both the state and the time evolution, and therefore preserves the ordered spectral-moment definition. In this quasistatic model,

$$A_{p,1}^{\mathrm{qs}}(\tau)=\frac{G\hbar R^2}{S_*^2Z_s\tau^{s+1}}F_{p,s}(\tau\boldsymbol{\Omega};R,S_*/B). \tag{71}$$

The response function $F_{p,s}$ includes the factor $32/3$ and the subtraction of $p/2$ times the relative variance correction. Equation (71) is a finite-model identity where the harmonic moments are positive; it does not assert that $F_{p,s}$ is constant.

For a geometric pole set, shifting the averaging time by the pole ratio shifts the dimensionless pole indices. If an interior group of poles dominates, with negligible endpoint and background changes, this gives the conditional envelope $A_{p,1}\propto\tau^{-(s+1)}$. The low, uniform, and high allocations consequently admit exponents $0,-1,-2$, respectively. A geometric lattice can also produce a periodic function of $\log_\rho\tau$. Dyadic widths at $\rho=2$ sample one phase of that function and cannot establish continuous scaling. Moreover, Z_s must be retained when R changes; for the infrared hierarchy, $Z_{-1}=2(1-2^{-R})/\Omega_{\min}$.

The marginal case $s=-1$ illustrates the distinction between a shape change and an anomalous exponent. A constant $A_{p,1}$ gives a nonzero first-order shape shift but no first-order excess slope: $\Delta_p=dA_p/d\log\tau=O(\beta^2)$. More generally,

$$\Delta_p=\beta\epsilon_s\left[-(s+1)F_{p,s}+\frac{dF_{p,s}}{d\log\tau}\right]+O(\beta^2) \tag{72}$$

in the quasistatic model. A periodic marginal response has zero mean first-order slope over one lattice period. Neither behavior alone establishes a persistent nonzero anomalous scaling exponent.

The quasistatic cosine model also obeys an exact scaling relation between different parameter sets, with a necessary normalization qualification. At fixed R and stiffnesses, the map $\Omega_r'=c\Omega_r$, $u_r'=u_r/c$, $q_r'=\sqrt{c}\,q_r$, and $p_r'=p_r/\sqrt{c}$, together with $v_{r,\mathrm{ref}}'=cv_{r,\mathrm{ref}}$, gives $H_{\mathrm{qs}}'(\beta)=cH_{\mathrm{qs}}(\beta)$ and $X_{\tau/c}'=cX_\tau$ between the corresponding calibrated quasistatic cosine Hamiltonians. The standardized shapes agree at all orders under this map. However, it changes G to G/c. A global pole rescaling with normalized weights and fixed G instead leaves u_r unchanged and does not have this symmetry. An interior index shift at $s=-1$ can approximate the required local law $u_r\propto1/\Omega_r$, subject to endpoint and background corrections.

Applying these identities to the full line requires additional tests. Its dynamic transfer, kinetic pressure, hard-mode virtual contributions, and effective pressure-source terms were omitted in defining Eqs. (68) and (69). Large τ alone does not bound all internal frequencies in a higher moment. Off-dyadic widths, spatial and mode refinement, a collapsed-pole control, and arithmetic resolution

are therefore needed to test the proposed interpretation. The perturbative remainder is at fixed regulator and need not be uniform in the window or hierarchy depth. This conditional scaling argument supplies discriminating predictions; it does not establish vacuum intermittency or a higher-cumulant continuum limit.

H. Response information and statistical closure

We next quantify the pressure information left undetermined by a specified nonlinear boundary calibration. For the fixed static budget, write $u_r = Gw_r/(1/4)^4$, with $w_r \geq 0$ and $\sum_r w_r = 1$. The first-order shape coefficient is $A_{p,1} = \sum_r c_{p,r} w_r$, where

$$c_{p,r} = \frac{G}{(1/4)^4}\left[\frac{m_{p,1}^{(r)}}{M_p^{(0)}} - \frac{p}{2}\frac{m_{2,1}^{(r)}}{M_2^{(0)}}\right]. \tag{73}$$

Here $m_{p,1}^{(r)}$ is the derivative of the increment moment per unit branch coupling u_r, rather than the coefficient after a nonlinear allocation has already been selected. The harmonic model, geometry, pressure, and averaging width are fixed within each comparison.

A normalized four-field response sample at Euclidean frequency ξ has the form

$$\frac{\Gamma_4(\xi, -\xi, 0, 0)}{G} = \sum_r w_r \left(\frac{\Omega_r^2}{\Omega_r^2 + \xi^2}\right)^2. \tag{74}$$

The zero-frequency row fixes the total budget. For additional frequencies let A be the matrix of the displayed response factors, including that static row, and let $y = Aw^{\text{true}}$ be the ideal response data of the specified allocation. The minimum and maximum of $\sum_r c_{p,r} w_r$ subject to $Aw = y$ and $w \geq 0$ define the sharp finite-family prediction interval. We evaluate these extrema numerically and retain the endpoint allocations and constraint residuals. The range measures incomplete response information, not an error bar on a chosen reconstruction or a model of experimental noise.

The positive frequencies are chosen before considering any pressure width or moment order. We select the smallest and largest branch poles, then successively bisect the widest remaining logarithmic pole interval, using the lower index to resolve ties. The retained sets are nested. They contain one, two, four, and eight positive frequencies where available, followed by every distinct pole frequency; for six branches the last such set contains six frequencies. Frequencies use the same inverse-time units as the line and branch spectrum.

With only the static row, the interval is exactly $[\min_r c_{p,r}, \max_r c_{p,r}]$. Each new response row can only restrict the allowed allocations. We compute an orthonormal nullspace basis N from a singular-value decomposition and write $w = w^{\text{true}} + Nz$, so the optimization imposes only $-Nz \leq w^{\text{true}}$. This parameterization represents the same ideal-data allocation set. If A has full column rank, its nullspace is zero and the interval is structurally the singleton $\{c_p \cdot w^{\text{true}}\}$; its zero width is not obtained by clipping a computed linear-programming width. The record retains the rank threshold, singular values, nullspace residuals, endpoint allocations, and original-constraint residuals. For nonzero nullspace dimension, objective rescaling and primal and dual solver tolerances 10^{-10} are used. Each witness must satisfy the original normalized response equations, total-weight constraint, and nonnegativity to absolute tolerance 10^{-10}. These are finite-dimensional numerical checks, not bounds on pressure-mode or finite-coupling errors.

Figure 6 selects uniform allocation $w_r = 1/R$ and $\tau = 0.8$. The raw interval width $W_{p,1}$ is dimensionless per unit of the dimensionless coupling multiplier β; the lower panels divide it by its static-only value. Every branch count retains its own harmonic reference. Mode counts are $M = 16$ for $R = 6$, $M = 24$ for $R = 10$, $M = 24$ for $R = 14$. The common static branch loading is $3/16$ and the quartic budget is $G = -1.00000 \times 10^{-7}$.

The intervals use completed per-branch first derivatives, not a fit to the pressure data. The saved analysis reconstructs the supplied allocation values from these branch coefficients and checks the static simplex range, nesting, and endpoint constraints. It carries unresolved small harmonic moments explicitly. The first-order comparison does not impose second-order two-point matching or control the finite-β remainder. Numerical mode refinement and measurement uncertainty are also separate from the ideal response-information range.

I. Fixed-calibration geometry comparison

At each complete finite-element regulator, the counterterm is fixed at $a_{\text{ref}} = 1$ and held unchanged at $a = 0.8$ and $a = 1.25$. Writing $v_r(a) = \langle q_r^2 \rangle_{0,a}$ and $\delta v_r = v_r(a) - v_r(a_{\text{ref}})$, the same branch perturbation becomes

$$V_{r,\text{ref}} = \frac{u_r}{24} : q_r^4 :_a + \frac{u_r \delta v_r}{4} : q_r^2 :_a + \frac{u_r \delta v_r^2}{8}. \tag{75}$$

The quadratic term contributes to the ordered pressure cumulants; dropping it would recalibrate the boundary at each length. The scalar shifts the vacuum energy but changes neither the ground-state vector nor Heisenberg evolution at fixed length, so it does not change any spectral pressure moment. The corresponding first-order energy shift is $E_1(a) = \sum_r u_r \delta v_r^2/8$, up to the fixed reference constant. Its derivative obeys

$$-\partial_a E_1 = -\sum_r \frac{u_r}{4} \delta v_r \partial_a v_r. \tag{76}$$

This was checked against an independent generalized-force pair contraction. The largest absolute discrepancy

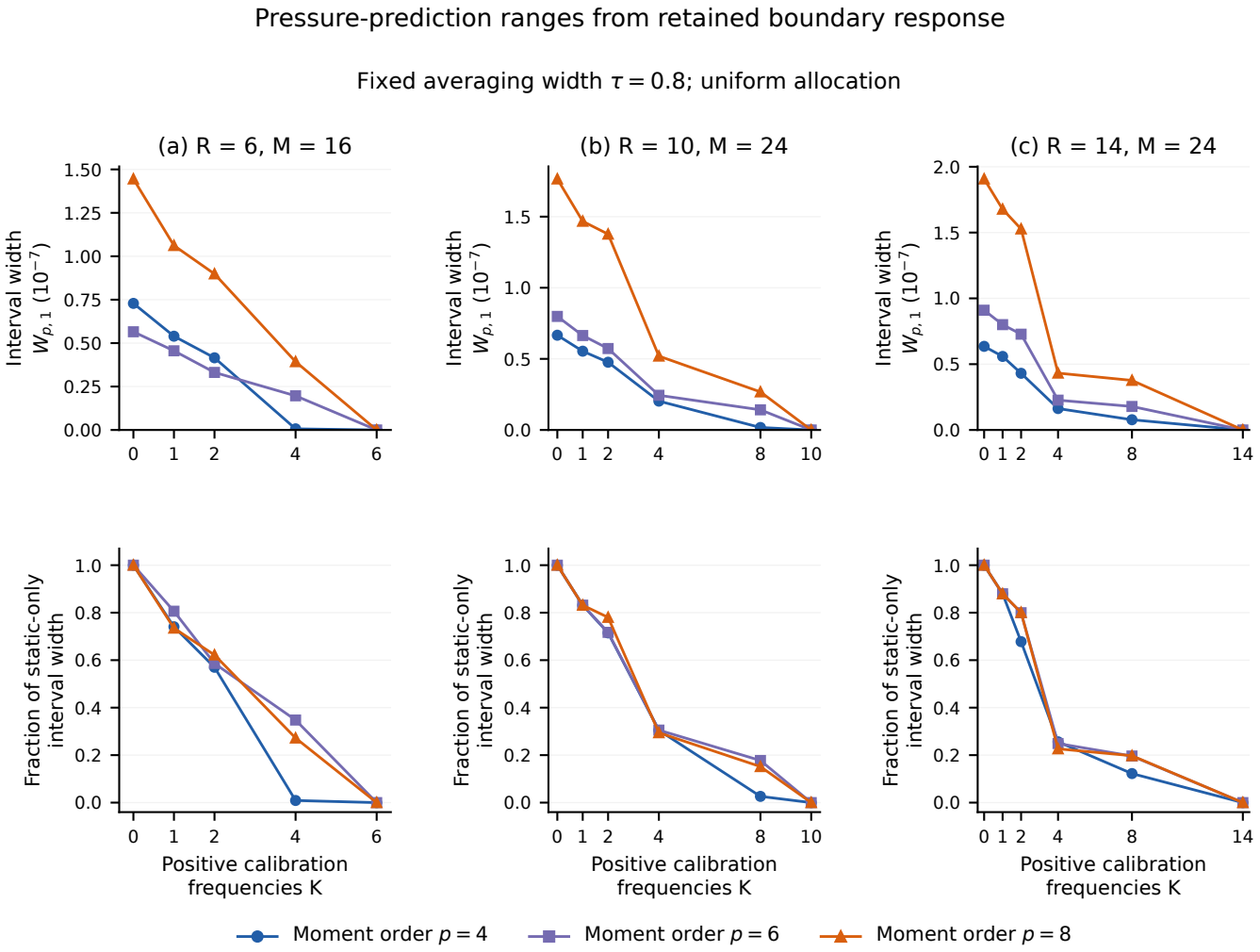


FIG. 6. Dependence of the first-order pressure-shape prediction range on retained four-field boundary data. The upper panels show the width of the allowed $A_{p,1}$ interval in the displayed dimensionless units; the lower panels show its fraction of the static-only interval. Colors identify $p = 4, 6, 8$. The horizontal coordinate counts positive-frequency samples, with the zero-frequency budget constraint always included. Lines join nested calibration sets, rather than averaging-scale samples. Thus this figure measures response-information closure, not an intermittency exponent. Unresolved moments or flagged numerical intervals are omitted, without clipping or introducing a positive floor.

across the completed meshes, lengths, and three allocations is 4.3506×10^{-18}. The derivative step was independently halved; this comparison is a numerical consistency test at each regulator.

Figure 7 shows the first-order local-pressure corrections at $\tau = 0.4$, $\eta = 0$, and $G = -10^{-4}$ on the complete mesh sequence 4, 8, 12, 16, 24. All branch and line modes of each mesh are retained. These are spatial regulator comparisons, with reference matching imposed at each regulator; they are not evaluations at a common finite mode projection. Table VIII separates the additional quadratic contribution at the finest mesh. The change between the two finest meshes is reported without extrapolating it into a continuum error bound. The relative mesh sensitivity depends on length and cumulant order; this sequence supplies no uniform continuum bound for the eighth cumulant.

The generalized length force used in the energy check is distinct from the last-element local stress at finite mesh size. Their means must not be equated by definition. Likewise, $\kappa_p(a) - \kappa_p(a_{\mathrm{ref}})$ compares two local pressure laws; subtraction alone does not make it the cumulant of a positive interaction-only law. These distinctions preserve the specified Casimir geometry while keeping the pressure observable and calibration explicit.

a. Independent arithmetic recheck. The high-order covariance calculation is also evaluated with the direct recurrence described above. All 15 ordinary-pressure cases at $\tau = 0.4$—five complete FEM meshes and three lengths—pass the four/eight-node and mode-order com-

TABLE VII. Selected eighth-moment shape intervals for the allocation and averaging width of Fig. 6. For each branch count the static-only, four-frequency, and all-pole sets are shown. Here K counts the positive-frequency samples; the zero-frequency row is additional. Rank includes the zero-frequency row. The final column is the larger endpoint residual $\|Aw - y\|_\infty$; it is not a bound on the pressure-mode error.

R	K	rank	lower $A_{8,1}$	upper $A_{8,1}$	interval width	witness residual
6	0	1	-5.1525×10^{-8}	9.2966×10^{-8}	1.4449×10^{-7}	1.1102×10^{-16}
6	4	5	-7.8517×10^{-9}	3.1450×10^{-8}	3.9302×10^{-8}	1.1102×10^{-16}
6	6	6	1.0767×10^{-8}	1.0767×10^{-8}	0	0
10	0	1	-8.6872×10^{-8}	8.9623×10^{-8}	1.7649×10^{-7}	0
10	4	5	-2.6389×10^{-8}	2.5554×10^{-8}	5.1943×10^{-8}	2.2204×10^{-16}
10	10	10	2.6416×10^{-9}	2.6416×10^{-9}	0	0
14	0	1	-1.0265×10^{-7}	8.8067×10^{-8}	1.9072×10^{-7}	2.2204×10^{-16}
14	4	5	-2.4409×10^{-8}	1.8786×10^{-8}	4.3195×10^{-8}	3.3307×10^{-16}
14	14	14	7.1220×10^{-10}	7.1220×10^{-10}	0	0

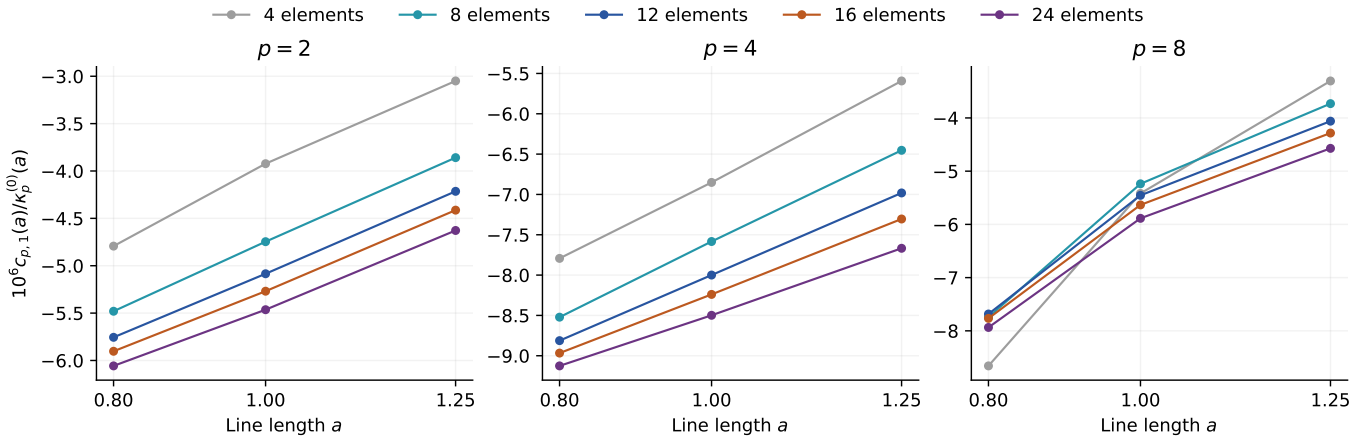


FIG. 7. Fixed-calibration geometry dependence of the local inside-pressure cumulants. The first-order correction is normalized by the harmonic cumulant at the same length and mesh. Parameters are $\tau = 0.4$, $\eta = 0$, $G = -10^{-4}$; the physical counterterm for each mesh stays fixed at $a_{\mathrm{ref}} = 1$. Lines connect the three calculated lengths.

parisons for each of the three branch allocations. The largest absolute difference between the new and archived eighth-cumulant coefficients across these cases is 2.602×10^{-13}. The largest observed node/permutation difference over the checked series is 3.173×10^{-18} in the declared harmonic comparison units. The quadratic detuning is independently replayed with the original fixed reference. These checks support the finite-mesh geometry coefficients shown here; they do not supply a continuum error bound. The separate long-window ordinary derivative problem requires the stable recurrence: successful geometry replay is not a general accuracy guarantee for the earlier representation.

IX. FINITE-STRENGTH PRESSURE STATISTICS

We evaluate the equal-reference-phase family at $R = 10$, $M = 20$, and $\tau = 51.2$, with the calibration fixed at $M_{\mathrm{ref}} = 24$. The full cosine is retained. The ordinary averaged pressure and the increment $X_\tau = O_\tau - O_{2\tau}$ are each centered in the actual ground state of their finite Hamiltonian. This fixes a source constant without reconstructing an absolute physical mean. The results below distinguish finite-model statistics from the separate bounds on omitted states and source information.

A. Cumulants and independent refinements

Table IX reports every pressure cumulant from second through eighth order. The harmonic columns are obtained from the exact ordered Gaussian recursion for the same twenty line modes; they are not evaluated in the nonlinear adapted basis. The table gives the fluctuation magnitudes, while Fig. 8 shows their standardized shape. Increasing β from one to three reduces the variance at this width while increasing several standardized higher cumulants. High standardized cumulants are already present in the harmonic pressure and are not, on

TABLE VIII. Fixed-calibration first-order pressure coefficients at 24 elements (26 complete harmonic modes), with parameters as in Fig. 7. The quadratic column is the contribution of the second term of Eq. (75); the last column is the absolute change of the full coefficient from the next-coarser mesh.

a	p	full $c_{p,1}$	quadratic part	mesh change
0.8	2	-3.4520×10^{-7}	-3.2135×10^{-9}	1.1979×10^{-8}
0.8	4	-4.2088×10^{-6}	1.5298×10^{-7}	7.4982×10^{-8}
0.8	8	-1.8115×10^{-2}	1.3944×10^{-3}	1.0198×10^{-4}
1	2	-3.2772×10^{-7}	0	1.2792×10^{-8}
1	4	-4.7879×10^{-6}	0	1.0210×10^{-7}
1	8	-2.1104×10^{-2}	0	9.5035×10^{-4}
1.25	2	-2.9427×10^{-7}	5.3247×10^{-9}	1.3414×10^{-8}
1.25	4	-5.4676×10^{-6}	-1.7803×10^{-7}	1.3754×10^{-7}
1.25	8	-2.4008×10^{-2}	-1.6736×10^{-3}	2.4402×10^{-3}

their own, evidence for a nonlinear intermittency mechanism. Table X varies the mode count, primitive oscillator resolution, bath selection, and excitation cutoff independently. Its entries are changes between specified calculations, not confidence intervals or certified truncation errors. The $Q = 12$, $\beta = 1$ baseline comes from the completed full-parity source calculation; the $\beta = 3$ baseline comes from the completed occupation comparison. Both retain the same ground-state pressure observable. The mode comparison retains the physical calibration at $M_{\mathrm{ref}} = 24$.

B. What the retained two-point reference removes

For each pressure increment define $A_p^{\mathrm{ref}} = \log[(M_p/M_2^{p/2})/(M_p^{\mathrm{ref}}/(M_2^{\mathrm{ref}})^{p/2})]$. Figure 9 compares the original harmonic reference with the Wick pressure built from the retained interacting D/Z transition spectrum. These answer different questions: the first includes changes of the two-point response, whereas the second removes the Wick contribution of the selected interacting two-point data. For $p = 8$, the latter contrast changes from 0.342254 at $\beta = 1$ to -0.543727 at $\beta = 3$. This statement concerns the retained spectrum. Missing odd transitions, source weight outside the retained tensor space, and approximate ground-state/source errors must be controlled before interpreting the difference as a complete connected-field measurement. At $\beta = 10$, A_8 relative to the original harmonic model reaches 14.0815, whereas the retained-Wick contrast is -0.35121. The magnitude of the latter is smaller than at $\beta = 3$, although the original-reference contrast is much larger. This comparison shows how strongly two-point retuning can affect the interpretation of the raw nonlinear contrast. Raising the primitive resolution from 40 to 48 changes the retained-Wick A_8 at $\beta = 10$ by 5.20×10^{-8}. The unfiltered upper bound on source covariance outside the retained tensor space remains 0.254312 in the declared source normalization. Primitive agreement alone therefore does not bound the omitted Gaussian pressure moments.

C. Averaging-width dependence

Earlier weak-strength fits, including a two-point interpolation, are preserved in the reproducibility records but are not used here as evidence for a resolved next-order law. The neighboring widths use common finite bases. For $\beta = 1$, the raw standardized-moment slope $d\log(M_8/M_2^4)/d\log\tau$ changes from 5.31829 to 16.7804 across the two adjacent intervals. The measured slopes vary across the current finite window. These three-window data do not supply a common widening scaling interval. Deeper-resonance calculations are reported separately; neither observation establishes or rules out intermittency in a broader boundary family.

D. Spectral tails, cross-scale ordering, and a declared instrument

Diagonalization of the saved finite increment produces positive spectral weights and threshold exceedances. The direct tail comparison in Sec. IX G uses common original-harmonic threshold units and explicitly changes both occupation allowance and retained selection. These are finite-operator spectral probabilities; moment agreement does not certify their omitted-state or continuum errors. Ordered cross-scale products and their reverse orderings are retained as separate diagnostics, rather than combined into an assumed classical joint law. The Gaussian Kraus instrument provides a distinct operational calculation. Its resolution and thresholds are fixed in common harmonic pressure units. For resolution one and a two-sided threshold eight, the first probe deposits energies 1.050×10^{-2}, 2.798×10^{-3} at $\beta = 1, 3$, respectively. Heating and correlated threshold outcomes also occur in the saved harmonic comparator. The sequential probabilities use the actual disturbed second marginal. They describe

TABLE IX. Ordered cumulants at $\tau = 51.2$. The nonlinear columns use $Q = 14$ and 32 primitive levels. Values have units of pressure to power n in the dimensionless model convention. These are finite-model values, not statistical estimates with inferred error bars.

	Ordinary averaged pressure			Increment, $b = 2$		
n	Harmonic	$\beta = 1$	$\beta = 3$	Harmonic	$\beta = 1$	$\beta = 3$
2	8.761×10^{-12}	9.146×10^{-12}	2.352×10^{-12}	8.604×10^{-12}	8.951×10^{-12}	2.341×10^{-12}
3	4.639×10^{-16}	4.921×10^{-16}	9.096×10^{-17}	8.250×10^{-17}	7.093×10^{-17}	8.058×10^{-18}
4	3.192×10^{-20}	3.403×10^{-20}	4.428×10^{-21}	3.419×10^{-21}	4.189×10^{-21}	5.615×10^{-22}
5	2.539×10^{-24}	2.789×10^{-24}	2.553×10^{-25}	7.946×10^{-26}	8.098×10^{-26}	4.442×10^{-27}
6	2.242×10^{-28}	2.613×10^{-28}	1.671×10^{-29}	3.502×10^{-30}	5.345×10^{-30}	3.097×10^{-31}
7	2.159×10^{-32}	2.751×10^{-32}	1.211×10^{-33}	1.161×10^{-34}	1.768×10^{-34}	3.336×10^{-36}
8	2.249×10^{-36}	3.229×10^{-36}	9.548×10^{-38}	5.665×10^{-39}	1.377×10^{-38}	2.345×10^{-40}

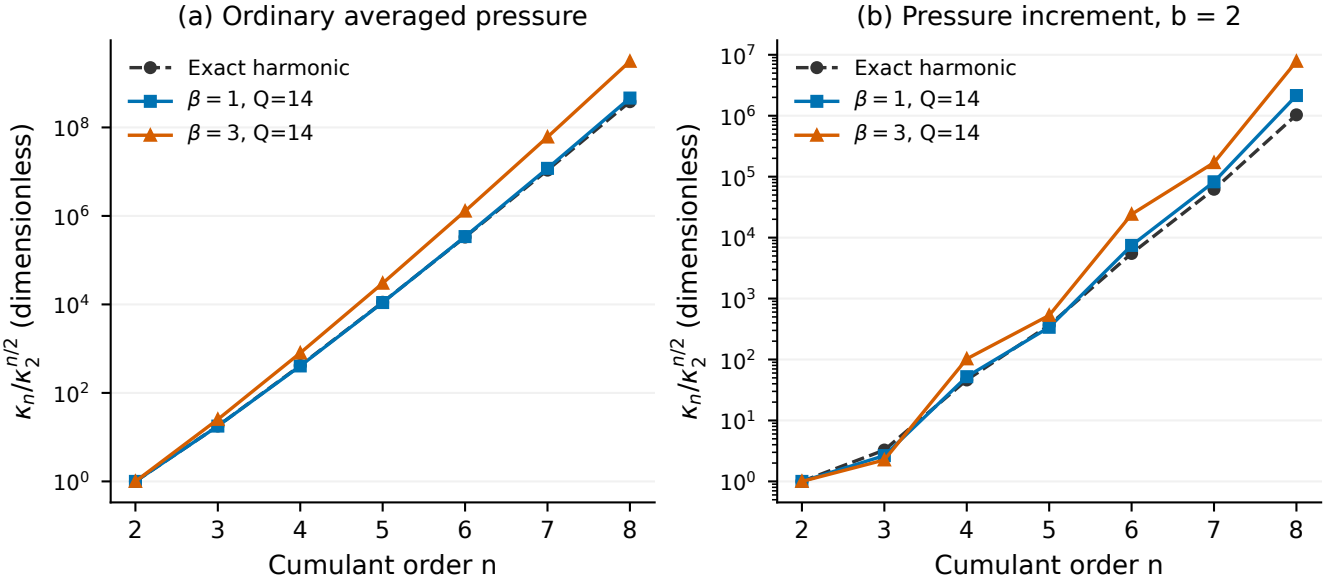


FIG. 8. Standardized cumulants for ordinary pressure and increments at $\tau = 51.2$. Harmonic statistics use the exact twenty-mode Riccati reference; nonlinear statistics use $Q = 14$. The horizontal coordinate is cumulant order, not an averaging scale. Growth along this axis is not a measured multiscaling exponent.

the declared finite measurement, not a local pressure detector or an undisturbed record of vacuum bursts.

E. Larger excitation space and deeper physical hierarchies

The $Q = 16$ refinement uses 1133 active configurations and 128 bath configurations. The target spectrum is selected by a fixed fraction 0.88 of the free bath threshold, before pressure moments are evaluated. High active configurations outside that target remain in the complementary tensor space. At $\beta = 1, 3$, the central A_8 values are 0.6268303851 and 1.8966494195, respectively. Their changes from $Q = 14$ are 9.20×10^{-5} and 1.26×10^{-5} as fractions of A_8. The conditional relative eighth-increment-moment path bounds at the central width are $e^{-11.57}$ and $e^{-61.53}$ for $\beta = 1, 3$. These concern omitted spectral paths inside the finite tensor model. They exclude the remaining graph, primitive, bath and line-mode errors. At the shorter width $51.2/\sqrt{2}$, the same bounds are inconclusive; agreement of adjacent-cutoff values is not a substitute for that missing certificate. At fixed reference phase variance, deeper hierarchies retain a finite shape contrast (Table XIII). These calculations follow the lowest resonance: $\tau\Omega_{\min} = 0.88$ for every row. Thus their averaging time grows by a factor 256 while their position relative to the infrared cutoff remains fixed. The apparent stability of A_8 is a statement about this moving cutoff regime, not an exponent over a common widening interval.

TABLE X. Fractional refinement changes at $\tau = 51.2$. For each observable, $\delta M = \max_{p=2,4,6,8} |M_p^{\mathrm{new}}/M_p^{\mathrm{base}} - 1|$. The last column is $|\kappa_8^{\mathrm{new}}/\kappa_8^{\mathrm{base}} - 1|$ for ordinary pressure. Bath counts refer to selected quartic seed configurations before downward closure, not the final bath dimension. Unlisted settings remain fixed within each comparison.

Comparison	β	δM ordinary	δM increment	$\delta\kappa_8$ ordinary
Modes $M : 20 \to 28$	1	7.72×10^{-6}	7.76×10^{-6}	7.72×10^{-6}
Modes $M : 20 \to 28$	3	4.84×10^{-5}	4.84×10^{-5}	4.84×10^{-5}
Primitive levels $32 \to 40$	1	7.85×10^{-13}	7.31×10^{-13}	7.55×10^{-13}
Primitive levels $32 \to 40$	3	2.94×10^{-11}	2.10×10^{-10}	1.86×10^{-11}
Quartic bath seeds $24 \to 48$	1	2.87×10^{-5}	2.87×10^{-5}	2.87×10^{-5}
Quartic bath seeds $24 \to 48$	3	4.12×10^{-4}	4.12×10^{-4}	4.12×10^{-4}
Excitation cutoff $Q : 12 \to 14$	1	1.10×10^{-3}	5.61×10^{-4}	1.11×10^{-3}
Excitation cutoff $Q : 12 \to 14$	3	6.49×10^{-3}	1.19×10^{-3}	6.51×10^{-3}

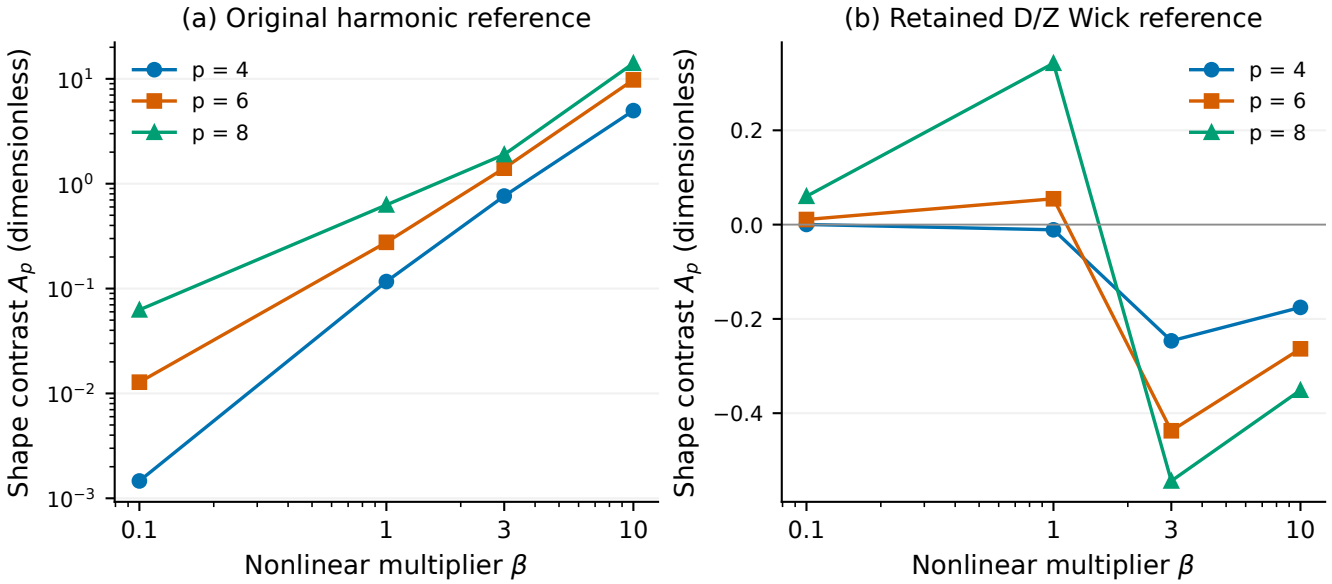


FIG. 9. Increment shape contrasts at $\tau = 51.2$, $R = 10$, $M = 20$, and $Q = 12$, with bath quanta fixed at four. The $\beta = 0.1, 10$ points use 40 primitive levels; $\beta = 1, 3$ use 32. Lines connect computed couplings. The left vertical axis is logarithmic; the right is linear. The right panel uses the retained D/Z Wick comparator and carries its source-completeness limitations; no omitted-source uncertainty bars are inferred from these curves.

F. An independent full-cosine response control

The isolated branch Hamiltonian $h_{\beta,\nu}$ defined in Sec. C provides a small calculation along the entire strength parameter. We evaluate its static susceptibility from all retained odd states and independently from an odd-sector resolvent. We also compute field cumulants and moments of the physical operator x^2, preserving intermediate oscillator states in powers through x^{16}. The test covers $\beta = 0, 0.001, 0.01, 0.03, 0.1, 0.3, 1, 3, 10, 30, 100$, $\nu = 1$ and $1 + 1/540$, and 32, 64 and 96 primitive levels. All 22 independent cosine-assembly checks and six exact harmonic controls pass; all final adjacent refinements satisfy the declared 10^{-8} scaled tolerance. The normalized field cumulants use a unit absolute scale floor in that comparison, so tiny connected values do not acquire a relative-accuracy guarantee. This is an isolated-branch control, not a replacement of the full-line reference calibration. At $\nu = 1$, the susceptibility decreases from 2 at $\beta = 0$ to 1.89991, 0.660495 and 0.0784313 at $\beta = 1, 10, 100$. The normalized fourth field cumulant takes values 0.0809601, 0.0134565 and 4.86374×10^{-6} at those three finite strengths. The computed sixth and eighth cumulants have additional sign changes. The curve therefore resolves a regime of non-Gaussian local shape followed by a return toward quadratic confinement as the calibration stiffness grows. It does not imply monotonic decay of every cumulant or determine the pressure hierarchy by itself.

TABLE XI. Two-probe events for the declared instrument at $\tau = t = 51.2$, $\sigma = 1$ and $h = 8$, in fixed original-harmonic pressure units. The harmonic rows use the same saved adapted basis as their nonlinear counterpart. p_2 includes the first measurement; $g = p_{11}/(p_1 p_2)$ therefore uses the actual marginals. These are finite-instrument probabilities, not a measurement of spontaneous pressure bursts.

β	Model	p_1	p_2	p_{11}	g
1	Full cosine	0.00340277	0.079732	0.00213143	7.85608
1	Saved-basis harmonic	0.00191821	0.0672272	0.000795795	6.17106
3	Full cosine	0.0006949	0.0163616	0.000148886	13.095
3	Saved-basis harmonic	0.0019071	0.0667319	0.000781205	6.13843

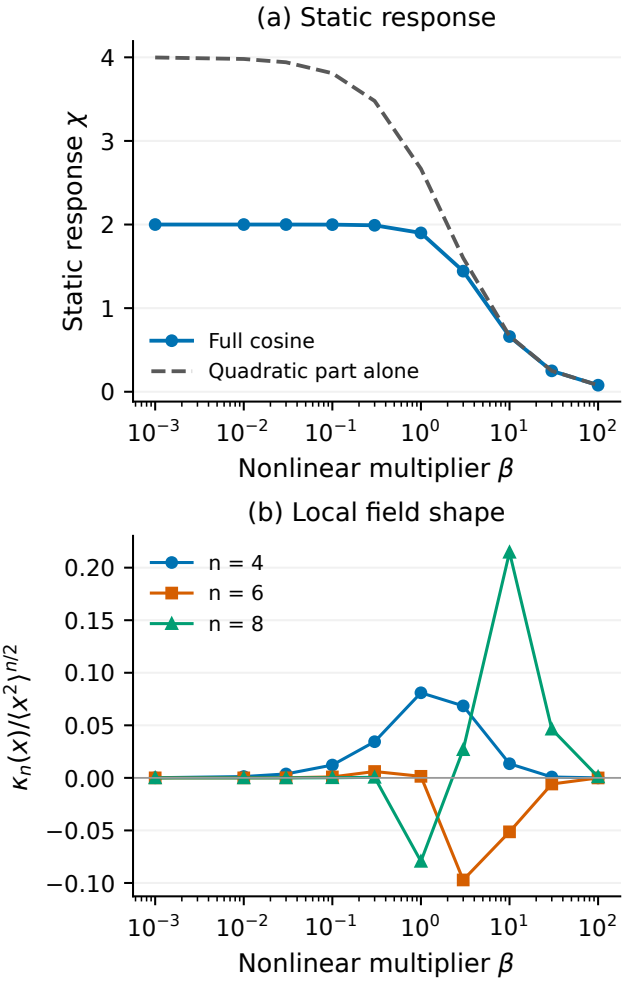


FIG. 10. Independent isolated full-cosine calculation at $\nu = 1$ with 96 primitive levels. Top: static susceptibility; the dashed curve is $8/(\beta + 2)$ for the quadratic part alone and serves as a strong-coupling comparison, not the calibrated $\beta = 0$ Hamiltonian. Bottom: normalized connected field moments; lines connect calculated strengths. Gaussian fields have zero values on this panel even though their squared-field pressure has nonzero higher cumulants.

TABLE XII. Further excitation refinement, $Q = 14 \to 16$, at $R = 10$, $M = 20$, $\tau = 51.2$. Entries are absolute fractional changes of each specified statistic. These changes combine occupation and retained-spectral selection changes; they do not isolate either contribution.

β	$M_8(O_\tau)$	$\kappa_8(O_\tau)$	$M_8(X_\tau)$
1	0.00012044241	0.00012617382	5.2058916×10^{-5}
3	0.00027094326	0.00027790966	1.7540687×10^{-5}

TABLE XIII. Finite-strength depth comparison at $\beta = 3$, $Q = 12$, with 32 primitive levels. The physical reference calibration is fixed within each row, using $M_{\mathrm{ref}} = 24, 24, 28$ respectively. All rows come from completed batches.

R	M	τ	A_8
10	20	51.2	1.8979057
14	24	819.2	1.8933952
18	28	13107.2	1.8913468

G. Finite-model tail sensitivity beyond moment agreement

We compare saved full-cosine pressure matrices at $R = 10$, $M = 20$, $M_{\mathrm{ref}} = 24$, $\tau = 51.2$ and $\beta = 1, 3$. The $Q = 14 \to 16$ comparison changes both occupation and retained spectral selection; the latter uses a bath-threshold fraction 0.88 at $Q = 16$. Write $O_\tau^c = O_\tau - \langle O_\tau \rangle_\beta I$ for the actual-ground-centered Gaussian pressure average and $X_\tau = O_\tau - O_{2\tau}$ for its width increment. For each observable Y, use one common unit $\sigma_0(Y)$ from the original distributed harmonic H_0 with all 20 declared modes. These units are $2.95984261 \times 10^{-6}$ and $2.93318955 \times 10^{-6}$ for O_τ^c and X_τ, respectively. The reference supplies units and moments here, not a calculated all-mode harmonic cumulative distribution. The hard effect is the strict event $|Y| > h\sigma_0$; equality belongs inside. For readout width $\eta > 0$, define $p_\eta(h) = \Pr(|Y/\sigma_0 + \eta Z| > h)$ with independent $Z \sim N(0, 1)$. This bounded spectral effect convolves a marginal law with readout noise; it does not specify sequential measurement back-action. At $\beta = 1, 3$, the increment eighth moments change by -0.0052059% and -0.0017541%, while the hard threshold-eight prob-

TABLE XIV. Threshold $h = 8$ in common original-H_0 units. $Y = O^c$ denotes centered ordinary pressure and $Y = X$ the width increment. $\eta = 0$ is the strict hard effect and $\eta = 1$ the stated Gaussian readout. $\delta p = 100(p^{Q=16}/p^{Q=14} - 1)$ is combined occupation/selection sensitivity, not a CDF error bound.

β	Y	η	$p^{Q=14}$	$p^{Q=16}$	δp (%)
1	O^c	0	0.001803474	0.001803926	+0.02508
1	O^c	1	0.002036868	0.002037002	+0.00657
1	X	0	0.001734939	0.002144290	+23.59456
1	X	1	0.003402774	0.003400081	-0.07911
3	O^c	0	0.000551751	0.000552117	+0.06643
3	O^c	1	0.000691927	0.000691985	+0.00837
3	X	0	0.000589481	0.000608203	+3.17592
3	X	1	0.000694900	0.000694909	+0.00135

abilities change by +23.5946% and +3.1759%. Thus agreement of moments through order eight does not certify hard tails. No requested threshold contains spectral mass within its numerical eigensolve edge band; the larger changes concern the discrete finite laws, rather than floating-point placement at a boundary. Across $h = 2, 4, 6, 8, 10, 12$ and $\eta = 0, 0.1, 0.25, 0.5, 1$, the largest relative hard-tail change is +44.4278% at $\beta = 1, h = 12$; for $\beta = 3$ it is +26.8414% at $h = 10$. The largest absolute change is 0.0018147707 at $\beta = 1, h = 2, \eta = 0.5$, exceeding the corresponding hard-effect difference. Smoothing therefore does not improve every threshold monotonically. The small $\eta = 1$ changes in Table XIV describe a different, smoother observable, not physical CDF convergence. All 17 independent controls pass, including known discrete laws, strict-boundary and zero-noise effects, degenerate-block rotations, and complete projected harmonic-pressure controls. All eight saved-law moment checks and 240 aggregate tail comparisons between two eigensolver drivers pass; the largest relative eigensolver tail discrepancy is 7.458×10^{-14}, and the smallest requested probability is 6.802377×10^{-6}, above the 10^{-10} relative-check floor. No filter-source deletion or overlap underflow is reported. These checks concern supplied finite matrices: they do not bound omitted-state, source, ground-state, primitive, bath or line-mode errors. The calculation neither establishes a rare-event limit nor resolves the growing common-interior hierarchy or vacuum intermittency.

a. Checking the numerical evaluation of the full cosine. We separately tested cancellation in the weak-coupling cosine matrices. An algebraically equivalent construction evaluates the complete projected cosine and sine remainders without numerically subtracting their leading quadratic and linear terms. It retains the full potential and the complete quadratic pressure contacts. Seven control groups include independently projected small Fock matrices and high-precision primitive matrix elements. For the physical $R = 10$, $M = 20$, $Q = 12$ calculation at $\beta = 0.0025, 0.005$, rebuilding the same finite space with this arithmetic gives a largest measured change in A_8 of 4.29×10^{-13} over three averaging widths, for both ordinary averages and increments. Separately recomputing the local orbitals at $\beta = 0.005$ changes A_8 by at most 2.98×10^{-13} relative to the production calculation. All original ground-state and retained-cluster checks remain satisfied. At the central increment, the measured pointwise changes divided by β^3 have magnitudes below 9×10^{-6}. This bounds the observed arithmetic difference for these two weak-strength rows. It does not bound a perturbative remainder or replace independent excitation, source, and higher-order refinements.

X. DISCUSSION

A. What closes the pressure statistics?

Within the specified branch families, the first-order calculations identify which missing boundary information changes the pressure prediction. A linear-field two-point calibration fixes neither the frequency dependence of the quartic boundary vertex nor the pressure statistics that sample it. The two-branch example separates this missing information from a change of linear response: at the reference geometry the two-point functions agree through first order, the leading static quartic coefficient is common, and the pressure-variance corrections nevertheless have opposite signs. The analytic mode-tail estimate protects this separation in the continuum coefficient. For larger branch families, the response-information calculation makes the same question quantitative: a set of vertex samples leaves a range of allowed pressure predictions until it removes every allocation direction to which that observable is sensitive. Closure is therefore relative to the observable, the averaging width, and the response information retained.

Higher pressure cumulants probe this information without requiring a corresponding increase of field-vertex order at first order in the coupling. Pressure is quadratic, so repeated pressure factors already sample a non-Gaussian distribution in a harmonic vacuum. A quartic insertion changes these ordered products through both preparation of the state and evolution of the smeared operator. Retaining both contributions is necessary for the sign and size of the correction. Their separate agreement with independent calculations is a stronger check than agreement of a sum whose errors might cancel.

At the next nonlinear order, the requirements change. A second quartic insertion contributes at the same order as the next microscopic potential coefficient. Normal ordering the latter changes lower vertices as well as producing a sextic term. The conditional six-field vertex in Sec. VI includes the corresponding exchange contribution; it is not simply the bare sextic contact. Within the occupation-refined three-mode regulator, the cosine comparison verifies the complete microscopic pressure coefficient through this order. It does not impose second-order equality of the two-point responses between allocations, nor does it replace an independent reduced real-time calculation with all pressure-source terms.

B. What does the intermittency test show?

A persistent nonlinear shape exponent would require A_p to acquire an approximately linear dependence on $\log\tau$, with a nonzero slope that survives widening the hierarchy and the relevant numerical controls. The resolved data show a different behavior. In the uniform fourteen-branch family, $A_{8,1}$ falls from 1.67622×10^{-5} to 1.14479×10^{-6} over $12.8 \le \tau \le 204.8$. Its adjacent scale derivative also decreases in magnitude. Allocating more quartic coupling to high-frequency branches makes the correction decay faster; low-frequency allocation gives a much flatter offset. This progression in the continuous allocation family is consistent with the conditional soft-branch envelope $\tau^{-(s+1)}$. A power law for this small correction is not an anomalous exponent of the full standardized moment.

Sampling between the dyadic widths reveals departures from a pure envelope that a dyadic fit can conceal. At still larger widths, exponentially small moments can produce enormous ratios from floating-point cancellation; the independent integration-rule check identifies examples that must be excluded. In the principal fourteen-branch window the tested mode and integration-rule changes are small, and changing the increment ratio from two to three preserves the decreasing correction. These checks support the reported first-order scale dependence. They do not supply a uniform bound on the full nonlinear remainder as the hierarchy depth or averaging width tends to infinity.

The full-cosine calculations extend this test beyond the perturbative remainder. They resolve changes of every pressure cumulant through eighth order, with independent numerical refinements at finite coupling. Their neighboring-window slopes vary substantially. The deeper-resonance series follows a moving infrared edge; the additional common-width comparison in Sec. D 3 instead keeps the averaging times fixed. At those times the ten-resonance device is beyond its infrared cutoff while the fourteen-resonance device remains within its nominal resonance band, and their eighth-moment shape contrasts differ substantially. Because their numerical regulators also differ and the fourteen-resonance omitted-path bound is inconclusive, this comparison measures sensitivity to the combined depth and regulator change. Neither series supplies a common widening interval with regulator-controlled moment estimates and a persistent anomalous exponent. The finite-strength effect itself remains: its size and interpretation depend on whether the reference retains the original harmonic response or the retained interacting two-point data. These are different closure questions, as illustrated in Sec. IX. The source-space enlargement in Sec. D 1 substantially reduces the measured field-source weight outside the tensor space, while changing the eighth pressure moments by about 0.002%. Most of the unfiltered linear-source norm nevertheless remains outside the retained odd-transition spectrum. The subsequent fixed-source refinement makes the limitation quantitative: neither the $64 \to 128$ nor the $128 \to 200$ comparison passes any of the 42 moment/shape criteria or the 12 signed coherent-source criteria. At the central increment window the reference A_8 changes by 0.5096, larger than the physical-minus-latest-reference contrast 0.3421. The source-only continuation in Sec. D 1 b improves from zero accepted matrix entries on $200 \to 256$ to four of nine on $256 \to 320$, and the shortest-width panel passes in full. The second edge is

markedly closer, but the two longer-width panels still fail the frozen tolerance, so the combined 24-criterion gate remains false. The retained interacting Gaussian comparison has therefore not attained the convergence needed to isolate a nonlinear pressure-shape claim.

The excited-sector test explains more specifically why variance agreement is insufficient. Write the centered averaged pressure in vacuum and excited blocks, with vacuum excitation vector g and excited block B. Its variance is $s = \|g\|^2$, whereas its eighth moment also contains the positive chain $q_6 = \|B^3 g\|^2$. Restricting excited energies to $4\hbar/\tau$ preserves more than 99.9999% of the variance in the three reported increments, but retains only about 11%, 43%, and 11% of q_6 at resonance counts 10, 14, and 18, respectively. Subsequent pressure actions therefore probe information barely visible in direct vacuum excitation. These are changes of the supplied matrices, not fractions of the total eighth moment or probability-error bounds. The near repetition of the propagation ratio at matched lowest-gap positions is consistent with a moving cutoff crossover; it does not establish a common-interior scaling exponent.

The fixed-coordinate theorem in Sec. C 5 identifies a further distinction. With the full confining cosine and every oscillator excitation retained, the ultimate moments at each fixed even order are still controlled by discrete energy-gap paths. Gaussian averaging makes their leading width dependence exponential in τ^2; the logarithmic standardized-moment contrast consequently approaches a constant plus a term proportional to τ^2, with exponentially smaller corrections. It has no nonzero asymptotic $\log \tau$ term at those fixed parameters. This result removes an occupation cutoff from that statement, but it does not show that the computed windows are asymptotic. It also leaves open a simultaneous increase of width, resonance depth and line-mode resolution: the rate separations and bounds can change as new soft resonances appear. The quasistatic analysis gives equal-time and fixed-projection controls for such sequences, while the uniform bound on higher ordered pressure paths along moving windows remains open.

The direct tail refinement in Sec. IX G exposes an additional convergence requirement. Small changes of moments through order eight coexist with appreciable changes of hard-threshold event probabilities and quantiles. The comparison changes both occupation and spectral selection, so it does not identify either contribution in isolation. Gaussian readout smoothing reduces some differences but can increase others; it defines a different effect and supplies no general omitted-state probability certificate. The distinction is physical as well as numerical: the resolution of a proposed pressure measurement must be specified before declaring its rare-event law stable.

Applying a common interacting energy ceiling in Sec. D 6 tests one part of this selection sensitivity. The ordinary-pressure tail remains stable in that comparison, whereas the increment tail depends strongly on both the excitation allowance and the energy ceiling. Its first four even moments remain nearly unchanged in the saved arithmetic as the ceiling moves. Agreement of those moments therefore does not resolve the missing probability information. The next test must restore the relevant pressure paths in a larger retained energy cluster at fixed occupation and calibration.

A spectral exceedance, a curved local moment-order spectrum, and clustered measurement outcomes remain worthwhile observables even when this stronger scaling criterion is unmet. The specified finite state and pressure operator, threshold, and reference unit determine the first; regulator-controlled moment estimates over a common widening interval test the second; a specified measurement instrument and its disturbance determine the third. The computed Gaussian instrument heats the initial ground state and changes the second event probability, including in the harmonic control. Its clustering consequently cannot be read as an undisturbed vacuum-burst record.

C. Geometry, scope, and measurable quantities

The connection to the Casimir effect is the quantized line spectrum and vacuum stress set jointly by the separated boundaries. The reproduced harmonic interaction energy provides the mean-force reference. The fixed-calibration length comparison is a first-order test of the same microscopic boundary within each complete finite-element regulator. Its counterterm is chosen at the reference separation and held fixed when the length changes, retaining the induced quadratic detuning. It demonstrates geometry-dependent local-pressure coefficients in those regulators; it is not a finite-strength geometry sweep or a continuum bound for every higher cumulant. This protocol is distinct from increasing resonance depth, which changes the circuit family. A net-force distribution would additionally require a specified balance of inside and outside stresses; subtracting two local cumulants does not construct such a distribution.

The scope of each numerical statement remains separate. The variance has an analytic continuum coefficient bound for finitely many positive branches. Higher first-order statistics have independent ordered checks and finite-mode or spatial refinement. The complete second-order cosine calculation is occupation-refined at three retained modes. The full-cosine calculations use refined finite tensor spaces, original pressure projections, and bounds on omitted interacting spectral paths. Those path bounds do not include configurations absent from the tensor factors. The Gaussian comparator additionally needs its own odd-transition and composite-source controls. The tables therefore distinguish measured refinement changes, finite-model spectral bounds, and the still unproved uniform nonlinear hierarchy limit.

Finally, each X_τ defines a positive spectral probability law, but different averaging widths need not define

commuting observables. The present moments consequently do not by themselves establish a joint classical process, time-resolved bursts, an energy cascade, or a detector counting law. Connecting the calculation to a measurement requires specifying that detector's coupling, bandwidth, ordering, and backaction. The closure result concerns the precisely defined pressure observable before those additional choices.

XI. CONCLUSION

The nonlinear boundary closure is incomplete when it retains a calibrated two-point response and only a static quartic coefficient. Ordered pressure statistics also sample nonlinear response at nonzero frequencies and the correctly reduced pressure source. The separated variance corrections make that statement quantitative in a Casimir transmission-line model, with a controlled continuum mode tail. Higher-cumulant calculations and the first-order fixed-calibration geometry comparison provide further finite-regulator tests. At the next coupling order, both repeated quartic insertions and the complete next potential term are required. In the tested finite Hamiltonians, calculations with the unexpanded cosine then resolve finite-strength changes beyond that expansion, including all ordinary and increment pressure cumulants through eighth order.

The intermittency question has several distinct outcomes. Nonlinear changes of pressure shape are present in the tested finite models. Their comparison with Gaussian pressure constructed from the retained interacting two-point spectrum differs substantially from comparison with the original harmonic device. Extending the necessary source measure through dimensions 256 and 320 substantially reduces the successive source-matrix discrepancy, but the two-edge gate still fails at the longer averaging widths. The tested averaging windows and resonance hierarchy therefore do not establish a persistent anomalous exponent. Direct rare-event calculations reveal an additional limitation: close agreement of moments does not guarantee stable hard-threshold probabilities under the tested occupation and selection change. Pressure propagation within the excited sector also matters even when nearly all of the variance is retained; that diagnostic does not by itself bound probabilities. Correlated outcomes of a declared measurement are also computed, with their separate source and backaction limitations stated. These results preserve the physical question: determining intermittency requires a specified pressure observable, a controlled extending scale interval or event law, and a boundary description that retains the information those observables need.

For each fixed finite coordinate representation of the confining cosine model, the long-window theorem holds without an excitation cutoff: even pressure moments ultimately follow discrete energy-gap rates in the squared Gaussian width. The theorem neither identifies the sampled regime nor interchanges the long-window, increasing-hierarchy and continuum limits. It therefore sharpens the scale test while leaving the physical question of intermittency along an extending resonance hierarchy open.

Appendix A: Finite-coupling dynamics with a correlated bath

Long averaging times emphasize low transition frequencies, but fast modes can still change those transitions through virtual excitation. We therefore divide the retained harmonic modes into a low-frequency group and a bath, while allowing the bath to dress every retained low-mode state. Individual oscillator orbitals are first adapted self-consistently to the anharmonic potential, following the coupled-oscillator strategy of Ref. [24]. A primitive basis of N_{prim} harmonic oscillator states resolves each local orbital. These orbitals provide a basis; correlations are subsequently determined by the full projected Hamiltonian. The final state is not restricted to a Gaussian state or a Hartree product.

Let $\mathcal{H}_L$ and $\mathcal{H}_B$ be finite spans of low-mode and bath configurations. The working space is their complete tensor product,

$$\mathcal{H}_T = \mathcal{H}_L \otimes \mathcal{H}_B, \qquad \Pi_T = \Pi_L \otimes \Pi_B. \tag{A1}$$

Every retained bath configuration thus accompanies every retained low configuration. In a separable control, the same interacting bath energy then cancels from all low transition gaps. Both parities are retained in each factor: odd–odd products belong to the total-even sector and can contribute to pressure paths. Completeness of the tensor product does not imply completeness of either factor.

The low-factor excitation cutoff is a weighted budget Q. If n_j labels the adapted local orbital in active mode j, the retained configurations satisfy

$$\sum_{j\in L} c_j n_j \le Q, \qquad c_j = \max\left(1, \frac{s_{\mathrm{soft}}}{\tau_{\mathrm{build}}\omega_j}, s_{\mathrm{hard}}\tau_{\mathrm{build}}\omega_j\right). \tag{A2}$$

The construction width τ_{build} sets this basis and is held fixed when its neighboring observation widths are evaluated. The ten-branch results use the lowest five modes, $s_{\mathrm{soft}} = 0$, $s_{\mathrm{hard}} = 1/2$, and $\tau_{\mathrm{build}} = 51.2$. Thus $Q = 12, 14, 16$ are weighted excitation budgets, not mode counts and not the unweighted occupation ceilings of the earlier three-mode checks. Bath configurations, primitive orbitals and the target spectrum below have separate cutoffs.

At finite dimensionless coupling β, the projected mi-

croscopic Hamiltonian is, up to a scalar,

$$H_T = \Pi_T \Bigg[H_0 + \sum_r \Bigg(U_r(q_r;\beta) - \frac{K_{Jr}}{2} q_r^2 - \frac{\beta u_r v_r^{\mathrm{ref}}}{4} q_r^2 \Bigg) \Bigg] \Pi_T, \tag{A3}$$

with the full cosine U_r of Eq. (12). The reference variance v_r^{ref} specifies the original stiffness calibration and is not replaced by the dressed variance. Splitting $q_r = q_{Lr} + q_{Br}$ permits exact tensor factorization through $\cos(\vartheta_r q_r) = \cos(\vartheta_r q_{Lr})\cos(\vartheta_r q_{Br}) - \sin(\vartheta_r q_{Lr})\sin(\vartheta_r q_{Br})$. These trigonometric operators are projected from their primitive oscillator matrix elements before changing basis; taking the cosine of a truncated coordinate would define a different operator. The quadratic pressure is likewise projected directly as $P_T = \Pi_T P \Pi_T$, including intermediate states required by its same-mode squares. Squaring an already projected linear field need not give this composite operator. Reference mean-field terms are subtracted when assembling the full interaction to avoid double counting.

1. Selected spectrum within the full tensor space

To obtain the low interacting spectrum without diagonalizing the entire tensor matrix, first diagonalize the correlated bath reference. Let $|b_0\rangle$ be its even ground state, and diagonalize $A_{\mathrm{ref}} = \langle b_0|H_T|b_0\rangle$ on the full active factor $\mathcal{H}_L$. A complete low-energy subset of its eigenstates $|a_j\rangle$ defines the target subspace $\mathcal{L} = \mathrm{span}\{|a_j\rangle \otimes |b_0\rangle : j \in \mathcal{I}_\sigma\}$ in total-parity sector σ. Its complement $\mathcal{Q}$ is taken within that same sector of $\mathcal{H}_T$. It contains both the bath-ground products with omitted high-active states and all excited-bath products of the allowed total parity. Thus reducing the target energy window does not discard active configurations from the variational tensor space. Conversely, increasing an occupation cutoff need not increase the retained spectral bandwidth. The original construction takes all active states of the chosen parity as targets; the selected construction keeps only the declared low-energy subset, without splitting a degenerate level.

In this sector decomposition write

$$H_T^{(\sigma)} = \begin{pmatrix} A & B^\dagger \\ B & D \end{pmatrix}, \qquad 0 = B + D\Omega - \Omega A - \Omega B^\dagger \Omega. \tag{A4}$$

The map $\Omega : \mathcal{L} \to \mathcal{Q}$ attaches complement amplitudes to each target state. The graph equation makes their span invariant under $H_T^{(\sigma)}$; it is solved iteratively without expanding the cosine or truncating its powers of β. With

$$S = I + \Omega^\dagger \Omega, \qquad U = \begin{pmatrix} I \\ \Omega \end{pmatrix} S^{-1/2}, \qquad H_{\mathrm{eff}} = U^\dagger H_T^{(\sigma)} U, \quad P_{\mathrm{eff}} = U^\dagger P_T^{(\sigma)} U, \tag{A5}$$

$U^\dagger U = I$. The metric factors normalize both the Hamiltonian and pressure. A small invariance residual alone does not identify the lowest spectral subspace. Acceptance also requires a lower bound for the full $\mathcal{Q}$ block above the target block, and separation above the highest retained interacting energy, with compression and residual margins. An even-sector ground state is used as the physical vacuum only after comparison with a lower bound for the full odd sector. Local-orbital stationarity and absolute gap accuracy are checked separately.

2. Ordered ordinary and increment cumulants

Let $|g_L\rangle$ be the ground eigenvector of H_{eff}. Center the supplied pressure in this actual interacting state,

$$P_{c,L} = P_{\mathrm{eff}} - c_L I, \qquad c_L = \langle g_L|P_{\mathrm{eff}}|g_L\rangle. \tag{A6}$$

This changes only the first ordinary cumulant; an increment is already insensitive to pressure constants. If the supplied pressure has lost an additive reference constant, c_L is not a calibrated Casimir mean. In the energy eigenbasis define either smeared operator by

$$\begin{aligned} (Y_{\tau,L})_{\mu\nu} &= (P_{c,L})_{\mu\nu} F_\tau(\omega_{\mu\nu}), \\ \omega_{\mu\nu} &= \frac{E_\mu - E_\nu}{\hbar}, \\ F_\tau^{\mathrm{ord}}(\omega) &= e^{-\tau^2\omega^2/2}, \\ F_\tau^{\mathrm{inc}}(\omega) &= e^{-\tau^2\omega^2/2} - e^{-b^2\tau^2\omega^2/2}, \end{aligned} \tag{A7}$$

where $b > 1$. Here Y denotes the centered ordinary average or the increment X, respectively. For $v_k = Y_{\tau,L}^k |g_L\rangle$, all raw moments through eighth order follow from

$$\begin{aligned} m_{2k,L} &= \langle v_k|v_k\rangle, \\ m_{2k+1,L} &= \langle v_k|Y_{\tau,L}|v_k\rangle. \end{aligned} \tag{A8}$$

For increments these are the moments $M_{p,L}$ used above. The even moments are positive norms. With $m_{0,L} = 1$ and $m_{1,L} = 0$, the spectral cumulants defined in Sec. IV obey

$$\kappa_{n,L} = m_{n,L} - \sum_{j=1}^{n-1} \binom{n-1}{j-1} \kappa_{j,L} m_{n-j,L}, \tag{A9}$$

for $n = 1, \ldots, 8$. This scalar recurrence acts on moments of one Hermitian operator and does not change their quantum ordering. Odd moments are retained: for example, the centered sixth cumulant contains $-10m_{3,L}^2$. Even moments alone therefore do not determine all the even cumulants. Independent diagonalization of $Y_{\tau,L}$ checks its spectral probabilities and reconstructs both moment and cumulant sequences. Neighboring widths use common factor spaces, orbitals, and Hamiltonians at each β; only the filter changes.

3. Conditional spectral-path bounds

Even an exact invariant subspace retains only low-to-low pressure matrix elements. Powers of that pressure omit paths through higher interacting states. To isolate this error, now let $\mathcal{L}$ be an exact spectral subspace of the same finite tensor Hamiltonian, containing its exact ground state $|g\rangle$. Use the common source $P_{c,T} = P_T - \langle g|P_T|g\rangle I$ in both spaces; its restriction then has exactly the same centering. Denote the unrestricted moment by m_p. If every omitted eigenenergy lies at least $\Lambda > 0$ above the ground energy, a p-step returning path visiting the complement obeys $\sum_j (\Delta E_j)^2 \geq 4\Lambda^2/p$. Since both filters satisfy $|F_\tau(\omega)| \leq e^{-\tau^2\omega^2/2}$,

$$|m_p - m_{p,L}| \leq \varepsilon_p, \qquad \varepsilon_p = e^{-2(\tau\Lambda/\hbar)^2/p}\, \|P_{c,T}\|_F^p, \tag{A10}$$

for every $p \geq 2$, odd or even; $\varepsilon_1 = 0$ under this common ground-state centering. The Frobenius norm is taken in the complete finite tensor space, or in its exact ground-state parity sector for this parity-even pressure, not in the retained effective space. A corresponding tensor-factor norm bound avoids storing the full matrix.

Intervals enclosing all m_p, including the odd moments, can be propagated through Eq. (A9) with outward rounding to enclose $\kappa_2, \ldots, \kappa_8$. Cancellation between moment products can make a cumulant poorly resolved even when the raw moments agree closely. These are conditional bounds for omitted interacting spectral paths. Errors in the computed ground state, eigenvalues, eigenvectors, source projection, and centering must be controlled separately before applying them to an approximate graph solution. Configurations excluded from either tensor factor, primitive oscillator levels, and line modes require independent refinements.

The path argument also extends to a Gibbs state of the finite tensor model, retaining the same fixed pressure centering in both spaces. Write $\epsilon_i = E_i - E_0$, $T_E = k_B T$, and let $\mathcal{I}_L$ be an exact retained spectral set containing the ground state. Its conditional Gibbs weights are $\pi_i^L = e^{-\epsilon_i/T_E}/Z_L$, with $Z_L = \sum_{i\in\mathcal{I}_L} e^{-\epsilon_i/T_E}$. Let $q_{\rm high}$ be the omitted population in the full Gibbs state, and assume each omitted excitation energy is at least Λ. A returning path from i that visits the omitted set satisfies $\sum_j (\Delta E_j)^2 \geq 4(\Lambda - \epsilon_i)_+^2/p$, where $(x)_+ = \max(x, 0)$. For $p \geq 2$,

$$|m_p(T_E) - m_{p,L}(T_E)| \leq \|P_{c,T}\|_F^p \left[2q_{\rm high} + \sum_{i\in\mathcal{I}_L} \pi_i^L e^{-2[\tau(\Lambda-\epsilon_i)_+/\hbar]^2/p}\right]. \tag{A11}$$

Here $m_{p,L}(T_E)$ uses the conditional Gibbs state and low-to-low pressure matrix. The population term bounds both omitted initial states and the normalization change. Both parities enter the thermal trace. The ordinary first moment need not vanish with this fixed centering; its difference is bounded by $2q_{\rm high}\|P_{c,T}\|_F$, whereas an increment has zero diagonal and hence zero Gibbs mean. This first-moment budget must also enter any thermal cumulant enclosure. As $T_E \to 0$, Eq. (A11) recovers the ground-state path bound. The same restrictions on approximate states and sources continue to apply.

Four small-model controls compare this reduction with full tensor diagonalization at $\beta = 0.1, 1$, for coupled and separable low–bath systems. The largest relative difference in the checked moments through eighth order is below 5.4×10^{-12}; the separable spectator invariance agrees within 3.3×10^{-15}. These checks concern the finite numerical construction. They neither specify which boundary vertices close the physical pressure statistics nor establish convergence of the resonance hierarchy.

4. Stable evaluation of ordered Gaussian insertions

The formal first-order calculation has an additional numerical requirement at long averaging widths. A smeared quadratic pressure may have very small vacuum pair-creation amplitudes while its number-conserving matrix elements remain large. Separately expanding exponentials of the latter can introduce large terms that should cancel before a small moment coefficient is extracted. Algebraically exact integration in the insertion fraction does not protect against this floating-point cancellation.

For the harmonic smeared operator, let $A = A^T$ and $B = B^T$ be its number-conserving and pair matrices, respectively. Introduce the formal moment variable s, the insertion fraction $t \in [0, 1]$, and

$$\mathcal{G} = \begin{pmatrix} -A & -B \\ B & A^T \end{pmatrix}, \qquad L(s) = \begin{pmatrix} 0 & I/2 \\ I/2 & Z(s) \end{pmatrix}. \tag{A12}$$

Here $Z(s)$ is the Riccati series of the harmonic vacuum amplitude. The ordered insertion uses the formal covariance

$$C(s, t) = e^{-ts\mathcal{G}} L(s) e^{-ts\mathcal{G}^T}. \tag{A13}$$

This is a generating-series identity for the specified ordered product; it does not replace the quantum law by a classical Gaussian process. Writing $L(s) = \sum_n s^n L_n$ and $C(s, t) = \sum_n s^n \sum_{k=0}^n t^k D_{k,n}$, its differential equation $\partial_t C = -s(\mathcal{G}C + C\mathcal{G}^T)$ gives

$$D_{0,n} = L_n, \qquad D_{k,n} = -\frac{\mathcal{G}D_{k-1,n-1} + D_{k-1,n-1}\mathcal{G}^T}{k}. \tag{A14}$$

We evaluate this recurrence before contracting the covariance with the unchanged quartic and quadratic insertion tensors. The state resolvent, reference calibration, and real-time ordering are unchanged.

The first coefficient provides a simple independent check: $C_1(t) = \mathrm{diag}(tB, (1-t)B)$. If $B = 0$, then

$Z = 0$ and all nonconstant covariance coefficients vanish, regardless of the size of A. Equation (A14) preserves this cancellation before high powers of A are formed. Controls also compare independent finite-Fock derivatives with nonzero quadratic calibration detuning, mode permutations, and two integration rules that are exact for the truncated formal polynomial. Agreement of the translated moments with their cumulants is checked separately; that internal identity alone cannot certify the accuracy of the underlying derivative.

Ordinary moments are centered about the interacting ground-state mean. If the original-mean calculation gives derivatives δM_n and μ', the actual-ground-centered derivative is $\delta M_n^c = \delta M_n - n\mu' M_{n-1,0}$. This scalar translation leaves cumulants of order two and higher unchanged. Finally, the independent all-occupation derivative at a fixed set of line modes need not equal the slope of a truncated nonlinear curve. A comparison of nonlinear orders must retain a possible linear truncation discrepancy after its measured zero-coupling offset is removed.

5. How excited-state pressure transitions feed high moments

A large normalized moment can arise when the pressure acts weakly on the vacuum but more strongly on states reached after that first action. We can separate these contributions without replacing the quantum observable by a classical process. Let X be either the ordinary smeared pressure centered in its actual ground state Ω, or the width increment. Within the supplied Hilbert space, write

$$X = \begin{pmatrix} 0 & g^\dagger \\ g & B \end{pmatrix}, \qquad s = \|g\|^2, \qquad q_k = \langle g, B^k g\rangle. \tag{A15}$$

Here g is the part of $X\Omega$ orthogonal to the vacuum, and B is pressure restricted to that orthogonal complement. This complement is inside the supplied space; it is distinct from states omitted by a numerical cutoff. Assume $s > 0$. The positive quantities $q_{2j} = \|B^j g\|^2$ measure subsequent pressure actions on the vacuum excitation. Direct block multiplication gives

$$\begin{aligned} M_2 &= s, \qquad M_4 = s^2 + q_2, \\ M_6 &= s^3 + 2sq_2 + q_1^2 + q_4, \\ M_8 &= s^4 + 3s^2 q_2 + q_2^2 + 3sq_1^2 \\ &\quad + 2q_1 q_3 + 2sq_4 + q_6. \end{aligned} \tag{A16}$$

The mixed term $q_1 q_3$ can be negative. Nevertheless,

$$\begin{aligned} s^3 + q_4 \le M_6 \le \tfrac{5}{2}(s^3 + q_4), \\ \tfrac{2}{3}(s^4 + q_6) \le M_8 \le 13(s^4 + q_6). \end{aligned} \tag{A17}$$

These constants do not depend on matrix size. For the lower eighth-order bound, complete the square in $3sq_1^2 + 2q_1 q_3$ and use $q_3^2 \le sq_6$. For the upper bounds, apply the moment inequalities of the positive spectral measure of B in g to each term of Eq. (A16).

Consequently, $R_{2m} = M_{2m}/s^m$ is bounded above and below by fixed positive multiples of $1 + J_{2m}$, where

$$J_4 = q_2/s^2, \qquad J_6 = q_4/s^3, \qquad J_8 = q_6/s^4. \tag{A18}$$

If the logarithmic decay exponents of s and q_{2m-2} exist along a specified limit, the exponent of M_{2m} is the smaller of the exponent of s^m and that of q_{2m-2}. This decay convention is opposite to the signed growth convention in Eq. (54); local slope limits require their own existence assumptions. A positive power growth of J_{2m} therefore identifies a corresponding standardized power anomaly. Logarithmic growth, finite-gap exponential growth and finite-window slope changes must be distinguished from that statement. The identities alone prove neither convergence of the supplied matrices nor existence of a physical limiting exponent.

The filter must act on both blocks in Eq. (A15). In an energy basis, ordinary Gaussian averaging multiplies g_i by $g_\tau((E_i - E_0)/\hbar)$ and the unfiltered B_{ij} by $g_\tau((E_i - E_j)/\hbar)$. Width increments use the difference of these filters. They remove every exactly zero-transfer block, including off-diagonal entries within degenerate energy eigenspaces. Removing a finite band of small transfers is a separately declared alteration of the observable.

6. A quantum control with a known anomalous spectrum

To test sensitivity to a quantum mechanism, consider a vacuum plus two excited spaces $L^2(0,1)$, with energy acting by multiplication by E. Set $\hbar = 1$ for this control. The unsmeared source couples the vacuum to $g_\alpha(E) = \sqrt{\alpha} E^{(\alpha-1)/2}$ in each channel and acts as cI inside the second channel. It vanishes inside the first channel. Take $\alpha_1 = 1$ and $\alpha_2 = 5/2$. After ordinary Gaussian averaging, its vacuum-generated subspace has the matrix

$$X_\tau = \begin{pmatrix} 0 & a_\tau & b_\tau \\ a_\tau & 0 & 0 \\ b_\tau & 0 & c \end{pmatrix}, \qquad a_\tau^2 \sim C_1 \tau^{-1}, \quad b_\tau^2 \sim C_2 \tau^{-5/2}, \tag{A19}$$

with $C_1, C_2 > 0$. Thus $s = a_\tau^2 + b_\tau^2$ and $q_k = c^k b_\tau^2$ for $k > 0$. For every fixed $c > 0$, the positive decay exponents of orders $2, 4, 6, 8$ are $(1, 2, 5/2, 5/2)$. The rarer second channel controls the higher moments. In contrast, width increments remove cI, give $B = 0$, and have $M_{2m} = s^m$. The ordinary $c = 0$ endpoint also has this latter form.

The saved calculation evaluates 33 widths $\tau = 2^k$, $0 \le k \le 32$, and $c = 0, 0.1, 0.3, 1$, for ordinary averages and ratio-two increments. Independent continuum-integral, endpoint and known-spectrum controls pass. The $c = 0.1$ case still has a sixth-order decay slope about 2.5503 on its last interval, although its limiting exponent is 2.5. Even this known mechanism can have a long crossover.

This bounded source is a mathematical quantum control, not a derived local Casimir pressure. In the harmonic transmission line, the pair-creation and number-conserving pressure kernels are built from the same endpoint and gradient coefficients. Their vacuum and excited-state couplings cannot be chosen independently as above. The full-cosine calculation must test Eq. (A18) using its actual interacting ground state and complete pressure source. Moreover, nonzero q_{2j} already occur for quadratic pressure in Gaussian fields. Their presence alone establishes neither a nonlinear field effect nor intermittency.

Appendix B: Physical mechanisms and finite-strength statistical tests

1. Which intermittency question is being tested?

A large pressure outcome, an anomalous moment spectrum, and a cluster of registered events answer different physical questions. We retain all three questions. For a single X_τ, diagonalization gives a positive spectral law, with weights $w_j = |\langle x_j|\Omega_\beta\rangle|^2$ at its eigenvalues x_j. Its exceedance probability is $\Pr(X_\tau > h) = \sum_{x_j>h} w_j$. A heavy or strongly asymmetric pressure distribution can already arise from a Gaussian field [3, 4]. Its presence is physical statistical information and is retained in the harmonic reference. Agreement of moments through eighth order does not, however, determine the far tail. Exceedances require their own threshold-resolution and cutoff checks.

For scale dependence we report the raw exponents ζ_p of Eq. (54), their shape combination $\zeta_p - p\zeta_2/2$, and the nonlinear excess Δ_p separately. A persistent curved moment-order spectrum is a stronger finding than a large value of one standardized moment. Conversely, the absence of a persistent nonlinear excess does not exclude scale dependence already present in the harmonic pressure. As the resonance range grows, comparisons must include a common interval of averaging widths before assessing whether that interval expands. Following only a moving lowest resonance can turn an infrared crossover into an apparent scaling law.

Marginal laws at different widths need not specify a joint classical process. We retain additional quantum cross-scale information through

$$S_{ab}(\tau,\sigma) = \langle X_\tau^a X_\sigma^{2b} X_\tau^a\rangle = \|X_\sigma^b X_\tau^a|\Omega_\beta\rangle\|^2, \qquad \text{(B1)}$$

with positive integers a, b and $2a + 2b \le 8$. The reverse ordering and $\|[X_\tau, X_\sigma]|\Omega_\beta\rangle\|^2$ test information lost by replacing these operators with commuting random variables. A closure for scale-dependent pressure statistics must state which of these observables it preserves. Event clustering in an actual record additionally requires a detector instrument and its disturbance to be specified.

2. Two paths through nonlinear strength

The original hierarchy holds the total leading static quartic coefficient $G = (1/4)^4 \sum_r u_r$ fixed, since each branch has zero-frequency transfer $h_r(0) = 1/4$. This definition excludes the multiplier β: the leading coefficient in the action is βG. Holding G fixed sets a total nonlinear budget, but need not keep the dimensionless strength of the soft branches fixed. For the hierarchy parameters $K_{Jr} = 1/(2R)$ and $C_r = 1/(R\Omega_r^2)$, the uncoupled soft-branch estimate makes this distinction explicit:

$$v_r \simeq \frac{\hbar R\Omega_r}{2}, \qquad \vartheta_r^2 v_r \simeq \beta|u_r|\hbar R^2\Omega_r. \qquad \text{(B2)}$$

Uniform allocation has $u_r = 256G/R$, so the estimated phase variance is $256\beta|G|\hbar R\Omega_r$. For inverse-frequency allocation, $u_r = 256G/(Z_{-1}\Omega_r)$ with $Z_{-1} = \sum_r \Omega_r^{-1}$, the estimate is branch-independent and equals $256\beta|G|\hbar R^2/Z_{-1}$. Along an octave-spaced extension with fixed highest frequency, both estimates decrease at the softest pole as R grows. These are estimates of the uncoupled soft-branch limit, not bounds on the interacting pressure statistics. Weak corrections along either path cannot settle the finite-strength question for all hierarchical boundaries.

We consequently define a second, explicit family. Choose the reference harmonic covariance v_r^{ref} before refining the line modes, and set

$$u_r = -\frac{K_{Jr}}{v_r^{\mathrm{ref}}}, \qquad \vartheta_r^2 v_r^{\mathrm{ref}} = \beta, \qquad \frac{\delta K_r(\beta)}{K_{Jr}} = \frac{\beta}{2}. \qquad \text{(B3)}$$

Here $\delta K_r(\beta) = -\beta u_r v_r^{\mathrm{ref}}/2$ is the actual quadratic stiffness adjustment. At each chosen β, all branches therefore have the same reference phase variance β; their actual interacting variances need not agree. The static quartic coefficient is now $G_R = -(1/4)^4 \sum_r K_{Jr}/v_r^{\mathrm{ref}}$, with the action coefficient βG_R. It changes with R and is recorded as part of this family; no equality with the original fixed-G family is asserted.

Increasing β indefinitely also has a specific meaning here. The calibration increases the positive quadratic confinement linearly in β, while the bounded cosine amplitude decreases as $1/\beta$. For $\beta > 0$ and fixed reference circuit parameters, the remaining cosine potential has operator norm at most $2\sum_{r:u_r<0} K_{Jr}^2/(\beta|u_r|)$ relative to the stiffened quadratic Hamiltonian. An interior maximum of non-Gaussian effects is therefore possible. A large change relative to the original harmonic model need not mean a large departure from every Gaussian-field description. This norm observation does not itself control unbounded pressure moments.

3. The stationary-well mechanism in this family

Switching between configurations localized near distinct classical minima is one possible source of slow event

clustering. The existence of such minima can be checked exactly for the present boundary. Minimize the static line energy at fixed endpoint and then eliminate that endpoint. The residual branch potential has quadratic matrix

$$L_{\rm pin} = \mathrm{diag}(K_{gr} + K_{cr} + \delta K_r) - \frac{kk^{\mathsf T}}{B_0}, \qquad k_r = K_{cr}, \qquad B_0 = K_b + (L_0 a)^{-1} + \sum_r K_{cr}. \tag{B4}$$

At the reference geometry of the hierarchy, $K_{cr} = K_{gr} = 1/(4R)$, $a = L_0 = 1$, and $K_b = 1/2$. The counterterm satisfies $\delta K_r \geq 0$ for the nonnegative couplings β considered here. Writing $K_J = \mathrm{diag}(K_{Jr})$, the smallest eigenvalue of $K_J^{-1/2} L_{\rm pin} K_J^{-1/2}$ is at least $13/14$. Since $\sin x/x \geq -1/\pi$ for real x, with its continuous value 1 at $x = 0$, the full static potential obeys

$$q \cdot \nabla U(q) \geq \left(\frac{13}{14} - \frac{1}{\pi}\right) \sum_r K_{Jr} q_r^2 > 0 \quad (q \neq 0). \tag{B5}$$

It has a single classical stationary point. This excludes switching between additional classical trapping wells in this parameter family. It does not exclude non-Gaussian quantum stress, cross-scale correlations, or clustering induced by a specified measurement.

4. Separating two-point retuning from connected field effects

At finite β, the first-order calibration does not guarantee preservation of the full linear response. A second Gaussian reference can separate two-point retuning from connected field effects. Let $F_D = \partial_x \Phi(a)$ and $F_Z = \dot\Phi(a)$ be the two fields entering pressure. For a nondegenerate even ground state and a complete set of odd excited states in the same source regulator, define

$$\nu_\alpha = (E_\alpha - E_0)/\hbar, \qquad f_{i\alpha} = \langle 0|F_i|\alpha\rangle, \quad i = D, Z. \tag{B6}$$

All ν_α are positive. The auxiliary vacuum oscillators $F_i^G = \sum_\alpha (f_{i\alpha} b_\alpha + f_{i\alpha}^* b_\alpha^\dagger)$, with frequencies ν_α, reproduce the entire two-source Wightman matrix:

$$\langle F_i^G(t) F_j^G(s)\rangle_G = \sum_\alpha f_{i\alpha} f_{j\alpha}^* e^{-i\nu_\alpha (t-s)} = \langle F_i(t) F_j(s)\rangle. \tag{B7}$$

This also matches their symmetrized two-point functions and retarded linear responses. It is an auxiliary realization of these selected data, not a claim that one finite-parameter harmonic circuit matches every spatial or nonlinear response of the original device.

For real time-reversal invariant dynamics write $f_{D\alpha} = d_\alpha$ and $f_{Z\alpha} = -i z_\alpha$, with real d_α, z_α. The corresponding Gaussian pressure has quadratic matrices

$$A_G = \frac{dd^{\mathsf T}}{L_0} + C_0 z z^{\mathsf T}, \qquad B_G = \frac{dd^{\mathsf T}}{L_0} - C_0 z z^{\mathsf T}. \tag{B8}$$

These are smeared at difference and sum frequencies, respectively, and evaluated with the same ordered quantum cumulant method. The imaginary mixed-field correlations must be retained. Provided the same complete quadratic pressure source is used, a controlled pressure-variance difference measures the integrated connected four-field function left after all Wick pairings of these matched two-point data are removed. Higher pressure differences also sample higher connected field functions and products involving them.

This comparison has a separate source-completeness requirement. Both odd-state spectral truncation and primitive occupation truncation must be controlled. The microscopic pressure is projected before restricting to a low subspace. In general $\Pi F_i^2 \Pi \neq (\Pi F_i \Pi)^2$; the missing term $\Pi F_i (1-\Pi) F_i \Pi$ can affect pressure statistics. A discrepancy created by dropping this term is not evidence for physical field non-Gaussianity. Even-sector pressure convergence alone does not certify the odd spectrum or this composite-source identity.

5. A specified sequential measurement

To distinguish event correlations from marginal pressure laws, we specify an ideal measurement mathematically. Set $A = X_\tau/s_0$, where $s_0 = \sqrt{\langle X_\tau^2\rangle_0}$ is the original harmonic standard deviation at the same window. This one pressure unit is kept fixed across couplings and comparators. A pointer outcome y with imprecision $\sigma > 0$ is represented by the positive Gaussian Kraus operator

$$K_y = (2\pi\sigma^2)^{-1/4} \exp[-(y-A)^2/(4\sigma^2)], \qquad \int dy\, K_y^\dagger K_y = I. \tag{B9}$$

A Kraus operator specifies both the outcome probability and the conditional change of the state. Let $\mathcal{E}(\rho) = \int dy\, K_y \rho K_y^\dagger$ be the state when the outcome is ignored, and $\mathcal{J}_h(\rho) = \int_{|y|\geq h} dy\, K_y \rho K_y^\dagger$ the unnormalized state associated with a threshold event. Both maps preserve positivity; only $\mathcal{E}$ preserves trace. In the eigenbasis of A, $\mathcal{E}$ multiplies ρ_{ij} by $\exp[-(a_i - a_j)^2/(8\sigma^2)]$, so ignoring the readout still disturbs coherences.

Write $\mathcal{U}_t(\rho) = e^{-iHt/\hbar} \rho e^{iHt/\hbar}$ and start in the ground state ρ_0. The two-event probabilities are

$$p_1 = \mathrm{Tr}\, \mathcal{J}_h(\rho_0), \qquad p_2 = \mathrm{Tr}\, \mathcal{J}_h \mathcal{U}_t \mathcal{E}(\rho_0), \qquad p_{11} = \mathrm{Tr}\, \mathcal{J}_h \mathcal{U}_t \mathcal{J}_h(\rho_0), \qquad g = \frac{p_{11}}{p_1 p_2}. \tag{B10}$$

Here $g > 1$ indicates positively associated threshold events for this instrument. The first probe deposits $\Delta E_1 = \mathrm{Tr}[H\mathcal{E}(\rho_0)] - E_0$. We compare p_2 with the value obtained without the first probe, rather than assuming its marginal remains p_1. Varying σ varies measurement strength continuously; $\sigma \to \infty$ suppresses the disturbance. These formulas specify an ideal instrument of

the already averaged operator. They do not derive a local causal detector coupling to pressure during the averaging window, and do not identify its outcomes with a pre-existing classical record in the vacuum.

6. Temperature and the physical frequency window

Nonperturbative zero-point effects in a nonlinear junction coupled to many transmission-line modes have been observed[6]. This establishes a relevant physical platform; it does not establish intermittent Casimir-pressure records. Preparation of a wide hierarchy near its vacuum imposes a distinct constraint. With ordinary frequency f and temperature T, a harmonic mode has

$$\begin{aligned} n_B(f,T) &= \frac{1}{e^{hf/(k_BT)}-1}, \\ n_B < 0.01 &\iff \frac{hf}{k_BT} > \log 101. \end{aligned} \tag{B11}$$

For an illustrative highest frequency of 10 GHz, fourteen octave-spaced resonances require $T < 12.69\ \mu$K for this criterion at the lowest pole; eighteen require $T < 0.794\ \mu$K. At 10 mK, at most four octave-spaced poles fit between the corresponding near-vacuum threshold and 10 GHz. These are per-mode occupation estimates for this illustrative embedding, not bounds on errors in an eighth pressure moment. The actual coupled excitation spectrum and source matrix elements govern that error.

We therefore keep temperature as an independent parameter, written as the energy $T_E = k_BT$. For finite interacting matrices, both parity sectors enter the Gibbs trace, and

$$M_{2k}(T_E) = \sum_i p_i \|X_\tau^k|i\rangle\|^2, \quad p_i = \frac{e^{-(E_i-E_0)/T_E}}{\sum_j e^{-(E_j-E_0)/T_E}}. \tag{B12}$$

The vacuum counterterm stays fixed. The harmonic thermal reference is evaluated independently by Gaussian-state purification after applying the physical time filter. Small thermal populations may multiply large pressure matrix elements; a population check alone is insufficient. Thermal populations, occupation cutoffs, and pressure paths must therefore be controlled separately from their ground-state counterparts. The weighted path bound in Eq. (A11) addresses the last of these within a fixed finite model. A thermal Gaussian reference matched to the interacting two-point functions would additionally require transitions from every thermally occupied state; Bose reweighting of ground-state transitions does not provide that match.

Appendix C: Increasing the resonance range: distinct limits and proved bounds

Adding slower boundary resonances need not have the same effect at every observation time. At a fixed averaging width, the newly added resonances have progressively smaller instantaneous source amplitudes. A width that grows with the hierarchy instead follows slower dynamics while an increasing number of faster branches contribute their static response. Following the lowest resonance is also different from following an interior scale with resonances on both sides. We distinguish these limits before drawing a conclusion about scale-dependent pressure statistics.

1. The canonical-branch model and its reference calibration

For this analysis, statically eliminate the endpoint coordinate and retain the leading gradient pressure. At the stated static stiffness $B = K_b + (L_0a)^{-1} + 1/4 = 7/4$, and with branch frequencies $\Omega_j = \Omega_{\max}2^{-j}$, $j = 0, \dots, R-1$, the resulting quasistatic model is

$$\begin{aligned} K_R &= H_{\rm qs}/\hbar = \sum_j \Omega_j h_{\beta,\nu_j} - g_R S_R^2, \\ h_{\beta,\nu} &= \frac{y^2}{4} + a_\beta x^2 + \frac{\nu}{4\beta}\left[1 - \cos\left(\sqrt{\beta/\nu}\,x\right)\right], \\ S_R &= \sum_j \sqrt{\Omega_j}\,x_j, \qquad [x_j, y_k] = 2i\delta_{jk}. \end{aligned} \tag{C1}$$

The coefficients are $g_R = 1/(112R)$ and $a_\beta = (2+\beta)/16$.

Here $q_j = \ell_j x_j$, $\ell_j^2 = \hbar R\Omega_j/2$, and $\nu_j = v_j^{\rm ref}/\ell_j^2$ uses the prescribed harmonic calibration. The gradient pressure is $P_{D,R} = (C/R)Q_R$, where $Q_R = S_R^2$ and $C = \hbar/(64B^2L_0a^2)$. The continuous $\beta = 0$ limit is harmonic; finite β retains the full cosine. Equation (C1) is an analytical model in canonical branch coordinates. Its reduction from the distributed line requires control of frequency-dependent endpoint elimination and of the velocity pressure; it does not replace the full-line Hamiltonian used in the numerical sections.

For the complete reference covariance at $B_{\rm ref} = 7/4$, its Euclidean branch resolvent gives

$$1 \le \nu_j^{\rm complete} \le 1 + \frac{1}{54R}. \tag{C2}$$

A partial modal covariance need not satisfy the lower bound of one. If every omitted frequency is at least $\Lambda > \Omega_{\max}$ and $C_b > 0$, the branch transfer and endpoint resolvent instead give

$$\begin{aligned} 1 - \frac{C_{\rm tail}}{R} \le \nu_j^{\rm modal} &\le 1 + \frac{1}{54R}, \\ C_{\rm tail} &= \frac{\Omega_{\max}^3}{16(\Lambda^2 - \Omega_{\max}^2)^2\sqrt{C_b(27/16)}}. \end{aligned} \tag{C3}$$

Thus both calibrations approach one uniformly under their stated conditions. For a sequence retaining the first $R+10$ complete ordered line modes, min–max comparison with the Dirichlet line supplies the conservative $\Lambda = 9\pi v/a_{\mathrm{ref}}$, where $v = (L_0C_0)^{-1/2}$ and a_{ref} is the calibration separation; in the stated units it exceeds $\Omega_{\max} = 8.8$. This argument requires complete mode enumeration, not merely a large last returned frequency. It does not change a finite device's calibrated variance during refinement.

2. Equal-time control and a fixed-window theorem

Fix $\beta > 1/7$. The local potential has $V''(x) \geq \beta/8$. In the positive Euclidean time-sliced measure, the action Hessian is bounded below by

$$\begin{aligned} \mathsf{A}_R(\xi) &= \mathrm{diag}_j \frac{\xi^2 + \rho_\beta^2 \Omega_j^2}{2\Omega_j} - 2g_R u u^{\mathsf{T}} \\ &\geq d_\beta \ \mathrm{diag}_j \frac{\xi^2 + \rho_\beta^2 \Omega_j^2}{2\Omega_j}, \\ \rho_\beta &= \frac{\sqrt{\beta}}{2}, \qquad d_\beta = 1 - \frac{1}{7\beta}. \end{aligned} \tag{C4}$$

where $u_j = \sqrt{\Omega_j}$. Apply the Brascamp–Lieb variance inequality [25] to a linear source and its linear tilts. The Hessian is unchanged by the tilt, so integrating the variance bound gives a Gaussian upper bound on the source's moment-generating function. Passing first to the time continuum and then to the finite-R ground state yields, for every branch subset J,

$$\begin{aligned} &\langle S_J^2 \rangle \leq \mathcal{B}_J, \qquad \mathcal{B}_J = \frac{\sum_{j\in J} \Omega_j}{d_\beta \rho_\beta}, \\ &\langle S_J^{2m} \rangle \leq 2^{m+1} m! \mathcal{B}_J^m. \end{aligned} \tag{C5}$$

The mean vanishes by parity. Ground-state preparation precedes the hierarchy limit; the finite-temperature zero-frequency mode does not obey the same uniform estimate. The threshold $\beta > 1/7$ is sufficient for this proof, not a predicted instability below it.

Let $K_{0,R} = \sum_j \Omega_j h_{\beta,\nu_j}$ and let $E_{0,R}$ be its product ground energy. The variational inequality $\langle K_{0,R} - E_{0,R} \rangle \leq g_R \mathcal{B}_{\mathrm{all}}$ makes each fixed set of ultraviolet-index branches approach its independent full-cosine product ground state: that fixed set has a positive excitation gap even though the entire hierarchy does not. The geometric tail in Eq. (C5) then proves convergence of every fixed equal-time moment of Q_R to the corresponding infinite independent-product moment at $\nu = 1$. An independent full-cosine product need not be Gaussian.

Stationarity gives an additional ordered result without replacing unequal-time operators by random variables. Center Q_R in the actual ground state $|\psi_R\rangle$, and write $Y_w = \int dt\, w(t) Q_{R,c}(t)$. For either $w = f_\tau$ or $w = f_\tau - f_{b\tau}$, let $\eta = \|w\|_1$, which is respectively one or at most two. The triangle inequality for this vector integral gives

$$\langle Y_w^2 \rangle = \|Y_w \psi_R\|^2 \leq \eta^2 \,\mathrm{Var}(Q_R) \leq 16 \eta^2 \mathcal{B}_{\mathrm{all}}^2. \tag{C6}$$

Replacing S_R by a source restricted to frequencies above Ω_c, while retaining the same Hamiltonian and ground state, changes this variance by $O(\sqrt{\Omega_c})$ uniformly in R. This follows by writing $S_R^2 - S_A^2 = 2S_A S_T + S_T^2$ and applying Eq. (C5) at the same time. It bounds a source tail, not the change of state and dynamics when branches are removed.

A separate fixed-window theorem holds at any fixed common primitive projection Π_d in each canonical branch whose projected local ground is simple and has a positive gap near $\nu = 1$. Compress all operators consistently, including $q_d = \Pi_d x^2 \Pi_d$, which generally differs from $(\Pi_d x \Pi_d)^2$, and the full cosine. Geometric summability bounds the projected $Q_{R,d}$ uniformly. After adding the independent projected tail $K_{\mathrm{tail},d}$, calibration continuity gives

$$\|K_{R,d} + K_{\mathrm{tail},d} - K_{\infty,d}\| \leq c_d/R, \tag{C7}$$

where $K_{\infty,d}$ is the independent infinite sum at $\nu = 1$. Duhamel's formula controls the finite-time propagator error; the Gaussian time integral is finite. Combining this with the fixed-branch state argument proves convergence of the even ordered increment moments at each fixed τ and d. The constants depend on d. This theorem uses the specified product projection in branch coordinates; the production adaptive normal-mode tensors are a different regulator family.

3. The unbounded pressure-path question

The independent model also has unbounded-coordinate ordered-word bounds. Write $S_{0,R}(t) = e^{itK_{0,R}} S_R e^{-itK_{0,R}}$ for its source evolution. For each fixed word length n, local oscillator graph estimates give $\|(h+1)^s x (h+1)^{-s-1}\| < \infty$, uniformly for ν in a compact positive interval and $s < n$. They follow from the quadratic confinement and the bounded derivatives of the cosine. Local evolution preserves these graph norms. Expanding collective sources by branch labels, while retaining the order on each branch, consequently gives

$$\|S_{0,R}(t_n) \cdots S_{0,R}(t_1) |G_R\rangle\| \leq \left[J_n \sum_j \sqrt{\Omega_j} \right]^n, \tag{C8}$$

for an R-independent J_n and the independent product ground state $|G_R\rangle$. This controls the independent ordered moments and a single Duhamel propagator difference on such reference vectors.

The first change of the interacting ground vector is also controlled at finite nonlinear strength. Vary $K_R(\lambda) =$

$K_{0,R} - \lambda Q_R$ and differentiate its real normalized ground state at $\lambda = 0$. One-site even excitations arise from x_j^2; two-site odd excitations arise from $x_j x_k$. Their denominators give

$$\begin{aligned} \|\psi_R'(0)\|^2 &\le \frac{RW_\beta}{\delta_\beta^2} + \frac{4V_\beta^2}{\delta_\beta^2} \sum_{j<k} \frac{\Omega_j \Omega_k}{(\Omega_j + \Omega_k)^2} \\ &\le R \frac{W_\beta + 4V_\beta^2}{\delta_\beta^2}, \\ E_\beta &= \tfrac{1}{2} + \tfrac{\beta}{16}, \qquad V_\beta = E_\beta / a_\beta, \\ W_\beta &= E_\beta^2 / a_\beta^2 + 1/a_\beta, \\ \delta_\beta &= 3\sqrt{2+\beta}/4 - E_\beta. \end{aligned} \tag{C9}$$

The bound is asserted when $\delta_\beta > 0$, in particular at $\beta = 1, 3, 10$, where it supplies a sufficient local gap bound; the energy and moment bounds follow from a Gaussian trial state and integration by parts, without replacing the local cosine. The pair sum is at most R because its summand is bounded by $2^{-|j-k|}$. Thus $\|g_R \psi_R'(0)\| = O(R^{-1/2})$. This is the first collective-interaction correction, not an all-orders ground-angle theorem.

An energy estimate alone cannot supply that missing theorem or its pressure extension. Translate the local ground states of $-\partial_x^2 + a_\beta x^2$ by $m_j = A/(\sqrt{R}\sqrt{\Omega_j})$ with fixed $A \neq 0$. For these normalized states $\Phi_{R,m}$,

$$\begin{aligned} \langle Q_R \rangle_{\Phi_{R,m}} &= A^2 R + O(1), \\ \sup_R \|(K_R + 1)^k \Phi_{R,m}\| &< \infty. \end{aligned} \tag{C10}$$

The second bound holds for every fixed integer k. To verify it, translate the Hamiltonian back: its polynomial coefficients have bounded geometric sums, and every fixed derivative of each shifted cosine is bounded independently of the displacement. A fixed operator word acting on the product Gaussian is therefore bounded uniformly. Hence no R-uniform polynomial-energy graph bound controls the bare Q_R on all states. These translated states are not asserted to be reached with appreciable weight by pressure insertions on the ground state. They disprove that proposed bound, not the ground-state limit or a bound using the complete increment filter.

For the interacting unbounded oscillator sequence, the uniform primitive projection error of higher ordered pressure moments remains open. In a closed p-insertion ground-state path that reaches excitation frequency L, $\sum_i \Delta_i^2 \ge 4L^2/p$, so Gaussian filters contribute at most $e^{-2\tau^2 L^2/p}$. A uniform bound on the accompanying summed pressure matrix elements is still required. Neither the first derivative nor the equal-time inequality permits interchanging the primitive and hierarchy limits. A Gaussian Wick source bound addresses a different, explicitly Gaussian comparator.

4. Moving windows retain collective response

In the harmonic control, the collective response becomes independent at each fixed nonzero frequency as R grows. This convergence is not uniform toward zero frequency. The exact susceptibility is

$$\chi_{S,R}(i\xi) = \frac{2T_R(\xi)}{1 - 4g_R T_R(\xi)}, \qquad T_R(\xi) = \sum_j \frac{\Omega_j^2}{\xi^2 + \Omega_j^2}. \tag{C11}$$

If $\xi_R = \Omega_{\max} 2^{-\theta R} z$, with $z > 0$ and $0 < \theta < 1$ fixed, then $T_R(\xi_R)/R \to \theta$ and the denominator tends to $D_0(\theta) = 1 - \theta/28$. This is the moving-interior limit, with observation widths of order $1/\xi_R$. A fixed width instead has $\theta \to 0$; following the lowest resonance approaches the infrared edge, $\theta \to 1$, and need not sample an expanding interior interval.

The static part of harmonic elimination of N_H faster branches gives $g_{\text{eff}} = g_R/(1 - 4g_R N_H)$. Differentiation with respect to the coupling shows that the corresponding adiabatic low-energy source has factor $(1 - 4g_R N_H)^{-2}$. The exact response limit thus suggests D_0^{-2} for that source, but transferring it to a real-time pressure moment requires the hard-state path reduction just identified.

For an isolated full-cosine branch at $\nu = 1$, define

$$\chi_\beta = 2 \sum_{n>0} \frac{|\langle 0|x|n\rangle|^2}{e_n - e_0}, \qquad D_\beta(\theta) = 1 - \chi_\beta \theta/56. \tag{C12}$$

The second expression is a conditional nonlinear static-loading closure: it assumes uniform adiabatic response of the coupled hard sector and control of its higher response and pressure-source remainders. It is not an exact nonlinear response identity. Nevertheless its isolated susceptibility cannot produce a pole: with $\delta_0 = 3/(2\sqrt{2}) - 1/2$, $\chi_\beta \le 8/\delta_0$ for $0 \le \beta \le 1$, and $\chi_\beta \le 8/\beta$ for $\beta \ge 1$. Thus this conditional D_β remains above 0.745 for $0 \le \theta \le 1$.

If a common multiplicative pressure-source factor survives a controlled reduction, it cancels from $M_p/M_2^{p/2}$. Order-dependent modulation within an octave can still remain, as can non-Gaussian pressure in the independent nonlinear model. The higher unbounded ordered moment law along moving windows has not been proved. These results therefore distinguish mechanisms and limits; they establish neither an intermittency law nor its absence.

5. Long Gaussian windows with all excitations retained

A fixed number of coordinates does not imply a finite number of quantum states: each oscillator still has arbitrarily high excitations. We now remove that occupation cutoff while keeping the coordinate representation and the physical parameters fixed. The resulting theorem

applies to the full confining cosine Hamiltonian, rather than only to a numerical matrix. Its Gaussian dependence is on the averaging width; it does not assert that the pressure probability distribution becomes Gaussian.

Proposition 2 (Fixed-coordinate long-window law)**.** *On $L^2(\mathbb{R}^d)$, with $d < \infty$, let*

$$\begin{aligned} H &= H_{\mathrm{osc}} + W, \\ H_{\mathrm{osc}} &= \tfrac{1}{2}\boldsymbol{\pi}^{\mathsf{T}}\mathsf{M}^{-1}\boldsymbol{\pi} + \tfrac{1}{2}\boldsymbol{x}^{\mathsf{T}}\mathsf{K}\boldsymbol{x}, \end{aligned} \tag{C13}$$

where $\boldsymbol{x}, \boldsymbol{\pi}$ are canonical coordinates and momenta, M, K are real positive-definite matrices, $\hbar > 0$, and W is a bounded real multiplication operator. Fix a Hermitian quadratic polynomial P in $\boldsymbol{x}, \boldsymbol{\pi}$, with symmetric ordering of mixed terms. Choose a normalized ground state g, with energy E_0, and set $P_c = P - \langle g, Pg\rangle I$. For $\tau > 0$, define

$$\begin{aligned} f_\tau(t) &= \frac{e^{-t^2/(2\tau^2)}}{\sqrt{2\pi}\tau}, \\ Y_\tau &= \int_{\mathbb{R}} a_\tau(t) e^{iHt/\hbar} P_c e^{-iHt/\hbar}\, \mathrm{d}t, \\ m_p(\tau) &= \langle g, Y_\tau^p g\rangle, \end{aligned} \tag{C14}$$

where $a_\tau = f_\tau$ for ordinary averaging, or $a_\tau = f_\tau - f_{b\tau}$ for increments at a fixed ratio $b > 1$. The integral on $D(H)$ has a unique self-adjoint closure from the finite-energy core, and g lies in the domain of every power of that closure. If $m_2(\tau) > 0$, then for every fixed even $p \geq 2$ there are $C_p > 0$, $\gamma_p \geq 0$ and $\delta\gamma_p > 0$ such that

$$m_p(\tau) = C_p e^{-\gamma_p \tau^2}\left[1 + O(e^{-\delta\gamma_p \tau^2})\right], \qquad \tau \longrightarrow \infty. \tag{C15}$$

All Hamiltonian and pressure parameters are held fixed. If only one grouped rate survives, the remainder is exactly zero.

Proof. Choose an energy unit $E_* > 0$ and set $T = 1 + (H - E_0)/E_* \geq 1$. Oscillator ladder estimates and quadratic-form Cauchy–Schwarz give two separate bounds on $D(H)$,

$$\begin{aligned} \|P_c\psi\| &\leq C_{\mathrm{op}}\|T\psi\|, \\ |\langle\psi, P_c\psi\rangle| &\leq C_{\mathrm{form}}\langle\psi, T\psi\rangle. \end{aligned} \tag{C16}$$

Bounded W preserves the oscillator operator domain and its graph norm. These bounds define the integral in Eq. (C14) on $D(H)$. On the span of the energy eigenvectors, integration by parts gives

$$\begin{aligned} &\left|\langle T\psi, Y_\tau\psi\rangle - \langle Y_\tau\psi, T\psi\rangle\right| \\ &\qquad \leq \frac{\hbar}{E_*} C_{\mathrm{form}}\|a'_\tau\|_1 \langle\psi, T\psi\rangle. \end{aligned} \tag{C17}$$

Together with $\|Y_\tau\psi\| \leq C_{\mathrm{op}}\|a_\tau\|_1\|T\psi\|$, this is the Glimm–Jaffe–Nelson commutator criterion, specifically Theorem A(i) of Ref. [26]. It establishes essential self-adjointness on that core. The operator bound shows that its closure agrees with the integral on $D(H)$.

Bounded-perturbation min–max and oscillator counting give $N(E) \leq C(1+E/E_*)^d$ states up to excitation energy E, including multiplicities. Equation (C16) bounds pressure matrix elements by a polynomial in their endpoint energies. A returning path with p pressure insertions that reaches excitation energy L has total absolute energy variation at least $2L$. Its Gaussian rate therefore satisfies

$$\begin{aligned} \gamma &= \frac{1}{2\hbar^2}\sum_{j=1}^{p} c_j(\epsilon_{i_j} - \epsilon_{i_{j-1}})^2 \\ &\geq \frac{2L^2}{p\hbar^2}, \end{aligned} \tag{C18}$$

where $\epsilon_i = E_i - E_0$, $i_0 = i_p = 0$, and $c_j = 1$ for ordinary averaging. Expanding an increment gives $c_j \in \{1, b^2\}$ with signed coefficients. A ground-starting k-edge half path similarly has rate at least $L^2/(2k\hbar^2)$.

These Gaussian bounds dominate polynomial state counts and source amplitudes at every positive width, even after multiplication by any fixed power of the endpoint energy. Half-path sums thus converge in every T^s graph norm, for fixed nonnegative integer s. Applying the operator bound inductively identifies their limits with the actual vectors $Y_\tau^k g$ and proves the required power-domain statement. Returning moment sums are absolutely convergent.

For a bounded rate, Eq. (C18) bounds every visited energy. Only finitely many states and paths then contribute. Grouping exactly equal rates, including degeneracies and cancellations, gives a locally finite exponential series. Its first nonzero even coefficient is positive because $m_{2k} = \|Y_\tau^k g\|^2$. Absolute convergence at one positive width and the separation to the next surviving rate give Eq. (C15). □

For two such fixed models, labeled β and 0, use the same prescribed pressure, centering rule, averaging type and increment ratio, and require nonzero variance in both. Their standardized-moment contrast obeys

$$\begin{aligned} A_p(\tau) &= \log\frac{m_p^\beta/(m_2^\beta)^{p/2}}{m_p^0/(m_2^0)^{p/2}} \\ &= K_p + c_p\tau^2 + O(e^{-(\delta\gamma)\tau^2}), \\ c_p &= \tfrac{p}{2}(\gamma_2^\beta - \gamma_2^0) \\ &\quad - (\gamma_p^\beta - \gamma_p^0). \end{aligned} \tag{C19}$$

Here K_p is a constant and $\delta\gamma > 0$ is a rate separation for the two models. The rate is selected only after equal-rate cancellations are combined; it need not be the square of a single lowest excitation gap. There is no nonzero asymptotic term proportional to $\log\tau$ at these fixed parameters.

The calibrated circuit satisfies the confinement assumptions at each fixed injective spatial or modal coordinate projection. Here Φ is the line flux, $\phi = \Phi(a)$ its endpoint value and q_r a branch flux. Retain the grounded

end $\Phi(0) = 0$, positive $C_0, L_0, C_b, C_r, K_{cr}, K_{Jr}$, nonnegative K_b, K_{gr}, $\beta \geq 0$, $u_r \leq 0$ and $v_{\mathrm{ref},r} \geq 0$. For an active branch, $\theta_r^2 = -\beta u_r / K_{Jr} > 0$. Removing its Josephson harmonic term leaves the positive line-gradient energy and the couplings $K_{cr}(q_r - \phi)^2/2$; the fixed calibration adds the stiffness $\delta K_r = -\beta u_r v_{\mathrm{ref},r}/2 \geq 0$. Zero gradient energy forces $\Phi = \phi = 0$ through the grounded end, and the positive coupling then forces every $q_r = 0$. Thus the remaining quadratic potential is positive definite in the chosen coordinates. Each full cosine is bounded at its fixed nonzero phase scale. Branches with $u_r = 0$ retain their Josephson quadratic term, and $\beta = 0$ is the harmonic case.

The theorem removes the occupation cutoff, while keeping the number of coordinates finite. Its constants and rate separations may depend on that number, the resonance hierarchy, the calibration, β, p and the smallest confinement frequency. It does not locate the ultimate regime at the sampled widths, certify a finite tensor approximation, or establish a joint limit in which the averaging width, hierarchy depth and distributed-line regulator all increase. Such a moving-window limit can remain in crossover even though every fixed member of the family has the asymptotic form proved here.

Appendix D: Extended finite-strength convergence and tail diagnostics

1. Source-space enlargement and the missing field frequencies

The Gaussian comparison needs the states reached by the two linear fields $F_D = \partial_x \Phi(a)$ and $F_Z = \dot{\Phi}(a)$, as well as their excitation frequencies. We test the first requirement at $R = 10$, $M = 20$, $Q = 12$ and $\beta = 1$, keeping the twenty-four-mode calibration fixed. Five orthonormal bath directions constructed from the missing linear-field sources enlarge its dimension from 128 to 133. Every bath direction accompanies every one of the 421 active configurations. The full cosine Hamiltonian is solved again, and the pressure is projected with its complete quadratic contacts.

For a supplied ground state g and tensor projector Π_T, define the unfiltered missing source weight

$$W_{\mathrm{out}} = \frac{\|(1 - \Pi_T) F_D g\|^2}{L_0} + C_0 \|(1 - \Pi_T) F_Z g\|^2. \quad \text{(D1)}$$

The calculation evaluates this quantity by subtracting the projected source Gram matrix from the complete quadratic-contact matrix at the stated regulator. Its stored binary64 values are cancellation estimates, not interval enclosures. Table XV separates the change of source space at fixed old ground state from the subsequent change of ground state. The numerical original-Hamiltonian residual bound for the enlarged solution is 2.83×10^{-13}, and its retained-cluster and odd-ground separation checks pass.

TABLE XV. Measured missing field-source weight at fixed $\beta = 1$. Here $C_0 = L_0 = 1$. This unfiltered weight is neither a pressure probability nor an error bar on an eighth moment.

Ground state	Bath dimension	W_{out}
original	128	1.361455×10^{-3}
original	133	5.185466×10^{-7}
recomputed	133	5.284543×10^{-7}

TABLE XVI. Fixed-model Gaussian-reference refinement. A pass count of zero means that every frozen numerical criterion failed; none of these failures is an unresolved zero or arithmetic exclusion.

Source dimensions	moment/shape	effect	coherent
64–128	0/42	4/42	0/12
128–200	0/42	7/42	0/12

Across the three ordinary and three increment windows, $\tau = 36.2039, 51.2, 72.4077$, the largest relative change of the eighth pressure moment is 2.011512×10^{-5}, while the largest absolute change of its original-reference logarithmic shape contrast is 7.655437×10^{-8}. These are measured sensitivities to this particular source-space enlargement.

The frequency-completeness requirement remains separate. In the enlarged ground state, the retained odd transitions carry a pressure-weighted linear source norm of only 0.00058804788, whereas the full specified tensor source norm is 142.2151521. Thus the much smaller W_{out} does not complete the field-transition spectrum used by the Gaussian reference. This missing norm is not a pressure-error estimate: time averaging weights excitation frequencies, and pressure moments also involve transitions between excited states. Resolving the missing frequency distribution is necessary before the retained-transition contrast can be interpreted as the complete finite-tensor Gaussian comparison.

a. *Executed frequency-refinement test*

We next hold the full nonlinear tensor model, ground state, two source operators, and all six averaging windows fixed while enlarging only the joint source Krylov space. Its dimension is a numerical transition-space regulator, not a count of boundary resonators. Table XVI reports the two successive refinements. A moment/shape test compares four even moments and three standardized log shapes in each of six windows. The effect screen additionally requires the reference change to be at most ten percent of the corresponding physical–reference contrast. The coherent test compares the signed DD, $D(Z/i)$, and $(Z/i)(Z/i)$ source weights at three long windows, including both relative and local-scale tests for the mixed entry.

The later changes are generally smaller, but no mo-

ment or shape criterion passes on either edge, and no one of the 42 effect screens passes both successive refinements. The lowest retained source frequency moves from 0.03852 to 0.02125 and then 0.01760; agreement of the unfiltered source Gram matrix therefore does not establish completeness for the low frequencies selected by long averaging.

The central increment window illustrates the size of the unresolved reference dependence. At $\tau = 51.2$, its Gaussian-reference eighth moment changes from 6.018655×10^{-40} to 3.887233×10^{-39}. The max-normalized discrepancy is 0.845169; this is not a relative increase with the smaller value as denominator. The corresponding standardized log-shape $A_8 = \log M_8 - 4 \log M_2$ changes by 0.509581, whereas the physical-minus-latest-reference contrast has magnitude 0.342070. The reference change is 1.490 times that contrast and fails the predeclared ten-percent separation test.

Consequently, the retained interacting-response Gaussian comparison has not converged at the tested source dimensions. These data neither erase the finite-model nonlinear pressure-shape changes nor isolate them from the changing Gaussian reference. They therefore provide no positive evidence for physical intermittency. The separately staged $200 \to 256 \to 320$ source-only continuation tests the necessary signed low-frequency weights before any further high-order Gaussian pressure calculation is admitted.

b. *Source-only continuation through dimension 320*

The preceding pressure-reference test leaves open whether its failure is a transient small-source effect. We therefore extend the same interacting Hamiltonian and the same two source vectors from dimension 200 to 256 and 320. This continuation recomputes only the source-projected transition measure. It neither invokes the Gaussian pressure hierarchy nor changes the boundary-resonator count, calibration, ground state, or nonlinear strength. The independently reloaded 200-dimensional archive exposed a list-versus-array reporting error in the first version of the extraction. That failed receipt is preserved. A versioned correction passed all fourteen control groups, including bitwise-equal source panels from list-backed and array-backed copies, before the continuation was replayed in a separate cache.

For each refinement we compare the three independent entries of the signed smoothed source matrix at $\tau = 36.2039$, 51.2, and 72.4077. The mixed $D(Z/i)$ entry must pass both a relative and a local-scale criterion, giving twelve criteria per edge and twenty-four for the two-edge gate. The tolerance remains 10^{-6}.

The second edge is markedly closer: its shortest-width panel passes in full, and four of nine entries pass, whereas no entry passes on the first edge. The two longer-width panels nevertheless remain outside tolerance. The lowest positive Ritz frequency changes only from 0.01759693 to 0.01759687, a relative change of 3.488×10^{-6}, but stability of that single frequency does not control the signed weights selected by long-time averaging. All Decimal-80/Decimal-120 replays pass, and neither edge contains an unresolved arithmetic case.

TABLE XVII. Executed source-only continuation. "Panels" counts averaging widths at which the complete signed matrix passes; "entries" counts the three independent matrix entries separately. The last column is the largest frozen discrepancy on that edge.

Source dimensions	panels	entries	maximum discrepancy
200–256	0/3	0/9	9.434×10^{-5}
256–320	1/3	4/9	3.309×10^{-6}

Thus the second edge is markedly closer, but the measured sequence does not attain the predeclared two-edge criterion at dimension 320. The necessary source gate is false, so no further high-order Gaussian-reference calculation is admitted from this ladder. This result does not remove the finite-model nonlinear shape changes. It shows that their separation from a retained interacting Gaussian reference, and hence any positive claim of persistent Casimir-pressure intermittency, is not established by the present source hierarchy.

2. Fixed-width probes beyond the finite infrared cutoff

We re-evaluate the saved $R = 10$, $\beta = 1$ full-cosine matrices at three fixed absolute widths, keeping $M = 20$, $M_{\mathrm{ref}} = 24$, 32 primitive levels and the same five-mode active prefix. Here O_τ is the actual-ground-centered Gaussian average and $X_\tau = O_\tau - O_{2\tau}$. For either observable Y, the reported contrast is

$$A_8^{(0)}(Y) = \log\left[\frac{M_8(Y)/M_2(Y)^4}{M_8^{(0)}(Y)/(M_2^{(0)}(Y))^4}\right]. \tag{D2}$$

The superscript (0) denotes the original distributed harmonic H_0, using ordered quadratic algebra for all 20 declared modes. It does not denote the retained interacting Wick comparator. Thus $A_8^{(0)}$ combines altered two-point response with higher information; it does not isolate an irreducible non-Gaussian field contribution. These widths have $\tau\Omega_{\min} = 2.489, 3.520, 4.978$, placing $R = 10$ beyond its lowest-resonance scale. Together with $A_8^{(0)}(X_{51.2}) = 0.626830$, the increment values show a sign change followed by a turn back toward zero; all three values in this panel remain negative. The longer-width extension in Sec. D 4 resolves the subsequent crossing. The ordinary contrast continues to decrease. This observable-dependent finite-R cutoff crossover supplies a baseline for a later same-width depth comparison; it supplies neither an interior scaling plateau nor a rare-tail result. At these widths the recorded nonlinear $Q = 16$

TABLE XVIII. Fixed-width $R = 10$ results at $\beta = 1$. The contrasts use $Q = 16$; $\delta_8 = M_8^{Q=16}(X_\tau)/M_8^{Q=14}(X_\tau) - 1$ measures this occupation/selection refinement alone. A dagger marks widths where binary64 spectral reconstruction fails its stated consistency check for both $Q = 16$ observables; the displayed moments use the normalized vacuum-vector recurrence.

τ	$A_8^{(0)}(O_\tau)$	$A_8^{(0)}(X_\tau)$	$10^6\delta_8$
144.815	-1.536029	-1.638610	+4.079
204.800†	-3.777832	-2.153411	+5.474
289.631†	-8.357968	-0.241262	-12.419

fractional variance differences $|M_2(O_\tau)/M_2(X_\tau) - 1|$ are 3.426×10^{-16}, 2.356×10^{-17}, and 6.023×10^{-17}, respectively, while the higher moments differ strongly. Ordered returning paths permit this difference: the ordinary filter retains zero-energy-gap entries between intermediate excited states, whereas the difference filter annuls those entries even when the ground-to-excited variance filters have become nearly equal. The recurrence preserves the order in powers of one smeared Hermitian operator, sets its actual ground diagonal to zero, normalizes each vacuum vector, and accumulates scalar amplitudes and cumulants with 80-digit Decimal arithmetic. Four small arithmetic controls pass, and the saved $\tau = 51.2$ ordinary/increment checks agree within 7.3×10^{-14} on their declared scales. Necessary positive-moment checks pass through order eight. At the two longer widths, the direct spectral weights, and hence a direct cumulative distribution from them, remain unresolved. Decimal accumulation does not improve the binary64 Hamiltonian eigenpairs or projected pressure entries. The conditional relative eighth-moment omitted-path envelopes are smaller than e^{-1171} for these rows. This statement assumes an exact finite spectral restriction with a common exact ground state; it excludes archived subspace, state, source, primitive, bath and line-mode errors. The small $Q = 14 \to 16$ changes in Table XVIII quantify only that refinement and do not complete those separate error controls.

3. A common-width comparison across resonance depths

We compare the $\beta = 1$ equal-reference-phase full-cosine models at the same three absolute averaging widths. The $R = 10$ result uses $M = 20$, $Q = 16$ and construction width 51.2; the $R = 14$ result uses $M = 24$, $Q = 10$ and construction width 204.8. Both use $M_{\mathrm{ref}} = 24$, 32 primitive levels, and a five-mode active prefix. Their bath spaces apply the same 24-seed downward-closure prescription, producing 128 configurations at $R = 10$ and 146 at $R = 14$; both target spectra use a fraction 0.88 of the free bath threshold. These are different physical depths and numerical regulators, so their difference measures sensitivity to this combined change. Here

TABLE XIX. Common absolute widths at two resonance depths. Entries are $A_8^{(0)}$. A dagger means that direct spectral reconstruction is unresolved; the displayed moments come from the controlled normalized vacuum-vector recurrence.

	Ordinary O_τ		Increment X_τ	
τ	$R10, Q16$	$R14, Q10$	$R10, Q16$	$R14, Q10$
144.815	-1.536029	0.417051	-1.638610	-0.013678
204.800	-3.777832†	0.482050	-2.153411†	0.413784
289.631	-8.357968†	0.533107	-0.241262†	0.450437

O_τ is the actual-ground-centered Gaussian average and $X_\tau = O_\tau - O_{2\tau}$. With $M_p(Y) = \langle Y^p \rangle$, define $A_8^{(0)}(Y) = \log[(M_8/M_2^4)/(M_8^{(0)}/(M_2^{(0)})^4)]$. At each depth, (0) means its own original distributed harmonic H_0, evaluated using ordered quadratic algebra for all declared M modes. It is neither the archived dressed harmonic proxy nor the retained interacting Wick comparator. Consequently this contrast includes altered two-point response and does not isolate irreducible non-Gaussian field information. The same widths correspond to $\tau\Omega_{\min} = 2.489, 3.520, 4.978$ for $R = 10$ and $0.156, 0.220, 0.311$ for $R = 14$. Thus $R = 10$ lies beyond its infrared cutoff while $R = 14$ lies inside its nominal resonance band. This is a common-width cutoff/depth comparison, not common-interior convergence. The three rows establish neither an extended scaling plateau nor a rare-tail law. The separately saved $R = 10$ occupation/selection refinement gives $10^6[M_8^{Q=16}(X_\tau)/M_8^{Q=14}(X_\tau) - 1] = +4.079, +5.474, -12.419$ at these widths. The largest corresponding absolute ordinary-moment change is 27.350 parts per million. No matching $R = 14$ occupation refinement is included in this panel; the $R = 10$ changes are not error bars for $R = 14$ or for the depth difference. The analysis retains the original odd-sector ground certificate and pressure-source diagnostics, and replays all three original $R = 14$ increment moments and all-mode harmonic references. The largest replay errors are 1.45×10^{-14} and 4.57×10^{-16}, respectively. The inherited independent arithmetic controls and necessary positive-moment checks through order eight pass. Actual ground centering is enforced, increment zero-gap entries are removed, and explicit filter-deletion bounds satisfy the frozen arithmetic gate. Scalar amplitudes and cumulants use 80-digit Decimal arithmetic, while vectors and eigenpairs remain binary64. Of the six ordinary/increment rows at each depth, direct spectral reconstruction is unresolved in 4 at $R = 10$ and 0 at $R = 14$. Those cases do not supply a direct spectral probability distribution or cumulative distribution, even though the vacuum recurrence passes its stated arithmetic checks. The largest conditional relative eighth-moment omitted-path envelopes are $e^{-1171.79}$ for $R = 10$ and $e^{162.05}$ for $R = 14$. These estimates assume an exact retained spectral subspace and a common exact ground state. They exclude errors in the original invariant subspace, ground

state, projected pressure source, primitive basis, bath selection and line-mode regulator; small path envelopes cannot substitute for those separate controls. An envelope at or above one is inconclusive as a relative-accuracy certificate and is retained as such.

TABLE XX. Extended fixed-R gap crossover. Contrasts use $Q = 16$; $\delta_8^Y = M_8^{Q=16}(Y)/M_8^{Q=14}(Y) - 1$ measures this occupation/selection refinement only. All rows use normalized vacuum-vector moments; direct spectral weights remain unresolved for both observables and both basis sizes.

τ	$A_8^{(0)}(O_\tau)$	$A_8^{(0)}(X_\tau)$	$10^6\delta_8^O$	$10^6\delta_8^X$
327.68	-10.922912	3.831327	-17.028	-14.778
368.64	-14.037487	10.047533	-13.574	-13.644
409.60	-17.518482	18.647866	-9.714	-11.541

4. Resolving the fixed-regulator gap crossover

We extend the saved $R = 10$, $\beta = 1$ full-cosine calculation to three longer widths with $M = 20$, $M_{\rm ref} = 24$, 32 primitive levels, five active modes and the 24-seed downward-closed bath of 128 configurations used in the saved $Q = 14$ and $Q = 16$ comparison. Let O_τ be the actual-ground-centered Gaussian average and $X_\tau = O_\tau - O_{2\tau}$. In this subsection $M_p(Y) = \langle Y^p\rangle$ and $A_8^{(0)}(Y) = \log[(M_8/M_2^4)/(M_8^{(0)}/(M_2^{(0)})^4)]$. The reference (0) is the original distributed harmonic H_0 evaluated with ordered quadratic algebra at all 20 declared modes. It is not the retained interacting Wick comparator, so the contrast combines altered two-point response with higher information. The widths have $\tau\Omega_{\rm min} = 5.632, 6.336, 7.040$, outside the $R = 10$ infrared interior. Together with $A_8^{(0)}(X_{289.631}) = -0.241262$, they establish a return through zero in the finite-regulator increment contrast. The increase is a normalized-shape effect: its raw eighth moment decreases throughout this extension. Neither a scaling plateau nor a rare-tail law follows. For the ordinary observable with a simple ground state, let Π_δ project onto the complete eigenspace at the smallest pressure-accessible excitation frequency δ, and set $v = \Pi_\delta P_c|0\rangle$, and $D_\delta = \Pi_\delta P_c \Pi_\delta$ on the entire excited block. When $D_\delta v \neq 0$, the leading returning paths give $M_{2k} \sim \|D_\delta^{k-1} v\|^2 e^{-\delta^2\tau^2}$: the two ground edges supply the cost and the intermediate zero-gap block supplies the prefactor. This condition holds in both saved nonlinear matrices and the harmonic reference across the declared source-threshold and near-degeneracy sensitivity checks. For $Q = 16$, the resulting conditional prediction is

$$A_8^{(0)}(O_\tau) \sim 0.80254504472 - 1.09201829768 \times 10^{-4}\,\tau^2. \tag{D3}$$

Its coefficient is $3(\delta_\beta^2 - \delta_0^2)$, with $\delta_\beta = 0.03377680165841$ and $\delta_0 = 0.03431141122418$. The new values agree with the full prefactor-and-rate prediction within 5.31×10^{-13} in A_8; the final measured τ^2 secant is $-1.09201829768 \times 10^{-4}$. This agreement concerns the saved matrices and their stated source. The increment annuls every zero-gap block, so higher-order paths traverse additional excited-state gaps. Its supported rates remain stable under the tested thresholds, but the leading coefficient is sensitive to numerical grouping of harmonic equal-cost paths. In the saved harmonic sensitivity rows, changing the relative tie tolerance from zero to 10^{-12} multiplies the sixth- and eighth-order candidate coefficients by two and four, respectively, without changing their rates; the nonlinear coefficients are unchanged. The states $|2,0\rangle$, $|1,1\rangle$ and $|0,2\rangle$ denote occupations of the two lowest harmonic normal modes, $\omega_0 < \omega_1$, with all other modes in the vacuum. They have successive gaps $\omega_1 - \omega_0$ and equal adjacent quadratic-source amplitudes. Their equal-cost half paths explain why coefficients must be grouped before squaring. Thus the strict-floating-tie increment intercept is not established as a unique physical asymptotic coefficient: the alternative grouping shifts the predicted $A_6^{(0)}$ and $A_8^{(0)}$ by $-\log 2$ and $-\log 4$. Actual finite-width values in the table use the all-mode harmonic reference and are unaffected by this path diagnostic. The harmonic leading-rate check includes every even Fock state below $8\omega_{\rm min}$ and excludes omitted-energy paths by a positive rate margin at orders two through eight. This is a rate-completeness check in the declared floating model, not a finite-width amplitude truncation. The nonlinear eighth-order supported increment path includes a pressure entry of about 1.80×10^{-9}; it survives the tested source thresholds, but those thresholds and the transform-roundoff scale are sensitivity diagnostics rather than a certified physical source-error bound. All 13 new rate, cancellation and rejection controls pass, and the earlier controlled arithmetic tests remain applicable. Replaying the prior long width agrees within 2.42×10^{-12} in raw moments. Necessary positive-moment checks pass through order eight; actual ground centering and increment zero diagonals are enforced. Nonzero reported statistics remain representable, and the explicit filter-deletion relative envelopes are below e^{-400}. All 12 new direct spectral reconstructions remain unresolved, so no direct probability distribution or cumulative distribution is inferred. Decimal scalar accumulation preserves small amplitudes without improving the binary64 eigenpairs or projected source. The conditional relative even-moment omitted-path envelopes are below e^{-6600}, assuming an exact retained spectral subspace and a common exact ground. They exclude the original subspace, state, source and physical-regulator errors. Together with the small $Q = 14 \to 16$ changes, these checks support the fixed-regulator crossover while leaving increasing-depth and continuum conclusions separate.

5. Executed pressure-chain tests in the nonlinear hierarchy

The executed finite-matrix test separates exciting the vacuum from the subsequent action of pressure within the excited sector. With the notation of Eq. (A16), $s = \|g\|^2 = M_2$ is the variance, whereas $q_6 = \|B^3 g\|^2$ is one positive excited-chain contribution to M_8. The latter measures three further pressure actions on the initial vacuum excitation. All calculations below use the actual interacting ground and original composite pressure projected into the supplied finite space at $\beta = 1$.

As a sensitivity test, retain only excited states satisfying $\tau(E_i - E_0)/\hbar \le 4$, replacing g by Πg and B by $\Pi B \Pi$. At the three specified central increments, Table XXI retains more than 99.9999% of the variance, but only about 11%, 43%, and 11% of q_6. Small direct vacuum excitation at higher energies therefore does not make subsequent excited-state propagation negligible. The time filter depends on energy *differences*; pressure can still connect excited states with nearby energies. These are ratios for a changed matrix. They are neither probability-error bounds nor fractions of the total M_8, which also contains the other terms and their interference in Eq. (A16).

The propagation ratio $J_8 = q_6/s^4$ also exposes a repeating crossover near the moving lowest gap. Table XXII uses the native $R = 10$ widths, $R = 14$ widths centered at $\tau = 819.2$, and native $R = 18$ widths. At each matched $\tau\Delta_e$ row the values nearly repeat across $R = 10, 14, 18$; as $\tau\Delta_e$ increases down the three rows, $\log J_8$ grows. At the still longer $R = 10$ width $\tau = 289.63094$, the positive norm recurrence gives $\tau\Delta_e \simeq 9.78281$ and $\log J_8 \simeq 229.56$. Direct spectral weights are unresolved in this last case; the recurrence does not certify its pressure probability law. The observed growth follows a finite-cutoff crossover and does not establish a persistent moment-order exponent. The modal dimensions are $M = 20, 24, 28$ and the fixed reference mode counts are $24, 24, 28$; these records establish neither a common-interior limit nor an unchanged-calibration depth limit.

Deleting a band of small energy transfers from B while keeping g fixed is a second, distinct matrix alteration. It is not additive attribution: for the $R = 10$, $Q = 16$ increment at $\tau = 51.2$, deleting $\tau|E_i - E_j|/\hbar \le 0.1, 0.5, 1$ gives respective q_6 ratios 0.999999081, 1.00160184, and 0.24678378. Wider deletion can first increase the chain norm because its matrix paths interfere. It cannot be read as a monotone accounting of independent events.

Six independent control groups pass, and all 42 ordinary/increment rows satisfy the stored positive-moment checks. Independently evaluated block identities agree with the full vector recurrences to better than 9.53×10^{-16}; original saved increment moments replay to 2.44×10^{-12}. The $R = 10$ changes from $Q = 14$ to 16 are at most 1.32×10^{-4} relatively in q_6 and 1.37×10^{-4} absolutely in $\log J_8$ across the tested rows. These measure combined occupation and selection sensitivity, not error bars. Direct pressure spectral reconstruction resolves 35 rows; seven longer-width $R = 10$ rows retain unresolved small spectral weights, so their norm recurrences do not certify a full spectral CDF. The deleted-entry estimates are small within their stated scope and exclude retained-entry, eigenpair and physical source errors. Outside-source, primitive, bath, mode and hierarchy refinements remain separate. Nonzero q_6 can also occur for quadratic pressure in Gaussian fields; this test alone identifies neither irreducible nonlinear field statistics nor measured temporal bursts.

6. Tested common energy ceilings do not stabilize the finite-matrix increment tail

At fixed $R = 10$, $M = 20$, reference mode count 24, $\beta = 1$ and $\tau = 51.2$, we applied the same interacting excitation-energy ceiling to the saved $Q = 14$ and $Q = 16$ models. The construction retains every represented even state below the ceiling and replaces the original composite pressure, projected into the supplied space, by its further compression $\Pi_E P \Pi_E$. This changes the operator whose vacuum spectral law is evaluated; it is not conditioning the original pressure law. The native models also differ in their historical target selection. The original all-M harmonic H_0 supplies a common standard deviation σ_0 for each observable, not a computed all-mode reference CDF.

The ordinary hard tail changes by only +0.02508% between $Q14$ and $Q16$, at all three ceilings; its probabilities at thresholds 8 and 9 coincide. It passes both prespecified 1% finite-matrix sensitivity targets. The increment fails both: the common-ceiling difference at threshold 8 ranges from −3.02384% to +60.89596%. Within $Q16$, raising the ceiling from 0.48 to 0.50 changes this probability by +33.68464%. Smoothing is a different resolution-dependent effect and is not uniformly stabilizing: even noise width σ_0 leaves a +1.33026% difference at ceiling 0.48.

Within each Q, all four positive even moments through order eight change by at most 8.88×10^{-16} relative when the ceiling changes. Nevertheless the hard increment tail moves substantially. The spectral mass in $8 < |X|/\sigma_0 \le 9$ accounts for 90.41% of the native threshold-8 difference and 99.48% at ceiling 0.50. This identifies a probability band without matching individual eigenvectors across dimensions. The stored signed telescoping comparison is not a unique causal attribution.

All 16 laws pass the recorded positivity, normalization and independent moment-reconstruction checks; all 160 threshold/noise comparisons resolve across the pressure eigensolvers. The largest independent moment discrepancy is 1.04×10^{-14} in the declared scale; the largest relative tail change across Hamiltonian eigensolvers is 3.76×10^{-13}. No filter-product or squared-overlap underflow occurs, and no probability mass lies in the numerical threshold-edge clusters. These arithmetic checks do not

Supplied model	τ	$\tau\Delta_e$	Variance kept (%)	q_6^{cut}/q_6
$R = 10$, $Q = 16$	51.2	1.72937	99.9999774	0.11008
$R = 14$, $Q = 10$	204.8	0.43258	99.9999305	0.43380
$R = 18$, $Q = 12$	13107.2	1.73085	99.9999779	0.11095

TABLE XXI. Joint excited-energy restriction for ratio-two pressure increments. Here $\Delta_e = (E_1 - E_0)/\hbar$ is the first gap in the supplied even sector. The three widths are stated separately; they do not constitute a common-window convergence test. No selected cut edge lies within the recorded eigenvalue uncertainty diagnostic.

Range of $\tau\Delta_e$	$\log J_8$, $R = 10$	$\log J_8$, $R = 14$	$\log J_8$, $R = 18$
1.2229–1.2239	12.87893	12.87884	12.88258
1.7294–1.7309	14.72826	14.73279	14.73789
2.4457–2.4478	20.55087	20.56554	20.57680

TABLE XXII. Natural logarithms of the increment propagation ratio at nearly matched lowest-gap positions. The widths are $(36.20387, 51.2, 72.40773)$ for $R = 10$, $(579.26188, 819.2, 1158.52375)$ for $R = 14$, and $(9268.19000, 13107.2, 18536.38000)$ for $R = 18$. Changing the averaging window with the lowest gap can repeat a crossover; it does not by itself establish a scale-independent law.

bound errors outside the supplied H/P matrices.

The same energy rule therefore does not remove the increment-tail uncertainty. A further retained-cluster test at fixed occupation and calibration would restore pressure paths absent from the saved native cluster; no such extension receives credit here. Bath, primitive, source and mode errors, the growing common-interior scaling limit and measurement interpretation remain separate. Neither moment agreement nor this finite tail test establishes a CDF certificate or persistent intermittency.

Ceiling	States $Q14/Q16$	$p_8(Q14)$	$p_8(Q16)$	Δp_8 (%)	Δp_9 (%)	$\Delta p_{8,1}$ (%)
Native	354/508	1.73494×10^{-3}	2.14429×10^{-3}	+23.59456	+2.64693	-0.07911
0.45	317/401	2.59639×10^{-3}	2.51788×10^{-3}	-3.02384	-1.12652	-0.30996
0.48	342/457	1.73292×10^{-3}	2.08387×10^{-3}	+20.25225	+12.74706	+1.33026
0.50	349/494	1.73144×10^{-3}	2.78582×10^{-3}	+60.89596	+0.36730	-0.01722

TABLE XXIII. Increment tails in the supplied finite matrices. Here $p_h = \Pr(|X| > h\sigma_0)$, with equality inside, and $p_{8,1} = \Pr(|X/\sigma_0 + Z| > 8)$ for independent standard-normal readout noise Z. Relative differences are $100[p(Q16)/p(Q14) - 1]$. A common energy ceiling changes the compressed pressure operator; these differences are sensitivity measurements, not probability error bounds.

Appendix E: Independent checks and ordering

An efficient independent implementation uses normal-ordered monomials $M_{pq} = (a^\dagger)^p a^q$, with multi-indices p, q. They are harmonic Liouvillian eigenoperators:

$$[H_0, M_{pq}]/\hbar = \nu_{pq} M_{pq}, \qquad \nu_{pq} = (p-q)\cdot\omega. \tag{E1}$$

Write $P = \sum_b P_b M_b$ in this monomial basis. If the commutator $[V, M_b]$ contains coefficient L_{ab} of M_a, then

$$D_a = \sum_b \frac{L_{ab}}{\hbar} g_\tau^{[1]}(\nu_a, \nu_b) P_b, \tag{E2}$$

where $g^{[1]}(x,y) = [g(x) - g(y)]/(x-y)$ and $g^{[1]}(x,x) = g'(x)$. With $\mathrm{ad}_H(X) = [H, X]$, this follows by differentiating $g_\tau(\mathrm{ad}_H/\hbar)$, or directly integrating the real-time commutator. It retains degenerate frequencies without dividing by an excited-state level splitting. Ordered Wick multiplication, rather than Fock-space diagonalization, then evaluates Eq. (22).

For normal-ordered monomials, the multiplication rule is

$$(a^\dagger)^p a^q (a^\dagger)^r a^s = \sum_{k \le \min(q,r)} \left[\prod_m \binom{q_m}{k_m}\binom{r_m}{k_m} k_m!\right] \times (a^\dagger)^{p+r-k} a^{q+s-k}. \tag{E3}$$

Only annihilators on the left contract with creators on the right. The implementation uses this ordered rule, with no occupation or energy cutoff, for up to three selected continuum modes. It agrees through fourth cumulants with a separate matrix derivative, including state and dynamics terms separately, for ordinary and increment filters. Analytic controls include $[a, a^\dagger] = 1$, vanishing harmonic-energy vacuum noise to first order, and an oscillator with $\omega = \hbar = 1$, $P = q^2 - 1/2$, $V = u : q^4 : /24$: its equal-time variance shift is $-u/32$ and harmonic fourth cumulant is 3.

The quadratic pair part of the first-order smeared operator has coefficient

$$(\delta B)_{mn} = \sum_r \frac{u_r c_{rm} c_{rn}}{2\hbar} \sum_{ij} W_{r,(i,j)} \times \left[g^{[1]}(-\Omega_{mn}, \Omega_{ij}) - g^{[1]}(-\Omega_{mn}, -\Omega_{ij})\right], \tag{E4}$$

with $\Omega_{mn} = \omega_m + \omega_n$. Since the harmonic smeared matrix is $B_{mn} = (B_0)_{mn} g(\Omega_{mn})$, the dynamics contribution to the variance is $\mathrm{Tr}(B\delta B)$. Here δB is part of the change in the smeared operator D, while the bare canonical pressure matrix B_0 remains fixed. The vacuum-preparation contribution is

$$-\sum_r \frac{u_r}{2\hbar} \sum_{p,q} \frac{W_{rp} W_{rq} g(\Omega_p) g(\Omega_q)}{\Omega_p + \Omega_q}. \tag{E5}$$

Combining these expressions and symmetrizing the ordered pair labels yields Eqs. (37) and (39). Thus the relative state/dynamics normalization can be checked without reference to a numerical implementation.

The matrix method evaluates the Fréchet derivative of the spectral filter. Degenerate excitation subspaces retain the g' term rather than a singular eigenvector rotation. Close nondegenerate gaps can use a direct divided-difference implementation. Powers of projected normal-ordered polynomials are distinguished from first projecting individual coordinates before forming those polynomials. Total occupation six suffices for the vacuum paths contributing through the first-order fourth pressure cumulant when all allowed energy states are kept; an additional energy cutoff must still be refined.

For branch elimination we retain all three modes of a one-element consistent-mass finite-element model. With the real-frequency convention $h_r(\omega) = K_{cr}/(K_{cr} + K_{gr} + K_{Jr} - C_r\omega^2)$, the explicit-branch calculation and a separate Wick calculation whose vertex legs are reconstructed as $h_r(\omega_m) T_m(a)$ agree. The Euclidean resolvent identity $G_{r\phi} = h_r G_{\phi\phi}$ is independently checked at six frequencies. This checks the harmonic branch-transfer reconstruction and its use in the microscopic ordered insertion calculation. It is distinct from an independently evaluated source-dependent reduced contour calculation, and does not identify omitted normal modes with eliminated branch variables.

Appendix F: Mode tails and scope of the continuum limit

The variance result admits a controlled exact-mode limit for $C_0, L_0, a, C_b, C_r, K_{cr}, K_{Jr} > 0$ and $K_b, K_{gr} \ge 0$, with finitely many branches and no coincidences among the combined branch and Dirichlet poles and a complete frequency-ordered exact-mode cutoff. The assumptions matter: the following estimates are not uniform when a boundary capacitance tends to zero or an infinite hierarchy of branches is added.

Write $x_{rm} = c_{rm} d_m$ and $y_{rm} = c_{rm} z_m$, so $W_{r,(m,n)} = x_{rm} x_{rn}/L_0 - C_0 y_{rm} y_{rn}$. Above all branch poles the secular equation and capacitance normalization imply

$$T_m(a) = \mathcal{O}(\omega_m^{-1}), \quad T_{rm} = \mathcal{O}(\omega_m^{-3}), \quad T_m'(a) = \mathcal{O}(\omega_m), \tag{F1}$$

$$x_{rm} = \mathcal{O}(\omega_m^{-3}), \quad y_{rm} = \mathcal{O}(\omega_m^{-4}). \tag{F2}$$

The branch variance converges since $c_{rm}^2 = \mathcal{O}(\omega_m^{-7})$. The harmonic pressure pair matrix after Gaussian averaging is Hilbert–Schmidt for $\tau > 0$, giving a finite harmonic vacuum variance.

For the Gaussian pressure window, let p, q denote positive pair frequencies and set $b_g = \tau^2$, so $F(p) = g(p)^2 =$

$e^{-b_g p^2}$. If $p \geq q > 0$, Eq. (39) gives

$$|K_g(p,q)| = \frac{F(q) + q[F(q) - F(p)]/(p-q)}{p+q} \leq \frac{1+2/e}{p+q}. \tag{F3}$$

The inequality follows from the mean value theorem and $2b_g s^2 e^{-b_g s^2} \leq 2/e$. Symmetry covers $p < q$ and the continuous limit covers equality. For an increment, g^2 is the sum of three Gaussians with coefficients $1, -2, 1$ and rates $b_g, 5b_g/2, 4b_g$; therefore $4(1+2/e)$ suffices as the numerator bound. The state kernel is bounded by $1/(p+q)$, and the dynamics term by the sum of these constants.

For finitely many branch poles, the high-frequency mode bound given below grows linearly with the mode index. Together with the amplitude estimates, it makes $X_r = \sum_m |x_{rm}|$ and $Y_r = \sum_m |y_{rm}|$ finite. Then $\sum_p |W_{rp}| \leq Q_r = X_r^2/L_0 + C_0 Y_r^2$. Since all pair frequencies are at least $2\omega_1 > 0$, these bounds prove absolute convergence of the state, dynamics, and total first-order variance sums separately. No subtraction between divergent state and dynamics terms is needed.

An explicit omitted-mode estimate is useful for a certified separation test. For retained modes $m \leq N$, define

$$X_{r,N}^{(w)} = \sum_{m>N} \frac{|x_{rm}|}{\omega_m}, \qquad Y_{r,N}^{(w)} = \sum_{m>N} \frac{|y_{rm}|}{\omega_m}, \tag{F4}$$

$$T_{r,N} = 2\left[\frac{X_r X_{r,N}^{(w)}}{L_0} + C_0 Y_r Y_{r,N}^{(w)}\right]. \tag{F5}$$

For one mode pair, either mode index can exceed the cutoff; summing those two contributions gives the factor two in $T_{r,N}$. In the double pair sum, either pair can contain an omitted mode. Bounding both contributions and combining them with the prefactor $1/(2\hbar)$ in Eq. (37) gives

$$|\delta\kappa_{2,\infty} - \delta\kappa_{2,N}| \leq \frac{C_g}{\hbar} \sum_r |u_r| Q_r T_{r,N}. \tag{F6}$$

Here C_g satisfies $|K_g(p,q)| \leq C_g/(p+q)$ for all positive pair frequencies in the sum. This is an absolute branch-wise estimate, so cancellations between branches cannot hide its error.

For completeness, all infinite sums in this estimate can be bounded from a finite prefix. Put $s_r = K_{cr} + K_{gr} + K_{Jr}$, $S = K_b + \sum_r K_{cr}$, $\Delta_\omega = \pi v/a$, $J = N - R - 1$, and $\Omega = \Delta_\omega J$, with R branches. For a cutoff above the branch poles set

$$I_* = a/2 - v/(4\Omega), \qquad \rho_r = s_r/(C_r\Omega^2), \tag{F7}$$

$$e_* = 1 - \frac{S}{C_b\Omega^2} - \sum_r \frac{K_{cr}^2}{C_b\Omega^2(C_r\Omega^2 - s_r)}. \tag{F8}$$

When $I_*, e_* > 0$ and $\rho_r < 1$, the exact sine norm and secular equation give envelopes

$$|x_{rm}| \leq A_r^* \omega_m^{-3}, \quad A_r^* = \frac{\hbar K_{cr}}{2C_r(1-\rho_r)C_b e_* I_*}, \tag{F9}$$

$$|y_{rm}| \leq B_r^* \omega_m^{-4}, \quad B_r^* = \frac{\hbar K_{cr}}{2C_r(1-\rho_r)L_0 C_b^2 e_*^2 I_*}. \tag{F10}$$

The ordered poles imply $\omega_m > \Delta_\omega(m - R - 1)$ at high mode number. Integral bounds on positive decreasing power sums therefore yield

$$\sum_{m>N} |x_{rm}| \leq \frac{A_r^*}{2\Delta_\omega^3 J^2}, \qquad \sum_{m>N} |y_{rm}| \leq \frac{B_r^*}{3\Delta_\omega^4 J^3}, \tag{F11}$$

$$X_{r,N}^{(w)} \leq \frac{A_r^*}{3\Delta_\omega^4 J^3}, \qquad Y_{r,N}^{(w)} \leq \frac{B_r^*}{4\Delta_\omega^5 J^4}. \tag{F12}$$

Combining these with the retained absolute sums evaluates Eq. (F6) and proves an $\mathcal{O}(J^{-3})$ remainder bound. A tighter Gaussian numerator uses $p_{\min} = 2\omega_1$, $x_0 = b_g p_{\min}^2$, and $z = \max(1, x_0)$: $C_g = e^{-x_0} + 2ze^{-z}$. For increments the three such constants are added with absolute weights $1, 2, 1$. The numerical package implements these formulas. They bound the exact omitted-mode sum conditional on the input modes; they are not interval-arithmetic certificates for root solving or floating-point accumulation.

The error need not be exponentially small in N. With q fixed and $p \to \infty$, $K_g(p,q) = -g(q)^2/p + \mathcal{O}(p^{-2})$. The vacuum-preparation term suppresses a high pair directly, but the dynamics term retains virtual high-frequency legs. This explains the algebraic tail and the inadequacy of inferring nonlinear convergence solely from the Gaussian external window.

This proof establishes the limit of the specified *first-order variance coefficient*. It does not establish an all-order interacting spectral law, a uniform weak-coupling expansion under regulator removal, or a theorem about every spatial discretization. The canonical pressure operator is held fixed as β changes, as in the finite regulators; the convergent coefficient requires no additional divergent contact subtraction in this prescription. A different eliminated-source observable requires its own matching. The isolated negative quartic Hamiltonian has no global stable ground state; throughout, its perturbation coefficients are interpreted through the stable microscopic completion rather than as a nonperturbative negative-quartic theory.

DATA AVAILABILITY

Code, exact inputs, executed data, validation records, and checksums for the reported numerical results are available in the reproducibility dataset [27] at https://doi.org/10.5281/zenodo.23215768, and in the accompanying reproducibility package.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

OpenAI Codex (GPT-6 Astra) assisted with derivations, implementation, literature checks, numerical validation, and manuscript preparation; the author reviewed the work and takes responsibility for its contents.

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